

1	1	/	FOCAL.ZZM	1
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7	7	/	F O C A L	7
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20	20	/	T H E F O R M U L A C A L C U L A T O R	20
21	21			21
22	22			22
23	23			23
24	24			24
25	25			25
26	26			26
27	27	/	FOR FORMULATING ON-LINE CALCULATIONS IN ALGEBRAIC LANGUAGE	27
28	28			28
29	29	/	ON THE FOLLOWING D.E.C. COMPUTERS:	29
30	30			30
31	31	/	5, 8, 8/I, 8/S, 8/L, LINC-8, LAB-8, TSS-8, PDP-12	31
32	32			32
33	33			33
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35	35			35
36	36			36
37	37			37
38	38			38
39	39	/	SOURCE RECREATION, TIDYING UP BY:	39
40	40			40
41	41	/	CHARLES J. LASNER P?S NY	41
42	42			42
43	43	/	LAST EDIT: 08-APR-1987 23:00:00 CJL	43
44				44
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12	56	/	F O C A L	AND	F L O A T	12
13	57					13
14	58					14
15	59					15
16	60					16
17	61					17
18	62	/	COPYRIGHT 1969 BY			18
19	63					19
20	64	/	DIGITAL EQUIPMENT CORPORATION			20
21	65					21
22	66	/	FOCAL IS A REGISTERED			22
23	67	/	TRADEMARK OF			23
24	68	/	DIGITAL EQUIPMENT CORPORATION			24
25	69					25
26	70	/	ALL RIGHTS RESERVED			26
27	71					27
28	72	/	3RD REVISION			28
29						29
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1	74	0000	FIXMRI	AND=	0000	/ACTUAL MEMORY REFERENCE INSTRUCTIONS	1
2	75	1000	FIXMRI	TAD=	1000		2
3	76	2000	FIXMRI	ISZ=	2000		3
4	77	3000	FIXMRI	DCA=	3000		4
5	78	4000	FIXMRI	JMS=	4000		5
6	79	5000	FIXMRI	JMP=	5000		6
7	80	5000	FIXMRI	FPOW=	5000	/PSEUDO-FLOATING POINT INSTRUCTIONS.	7
8	81	1000	FIXMRI	FADD=	1000		8
9	82	2000	FIXMRI	FSUB=	2000		9
10	83	4000	FIXMRI	FMUL=	4000		10
11	84	3000	FIXMRI	FDIV=	3000		11
12	85	0000	FIXMRI	FGET=	0000		12
13	86	6000	FIXMRI	FPUT=	6000		13
14	87						14
15	88	7000	FNOR=	7000			15
16	89	0000	FEXT=	0000			16
17	90	0000	FXIT=	0000			17
18	91	4407	FINT=	JMS I	7		18
19	92	7000	NOP=	7000			19
20	93	7200	CLA=	7200			20
21	94	7100	CLL=	7100			21
22	95	7040	CMA=	7040			22
23	96	7004	RAL=	7004			23
24	97	7020	CML=	7020			24
25	98	7010	RAR=	7010			25
26	99	7012	RTR=	7012			26
27	100	7006	RTL=	7006			27
28	101	7001	IAC=	7001			28
29	102	7500	SMA=	7500			29
30	103	7440	SZA=	7440			30
31	104	7510	SPA=	7510			31
32	105	7450	SNA=	7450			32
33	106	7420	SNL=	7420			33
34	107	7430	SZL=	7430			34
35	108	7410	SKP=	7410			35
36	109	7402	HLT=	7402			36
37	110	7041	CIA=	7041			37
38	111	6001	ION=	6001			38
39	112	6002	IOF=	6002			39
40	113	6031	KSF=	6031			40
41	114	6036	KRB=	6036			41
42	115	6041	TSF=	6041			42
43	116	6042	TCF=	6042			43
44	117	6044	TPC=	6044			44
45	118	6046	TLS=	6046			45
46	119	6026	PLS=	6026			46
47	120	6011	RSF=	6011			47
48	121	6012	RRB=	6012			48
49	122	6014	RFC=	6014			49
50	123	6101	SMP=	6101			50
51	124	6244	RMF=	6244			51
52							52
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1	126			/		* FOCAL * - BY RICK MERRILL - FOR THE FAMILY OF 8.	1
2	127						2
3	128			/		MISCELLANEOUS ITEMS	3
4	129						4
5	130		*0001		*1		5
6	131						6
7	132	000001	5403		JMP I .+2	/INTERRUPT PROCESSOR ENTRY.	7
8	133	000002	5403		JMP I .+1	/(USED BY PDP-5)	8
9	134	000003	2603		INTRPT		9
10	135						10
11	136	000004	0004	DDTJR,	DDTJR	/USED FOR DEBUGGING	11
12	137	000005	0013	P13,	13	/CONSTANT	12
13	138	000006	0100	C100,	100	/CONSTANT	13
14	139						14
15	140		0000		T= 00	/TEXT FIELD NO.	15
16	141		0000		P= 00	/PROGRAM FIELD NO.	16
17	142		7000		CDF= 7000	/(X-MEM) - OPR	17
18	143						18
19	144	000007	6400		FPNT	/ADDRESS OF FLOATING POINT INTERPRETER (LOC *7)	19
20	145						20
21	146			/		AUTO-INDEX REGISTERS - (START OF SAVE BY QUAD)	21
22	147						22
23	148	000010	0000	AXIN,	0	/STORAGE INDEX (LOC *10)	23
24	149	000011	0000	XRT,	0	/EXTRA XR	24
25	150	000012	0000	XRT2,	0	/EXTRA XR	25
26	151	000013	4370	PDLXR,	BEGIN-1	/PUSHDOWN LIST INDEX REGISTER.	26
27	152	000014	3117	FLT XR,	IOBUF-1	/XR FOR FLOATING POINT	27
28	153	000015	0000	FLT XR2,	0	/EXTRA FOR F.P.	28
29	154	000016	7402	TELSW,	HLT	/TELETYPE IN PROGRESS SWITCH	29
30	155						30
31	156		0017		TEXTP=.	/TEXT POINTERS (LOC *17)	31
32	157						32
33	158	000017	3215	AXOUT,	FRSTX	/OUTPUT INDEX	33
34	159	000020	0000	XCT,	0	/UNPACK SWITCH	34
35	160	000021	0000	GTEM,	0	/UNPACK STORAGE	35
36	161	000022	2407	PC,	FLTZER	/PROGRAM COUNTER	36
37	162						37
38	163	000023	0000	THISLN,	0	/LINE POINTER FROM 'FINDLN'	38
39	164	000024	0000	THISOP,	0	/CURRENT 'EVAL' OPERATION	39
40	165	000025	0000	LASTLN,	0	/BACK POINTER FROM 'FINDLN'	40
41	166	000026	0001	DEBGSW,	1	/DEBUG SWITCH ; NON-ZERO FOR LITERAL.	41
42	167	000027	0000	PACKST,	0	/RUBOUT PROTECTION	42
43	168	000030	0000	PT1,	0	/VARIABLE POINTER	43
44	169	000031	3217	LASTV,	BUFBEQ	/ADDRESS OF LAST VARIABLE	44
45	170	000032	0000	T1,	0	/TEMPORARY REGISTER - MAIN	45
46	171	000033	0000	T3,	0	/TEMP REGISTER FOR OUTPUT	46
47	172	000034	0000	INBUF,	0	/KEYBOARD INPUT BUFFER	47
48	173	000035	4370	BOTTOM,	BEGIN-1	/LAST LOCATION CURRENTLY AVAILABLE IN FIELD ZERO **	48
49	174	000036	0000	INSUB,	0	/0= GETC; #0 = READC	49
50	175	000037	0000	HINBUF,	0	/HIGH SPEED INPUT BUFFER	50
51	176						51
52	177			/		*40 = FLOATING POINT	52
53							53
54							54
55							55
56							56
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1	179	*0054		*54		
2	180					
3	181	/		VARIABLES - INITIALIZED FOR THE DIALOGUE		
4	182					
5	183	000054	0000	SORTCN,	0	/NUMBER IN TABLE FROM SORTC
6	184	000055	0000	LASTOP,	0	/LAST OPERATION FOR EVAL
7	185					
8	186	0056		EFOP=	.	/FUNCTION CODE.
9	187					
10	188	000056	0000	ATSW,	0	/ASK-TYPE SWITCH
11	189	000057	7760	CNTR,	-20	/DELETE AND ERROR COUNTER (USED BY F.P. ALSO)
12	190					
13	191	0060		STARTV=	.	/=END FOR 8K
14	192					
15	193	000060	3217	BUFR,	BUFBEQ	/NEXT LOCATION IN BUFFER = LAST LOCATION OF TEXT.
16	194					
17	195	000061	1354	ADD,	OUTL	/CHAR. BUF. IN. (DEBUG AIDS. SEE BELOW.)
18	196	000062	2414	XCTIN,	I33	/PACK SWITCH
19	197	000063	2676	OUTDEV,	XOUTL	/POINTER TO OUT. SUB. (OUTL)-FOR DEBUGGING
20	198	000064	2666	INDEV,	XI33	/POINTER TO IN. SUB. (I33)-FOR DEBUGGING
21	199					
22	200	000065	0001	NAGSW,	0001	/NOT ALL AND/OR GROUP SWITCH (4000=ONE; 1=ALL;
23	201					
24	202	000066	0215	CHAR,	"M&237	/THE MOST IMPORTANT REGISTER
25	203	000067	0000	LINENO,	0000	/LINE NUMBER READ BY GETLN; (0400)-FOR TSS-8
26	204	000070	0005	GINC,	WORDS+2	/=6 FOR 4-WORD - CONSTANT
27	205					
28	206	000071	0000	T2,	0	/TEMP REGISTER - FOR NEW INST. ROUTINES.
29	207					
30	208					
31	209					
32	210	/		FOR DEBUGGING, SET OUTL AND I33 INTO OUTDEV AND INDEV;		
33	211	/		ALSO PATCH THE ERROR ROUTINE = FOUR		
34	212	/		PATCHES PLUS TWO FOR THE HIGH SPEED READER.		
35	213					
36	214					
37	215	0072		LIST6=	.	/INPUT LIST FOR "SFOUND".
38	216					
39	217	000072	0214			"L&237
40	218	000073	0207			"G&237
41	219					
42	220	0074		LIST7=	.	
43	221					
44	222	000074	0203			"C&237
45	223	000075	0337	P337,	"^+1	/CONTROL-C FOR DEBUGGING AND TSS8
46	224	000076	0212	CLF,	"J&237	/LEFT ARR
47	225					
48	226	0077		LIST3=	.	/EXCRETION LIST
49	227					
50	228	000077	0215	CCR,	"M&237	/LIST BRANCHER.
51	229	000100	7402	DMP5W,	HLT	/((SEARCH CHARACTER)-VARIABLE; =0000 FOR TRACE ON.
52	230					
53	231					
54	232	/		THE REST OF PAGE ZERO IS PURE TO THE MULTI-USER SYSTEM		
55						
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58						

1	234	0101	M100=	.		1
2	235					2
3	236	000101	7700	P7700,	7700	3
4	237	000102	0256	PER,	"	4
5	238	000103	7701	M77,	-77	5
6	239	000104	7600	P7600,	7600	6
7	240	000105	7760	M20,	-20	7
8	241	000106	0177	P177,	177	8
9	242	000107	0017	P17,	17	9
10	243	000110	0277	P277,	"?	10
11	244	000111	7776	M2,	-2	11
12	245	000112	7477	MINUSA,	-"A	12
13	246	000113	0260	C260,	"0	13
14	247	000114	7540	M240,	-"	14
15	248	000115	7522	MPER,	-"	15
16	249	000116	7563	MCR,	-"M!100	16
17	250	000117	7775	MFLT,	-WORDS	17
18	251	000120	7773	M5,	-5	18
19	252	000121	7767	M11,	-11	19
20	253	000122	0077	P77,	77	20
21	254	000123	0200	C200,	200	21
22	255	000124	4000	P4000,	4000	22
23	256					23
24	257	000125	2030	FLARGP,	FLARG	24
25	258	000126	2155	PTCH,	CHIN	25
26	259	000127	5715	DOUBLE,	MULT2	26
27	260	000130	6000	FOUTPUT,	FLOUTP	27
28	261	000131	6200	FINPUT,	FLINTP	28
29	262	000132	3140	COMBUF,	COMEIN	29
30	263	000133	3206	CFRS,	FRST	30
31	264	000134	3140	END,	COMEIN	31
32	265	000135	3217	ENDT,	BUFBEQ	32
33	266	000136	2017	EFUN3I,	EFUN3	33
34	267	000137	2407	CFRSX,	FLTZER	34
35	268					35
36	269					36
37	270		/	'FINPUT'	USES CHAR AND GETC OR READC TO DEVELOP	37
38	271		/	A NUMBER	WHICH IS THEN STORED VIA PT1.	38
39	272					39
40	273	0003	WORDS=	3	/OR 4	40
41						41
42						42
43						43
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1	275		/	NEW INSTRUCTIONS:	1
2	276				2
3	277	4540		PUSHJ= JMS I .;XPUSHJ/RECURSIVE SUBROUTINE CALL	3
4	278	000140	0521		4
5	279		1413	POPA= TAD I PDLXR /RESTORE AC	5
6	280				6
7	281		5541	POPJ= JMP I .;XPOPJ /SUBROUTINE RETURN	7
8	282	000141	1565		8
9	283		4542	PUSHA= JMS I .;XPUSHA/SAVE AC	9
10	284	000142	0477		10
11	285		4543	PUSHF= JMS I .;PD2 /SAVE GROUP OF DATA	11
12	286	000143	0534		12
13	287		4544	POPF= JMS I .;PD3 /RESTORE GROUP	13
14	288	000144	0554		14
15	289		4545	GETC= JMS I .;UTRA /UNPACK A CHARACTER	15
16	290	000145	2274		16
17	291		4546	PACKC= JMS I .;PACBUF/PACK A CHARACTER	17
18	292	000146	2502		18
19	293		4547	SORTJ= JMS I .;SORTB /SORT AND BRANCH ON AC OR CHAR	19
20	294	000147	1314		20
21	295		4550	SORTC= JMS I .;XSORTC/SORT CHAR	21
22	296	000150	0721		22
23	297		4551	PRNTC= JMS I .;OUT /PRINT AC OR CHAR	23
24	298	000151	2465		24
25	299		4552	RDIV, READC= JMS I .;CHIN /READ DATA INTO CHAR AND PRINT IT	25
26	300	000152	2155		26
27	301		4553	PRNTLN= JMS I .;XPRNT /PRINT C(LINENO)	27
28	302	000153	2425		28
29	303		4554	GETLN= JMS I .;XGETLN/UNPACK AND FORM A LINENUMBER	29
30	304	000154	0302		30
31	305		4555	FINDLN= JMS I .;XFIND /SEARCH FOR A GIVEN LINE	31
32	306	000155	2242		32
33	307		4556	ENDLN= JMS I .;XENDLN/INSERT LINE POINTERS	33
34	308	000156	2360		34
35	309		4557	RTL6= JMS I .;XRTL6 /ROTATE LEFT SIX	35
36	310	000157	0413		36
37	311		4560	SPNOR= JMS I .;XSPNOR/IGNORE SPACES AND LEADING ZEROS	37
38	312	000160	1517		38
39	313		4561	TESTN= JMS I .;XTESTN/PERIOD; OTHER; NUMBER	39
40	314	000161	1533		40
41	315		4562	TSTLPR= JMS I .;LPRTST/SKIP IF 5<SORTCN<= 11 (I.E. AN L-PAR)	41
42	316	000162	2035		42
43	317		4563	TSTGRP= JMS I .;GRPTST/SKIP IF G(AC) = G(LINENO)	43
44	318	000163	0744		44
45	319		4564	TESTC= JMS I .;XTESTC/TERM; NUMBER; FUNCTION; LETTER- AND IGNORE SPACES.	45
46	320	000164	0700		46
47	321		4565	PSIN, DELETE= JMS I .;XDELET/REMOVE OLD TEXT LINE	47
48	322	000165	2062		48
49	323		4566	ERROR2= JMS I .; /EXCESS SOMETHING ERROR	49
50	325				50
51	326		4566	ERROR3= JMS I .; /MISCELLANEOUS ERROR	51
52	328				52
53	329		4566	ERROR4= JMS I .;ERR2 /FORMAT ERROR	53
54	330	000166	2726		54
55	331		/	USED BY 8K	55
56					56
57					57
58					58

1	333		/	FOCAL'S COMMAND/INPUT DRIVER	1
2	334				2
3	335	*0176		*176	3
4	336				4
5	337	000176	4371	BEGIN	5
6	338			/BECOMES (RECOVR+1) **	6
7	339	000177	7610	START, SKP CLA	7
8	340	000200	5576	JMP I .-2	8
9	341	000201	1137	TAD CFRSX	9
10	342	000202	3022	DCA PC	10
11	343	000203	7001	IAC	11
12	344	000204	3100	DCA DMPSW	12
13	345	000205	3026	DCA DEBGSW	13
14	346	000206	1226	TAD COMBOT	14
15	347	000207	3013	DCA PDLXR	15
16	348	000210	1225	TAD CSTAR	16
17	349	000211	4551	PRINTC	17
18	350	000212	1132	IBAR, TAD COMBUF	18
19	351	000213	3010	DCA AXIN	19
20	352	000214	3062	DCA XCTIN	20
21	353	000215	1132	TAD COMBUF	21
22	354	000216	3027	DCA PACKST	22
23	355	000217	4552	IGNOR, READC	23
24	356	000220	4547	SORTJ	24
25	357	000221	0073	LIST7-1	25
26	358	000222	0474	INLIST-LIST7	26
27	359	000223	4546	PACKC	27
28	360	000224	5217	JMP IGNOR	28
29	361				29
30	362	000225	0252	CSTAR, **	30
31	363	000226	3220	COMBOT, COMEOUT+12	31
32				/END OF COMMAND BUFFER, LESS PROTECTION COUNT.	32
33					33
34					34
35					35
36					36
37					37
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1	365		/	COMMAND/INPUT PROCESSOR			1
2	366						2
3	367	000227	4546	IRETN,	PACKC	/START TO PACK C.R.	3
4	368	000230	4546		PACKC	/FINISH C.R.	4
5	369	000231	1132		TAD	COMBUF	5
6	370	000232	3017	GONE,	DCA	AXOUT	6
7	371	000233	3020		DCA	XCT	7
8	372	000234	4545		GETC		8
9	373	000235	1035		TAD	BOTTOM	9
10	374	000236	3013		DCA	PDLXR	10
11	375	000237	4560		SPNOR		11
12	376	000240	4561		TESTN		12
13	377	000241	5362		JMP	GZERR	13
14	378	000242	5271		JMP	INPUTX	14
15	379	000243	2026		ISZ	DEBGSW	15
16	380	000244	4554		GETLN		16
17	381	000245	1124		TAD	P4000	17
18	382	000246	1065		TAD	NAGSW	18
19	383	000247	7640		SZA CLA		19
20	384	000250	4566		ERROR3		20
21	385	000251	1060		TAD	BUFR	21
22	386	000252	3010		DCA	AXIN	22
23	387	000253	3062		DCA	XCTIN	23
24	388	000254	1067		TAD	LINENO	24
25	389	000255	3410		DCA I	AXIN	25
26	390	000256	4560		SPNOR		26
27	391	000257	7410		SKP		27
28	392	000260	4545		GETC		28
29	393	000261	4546	SRETN,	PACKC		29
30	394	000262	1066		TAD	CHAR	30
31	395	000263	1116		TAD	MCR	31
32	396	000264	7640		SZA CLA		32
33	397	000265	5260		JMP	.-5	33
34	398	000266	4565		DELETE		34
35	399	000267	4556		ENDLN		35
36	400	000270	5177		JMP	START	36
37	401						37
38	402	000271	4540	INPUTX,	PUSHJ		38
39	403	000272	0611		PROC		39
40	404	000273	1422		TAD I	PC	40
41	405	000274	7450		SNA		41
42	406	000275	5177		JMP	START	42
43	407	000276	3022		DCA	PC	43
44	408	000277	1022		TAD	PC	44
45	409	000300	7001		IAC		45
46	410	000301	5232		JMP	GONE	46
47	411						47
48	412			/	TEXT LINE BUFFER FORMAT*		48
49	413						49
50	414			/	#1 : POINTER OR ZERO IN LAST		50
51	415			/	#2 : LINENO		51
52	416			/	#3 - #N+1 : TEXT		52
53	417			/	#N : C.R.		53
54							54
55							55
56							56
57							57
58							58

1	419		/	LINE NUMBER FORMATION		1
2	420					2
3	421	000302	0000	XGETLN, 0	/DEVELOP I.D. - "GETLN"	3
4	422	000303	4560	SPNOR	/IGNORE LEADING SPACES.	4
5	423	000304	1066	TAD CHAR	/"ALL" IS A SPECIAL ARGUMENT.	5
6	424	000305	1112	TAD MINUSA		6
7	425	000306	7650	SNA CLA		7
8	426	000307	5322	JMP TESTA		8
9	427	000310	3036	DCA INSUB	/CALL 'GETC' FROM 'INPUT' VIA 'DECON'	9
10	428	000311	4771	JMS I LCON	/((DECONV - IN FLOAT.)	10
11	429	000312	1047	TAD FLAC+3	/GROUP TOO LARGE?	11
12	430	000313	0372	AND P7740		12
13	431	000314	1046	TAD FLAC+2		13
14	432	000315	7640	SZA CLA		14
15	433	000316	4566	ERROR2	/GROUP NUMBER TOO LARGE	15
16	434	000317	1047	TAD FLAC+3		16
17	435	000320	4557	RTL6		17
18	436	000321	7004	RAL		18
19	437	000322	3067	TESTA, DCA LINENO		19
20	438	000323	4561	TESTN	/TEST3	20
21	439	000324	4545	GETC	/READ STEP NUMBER.	21
22	440	000325	4561	TESTN	/TEST4, OTHER	22
23	441	000326	5340	JMP GERR	/DOUBLE PERIODS	23
24	442	000327	5352	JMP GEXIT	/OTHER	24
25	443	000330	1054	TAD SORTCN	/NUMBER	25
26	444	000331	7106	CLL RTL		26
27	445	000332	1054	TAD SORTCN		27
28	446	000333	7004	RAL		28
29	447	000334	1067	TAD LINENO		29
30	448	000335	3067	DCA LINENO		30
31	449	000336	4545	GETC	/READ SECOND STEP NUMBER.	31
32	450	000337	4561	TESTN	/TEST4, OTHER	32
33	451	000340	4566	GERR, ERROR4	/DOUBLE PERIODS	33
34	452	000341	5352	JMP GEXIT	/OTHER	34
35	453	000342	1054	TAD SORTCN	/NUMBER	35
36	454	000343	1067	TAD LINENO		36
37	455	000344	3067	DCA LINENO		37
38	456	000345	4545	GETC	/TEST FOR CORRECT TERMINATOR	38
39	457	000346	4561	TESTN	/CHECK SIZE	39
40	458	000347	5340	JMP GERR	/.	40
41	459	000350	7410	SKP		41
42	460	000351	4566	ERROR2	/TOO LARGE A LINE NUMBER.	42
43						43
44						44
45						45
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47						47
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58						58

1	462	000352	7100	GEXIT,	CLL		/CLEAR LINK BIT	1
2	463	000353	1067		TAD	LINENO	/TEST FOR GROUP NUMBER.	2
3	464	000354	0104		AND	P7600		3
4	465	000355	7640		SZA	CLA		4
5	466	000356	7020		CML			5
6	467	000357	1067		TAD	LINENO		6
7	468	000360	0106		AND	P177	/REPAIRE "NAGSW"	7
8	469	000361	7460		SNL	SZA		8
9	470	000362	4566	GZERR,	ERROR2		/0.X = ERROR: ILLEGAL LINE NUMBER.	9
10	471	000363	7640		SZA	CLA		10
11	472	000364	1373		TAD	P2000		11
12	473	000365	7020		CML			12
13	474	000366	7004		RAL			13
14	475	000367	3065		DCA	NAGSW		14
15	476	000370	5702		JMP I	XGETLN		15
16	477							16
17	478	000371	5600	LCON,	DECONV			17
18	479	000372	7740	P7740,	7740			18
19	480	000373	2000	P2000,	2000			19
20	481							20
21	482			/		RANGE OF ACCEPTIBLE LINE NUMBERS = 1.01 TO 31.99		21
22	483			/		NAGSW:		22
23	484							23
24	485			/		GROUP=0000		24
25	486			/		LINE=4000		25
26	487			/		ALL=0001		26
27	488							27
28	489			/		LIST OF FUNCTION ADDRESSES. (NAMES ARE IN "FNTABL")		28
29	490							29
30	491		0374		FNTABF=	.		30
31	492							31
32	493	000374	2014		XABS		/ABS -ABSOLUTE VALUE	32
33	494	000375	2010		XSGN		/SGN -SIGN PART	33
34	495	000376	1160		XINT		/ITR -INTEGER PART	34
35	496	000377	1142		XDYS		/DIS -DISPLAY AND INTENSIFY	35
36	497	000400	1553		XRAN		/RAN -RANDOM NUMBER	36
37	498	000401	1343		XADC		/ADC -READ ANALOG TO DIGITAL CONVERTER	37
38	499	000402	5000		ARTN		-	38
39	500	000403	4620		FEXP		/EXP -EXPONENTIAL FUNCTIONS	39
40	501	000404	5040		FLOG		/LOG -	40
41	502	000405	5205		FSIN		/SIN -TRIG FUNCTIONS	41
42	503	000406	5200		FCOS		/COS -	42
43	504	000407	7400		XSQRT		/SQT -SQUARE ROOT	43
44	505	000410	2725		ERROR5		/NEW -USER DEFINED FUNCTIONS	44
45	506	000411	2725		ERROR5		/COM -	45
46	507	000412	2725		ERROR5		/X -	46
47	508							47
48	509	000413	0000	XRTL6,	0		/ROTATE AC LEFT SIX - "RTL6"	48
49	510	000414	7106		CLL	RTL		49
50	511	000415	7006		RTL			50
51	512	000416	7006		RTL			51
52	513	000417	5613		JMP I	XRTL6		52
53								53
54								54
55								55
56								56
57								57
58								58

1	515		/	RECURSIVE OPERATE, EXECUTE, OR CALL	1
2	516				2
3	517	000420	4554	DO, GETLN /EXECUTE ONE LINE, A GROUP, OR ALL	3
4	518	000421	1022	TAD PC /SAVE ADDRESS	4
5	519	000422	4542	PUSHA /OF CURRENT LINE	5
6	520	000423	4543	PUSHF /SAVE REST OF THIS LINE	6
7	521	000424	0017	TEXTTP /ADDRESS OF TEXT POINTERS	7
8	522	000425	4543	DGRP, PUSHF /SAVE NAGSW; CHAR; AND LINENO.	8
9	523	000426	0065	NAGSW	9
10	524	000427	1065	TAD NAGSW /CHECK DATA FROM GETLN.	10
11	525	000430	7710	SPA CLA /SKIP IF GROUP OR ALL	11
12	526	000431	5263	JMP DOONE /DO ONE LINE	12
13	527	000432	4555	FINDLN /INIT FOR GROUP AND SET THISLN	13
14	528	000433	7000	NOP	14
15	529	000434	1023	TAD THISLN /TEST FOR GOOD GROUP NUMBER.	15
16	530	000435	3011	DCA XRT	16
17	531	000436	1411	TAD I XRT /(X-MEM)	17
18	532	000437	4563	TSTGRP	18
19	533	000440	4566	ERROR2 /NO SUCH GROUP NUMBER	19
20	534	000441	4540	DGRP1, PUSHJ /EXECUTE OBJECT LINE AND SET PC.	20
21	535	000442	0606	PROCESS-2	21
22	536	000443	4544	POPF /RESTORE THE DATA	22
23	537	000444	0065	NAGSW	23
24	538	000445	1422	TAD I PC /CHECK FOR END OF TEXT (X-MEM)	24
25	539	000446	7450	SNA	25
26	540	000447	5271	JMP DCONT /ALL DONE	26
27	541	000450	7001	IAC	27
28	542	000451	3030	DCA PT1 /SAVE POINTER TO LINENO	28
29	543	000452	1065	TAD NAGSW /CHECK FOR GROUP	29
30	544	000453	7740	SMA SZA CLA	30
31	545	000454	5260	JMP .+4 /DO ALL	31
32	546	000455	1430	TAD I PT1 /TEST GROUP (X-MEM)	32
33	547	000456	4563	TSTGRP	33
34	548	000457	5271	JMP DCONT /NOT IN GROUP	34
35	549	000460	1430	TAD I PT1 /READ NEXT LINE NO. (X-MEM)	35
36	550	000461	3067	DCA LINENO	36
37	551	000462	5225	JMP DGRP /CONTINUE THE SUBROUTINE	37
38	552				38
39	553	000463	4555	DOONE, FINDLN /FIND THE LINE	39
40	554	000464	4566	ERROR2 /NO SUCH LINE NUMBER	40
41	555	000465	4540	PUSHJ /EXECUTE IT	41
42	556	000466	0610	PROCESS	42
43	557	000467	4544	POPF /RESTORE CHAR	43
44	558	000470	0065	NAGSW	44
45	559	000471	4544	DCONT, POPF /RESTORE TEXT POINTERS	45
46	560	000472	0017	TEXTTP	46
47	561	000473	1413	POPA /RESTORE ADDRESS OF CURRENT LINE.	47
48	562	000474	3022	DCA PC	48
49	563	000475	5676	JMP I .+1; PROC /CONTINUE PROCESSING THIS LINE.	49
50	564	000476	0611		50
51					51
52					52
53					53
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1	566			/		PUSHDOWN LIST CONTROLS	1
2	567						2
3	568	000477	0000	XPUSHA,	0	/PUSHDOWN THE AC - "PUSHA"	3
4	569	000500	3071	DCA	T2	/BACKUP POINTER	4
5	570	000501	7040	CMA		/AND THEN	5
6	571	000502	4310	JMS	PCHK	/CHECK CORE USAGE	6
7	572	000503	1071	TAD	T2	/OK	7
8	573	000504	3413	DCA I	PDLXR	/PUSH DOWN LIST POINTER	8
9	574	000505	7040	CMA		/BACKUP AGAIN	9
10	575	000506	4310	JMS	PCHK		10
11	576	000507	5677	JMP I	XPUSHA		11
12	577						12
13	578	000510	0000	PCHK,	0		13
14	579	000511	1013	TAD	PDLXR	/INC IN AC	14
15	580	000512	3013	DCA	PDLXR		15
16	581	000513	1013	TAD	PDLXR		16
17	582	000514	7141	CLL CIA			17
18	583	000515	1031	TAD	LASTV		18
19	584	000516	7630	SZL CLA			19
20	585	000517	4566	ERROR3		/STORAGE FILLED BY PUSH-DOWN LIST	20
21	586	000520	5710	JMP I	PCHK		21
22	587						22
23	588	000521	0000	XPUSHJ,	0	/RECURSIVE SUBROUTINE CALL - "PUSHJ"	23
24	589	000522	1721	TAD I	XPUSHJ		24
25	590	000523	3071	DCA	T2	/SAVE SUBR. ADDR.	25
26	591	000524	7040	CMA			26
27	592	000525	4310	JMS	PCHK		27
28	593	000526	1321	TAD	XPUSHJ		28
29	594	000527	7001	IAC			29
30	595	000530	3413	DCA I	PDLXR	/SAVE RETURN	30
31	596	000531	7040	CMA			31
32	597	000532	4310	JMS	PCHK		32
33	598	000533	5471	JMP I	T2	/TRANSFER CONTROL	33
34	599						34
35	600	000534	0000	PD2,	0	/SAVE A FLOATING POINT NUMBER - "PUSHF"	35
36	601	000535	7240	CLA CMA		/COMPUTE VARIABLE ADDR	36
37	602	000536	1734	TAD I	.-2		37
38	603	000537	3011	DCA	XRT		38
39	604	000540	2334	ISZ	PD2	/FIX RETURN	39
40	605	000541	1117	TAD	MFLT	/COMPUTE PUSH. POINTER	40
41	606	000542	4310	JMS	PCHK		41
42	607	000543	1117	TAD	MFLT		42
43	608	000544	3071	DCA	T2		43
44	609	000545	1411	TAD I	XRT		44
45	610	000546	3413	DCA I	PDLXR		45
46	611	000547	2071	ISZ	T2		46
47	612	000550	5345	JMP	.-3		47
48	613	000551	1117	TAD	MFLT	/RESET POINTER	48
49	614	000552	4310	JMS	PCHK		49
50	615	000553	5734	JMP I	PD2		50
51							51
52							52
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56							56
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1	617	000554	0000	PD3,	0	/RESTORE A FLOATING POINT NUMBER - "POPF"	1
2	618	000555	7240		CLA CMA	/GET VAR. ADDR.	2
3	619	000556	1754		TAD I PD3		3
4	620	000557	2354		ISZ PD3		4
5	621	000560	3011		DCA XRT		5
6	622	000561	1117		TAD MFLT		6
7	623	000562	3071		DCA T2		7
8	624	000563	1413		TAD I PDLXR	/MOVE	8
9	625	000564	3411		DCA I XRT		9
10	626	000565	2071		ISZ T2		10
11	627	000566	5363		JMP .-3		11
12	628	000567	5754		JMP I PD3	/EXIT	12
13	629						13
14	630		0570		INLIST= .	/INPUT CONTROL CHARACTERS	14
15	631						15
16	632	000570	2740		RECOVR	/C.C. = BREAK	16
17	633	000571	0212		IBAR	/B.A. = RESTART	17
18	634	000572	0217		IGNOR	/L.F. = IGNORE	18
19	635	000573	0227		IRETN	/C.R. = TERMINATE STRING	19
20	636						20
21	637	000574	1075	FLIST2,	FLIMIT	/,=STANDARD	21
22	638	000575	1137		FINFIN	/;=SHORT	22
23	639	000576	2725		ERROR5	/CR=DUMB	23
24	640						24
25	641	000577	1065	FLIST1,	FINCR	/,=STANDARD FORMAT	25
26	642	000600	0610		PROCESS	/;=SET; PLUS ,..	26
27	643	000601	0614		PC1	/C.R.=SET COMMAND.	27
28	644						28
29	645	000602	7472	MF,	-"F	/USED BY TESTC	29
30							30
31							31
32							32
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1	647			/	PRIMARY CONTROL AND TRANSFER		1
2	648						2
3	649	000603	4554	GOTO,	GETLN	/READ THE LINE NUMBER REQUESTED	3
4	650	000604	4555		FINDLN	/LOCATE IT AND RESET TEXTP	4
5	651	000605	4566		ERROR2	/NOT THERE	5
6	652	000606	1023		TAD THISLN	/SET PC	6
7	653	000607	3022		DCA PC		7
8	654	000610	4545	PROCESS,	GETC	/TEST FOR END OF LINE	8
9	655	000611	1066	PROC,	TAD CHAR	/FIRST CHARACTER READY = USE PROC	9
10	656	000612	1116		TAD MCR		10
11	657	000613	7650		SNA CLA		11
12	658	000614	5541	PC1,	POPJ	/EXIT "PROCESS"	12
13	659	000615	4550		SORTC	/IGNORE "SPACE", " ", AND ";".	13
14	660	000616	1376		GLIST-1		14
15	661	000617	5210		JMP PROCESS		15
16	662	000620	1066		TAD CHAR	/SAVE COMMAND CHARACTER	16
17	663	000621	0075		AND P337	/EXECUTE LOWER CASE ALSO	17
18	664	000622	4542		PUSHA		18
19	665	000623	4545		GETC	/GO TO TERMINATOR	19
20	666	000624	4550		SORTC		20
21	667	000625	1376		GLIST-1		21
22	668	000626	7410		SKP		22
23	669	000627	5223		JMP . -4		23
24	670	000630	1413		POPA		24
25	671	000631	4547		SORTJ	/GO DO COMMAND	25
26	672	000632	0773		COMLST-1		26
27	673	000633	0167		COMGO-COMLST		27
28	674	000634	4566		ERROR2	/ILLEGAL COMMAND	28
29	675						29
30	676		0614		COMMENT=PC1	/ALSO IS CONTINUE	30
31	677						31
32							32
33							33
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1	679		/	OUTPUT COMMAND TEXT			1
2	680						2
3	681	000635	4554	WRITE,	GETLN	/SET LINENO	3
4	682	000636	2026		ISZ DEBGSW	/DISABLE TRACE	4
5	683	000637	4555		FINDLN	/SEARCH FOR LINE NUMBER	5
6	684	000640	5267		JMP WTESTG	/NOT THERE OR GROUP	6
7	685	000641	1067		TAD LINENO		7
8	686	000642	7640		SZA CLA		8
9	687	000643	4553		PRNTLN	/PRINT LINE NUMBER AND A SPACE.	9
10	688	000644	4545		GETC		10
11	689	000645	4551		PRINTC	/PRINT TEXT OF A LINE.	11
12	690	000646	1066		TAD CHAR		12
13	691	000647	1116		TAD MCR		13
14	692	000650	7640		SZA CLA	/SKIP IF END OF LINE	14
15	693	000651	5244		JMP .-5		15
16	694	000652	1423		TAD I THISLN	/TEST FOR END OF TEXT (X-MEM)	16
17	695	000653	7450	WTEST2,	SNA		17
18	696	000654	5271		JMP WX-2	/EXIT; DO NEXT INDIRECT LINE.	18
19	697	000655	7001		IAC		19
20	698	000656	3030		DCA PT1	/SAVE POINTER TO LINENO OF NEXT	20
21	699	000657	1065		TAD NAGSW		21
22	700	000660	7700		SMA CLA		22
23	701	000661	1430		TAD I PT1	/(X-MEM)	23
24	702	000662	4563		TSTGRP	/TRY NEXT LINENO FOR GROUP.	24
25	703	000663	5273		JMP WX		25
26	704	000664	1430	WALL,	TAD I PT1	/SET LINENO (X-MEM)	26
27	705	000665	3067		DCA LINENO		27
28	706	000666	5237		JMP WRITE+2		28
29	707						29
30	708	000667	1023	WTESTG,	TAD THISLN	/INIT GROUP PRINTOUT	30
31	709	000670	5253		JMP WTEST2		31
32	710						32
33	711	000671	3026		DCA DEBGSW		33
34	712	000672	5541		POPJ		34
35	713						35
36	714	000673	1065	WX,	TAD NAGSW		36
37	715	000674	7750		SPA SNA CLA	/SKIP IF ALL	37
38	716	000675	5271		JMP WX-2		38
39	717	000676	4551		PRINTC	/PRINT C.R. AGAIN	39
40	718	000677	5264		JMP WALL		40
41							41
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1	720	000700	0000	XTESTC, 0		/TEST THE NATURE OF THE NEXT ALPHANUMERIC - "TESTC"	1
2	721	000701	4560	SPNOR		/IGNORE SPACES	2
3	722	000702	4550	SORTC		/TEST THE VARIABLE TERMINATORS	3
4	723	000703	1767	TERMS-1			4
5	724	000704	5700	JMP I	XTESTC	/YES - SORTCN IS SET	5
6	725	000705	1066	TAD	CHAR	/NO	6
7	726	000706	2300	ISZ	XTESTC		7
8	727	000707	1202	TAD	MF		8
9	728	000710	7650	SNA CLA		/TEST FOR "F"	9
10	729	000711	5317	JMP	XT3		10
11	730	000712	4561	TESTN			11
12	731	000713	5700	JMP I	XTESTC	/.	12
13	732	000714	7410	SKP		/OTHER	13
14	733	000715	5700	JMP I	XTESTC	/NUMBER	14
15	734	000716	2300	ISZ	XTESTC		15
16	735	000717	2300	XT3, ISZ	XTESTC	/RETURNS:T; N; F; A	16
17	736	000720	5700	JMP I	XTESTC		17
18	737						18
19	738	000721	0000	XSORTC, 0		/SORT CHAR AGAINST TABLE - "SORTC"	19
20	739	000722	1721	TAD I	XSORTC		20
21	740	000723	3012	DCA	XRT2	/1ST ARG IS LIST-1	21
22	741	000724	1412	TAD I	XRT2		22
23	742	000725	7510	SPA		/LIST IS ENDED BY A NEGATIVE NUMBER	23
24	743	000726	5340	JMP	SEXC	/2AND EXIT = NOT IN LIST	24
25	744	000727	7041	CIA			25
26	745	000730	1066	TAD	CHAR		26
27	746	000731	7640	SZA CLA		/COMPARE	27
28	747	000732	5324	JMP	.-6		28
29	748	000733	1721	TAD I	XSORTC	/COMPUTE INCREMENT : 0 - N	29
30	749	000734	7040	CMA			30
31	750	000735	1012	TAD	XRT2		31
32	751	000736	3054	DCA	SORTCN		32
33	752	000737	7410	SKP		/1ST EXIT = YES	33
34	753	000740	2321	SEXC, ISZ	XSORTC		34
35	754	000741	2321	ISZ	XSORTC		35
36	755	000742	7300	CLA CLL			36
37	756	000743	5721	JMP I	XSORTC		37
38							38
39							39
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1	758	000744	0000	GRPTST, 0	/AC VS LINENO - "TSTGRP"	1
2	759	000745	0104	AND P7600		2
3	760	000746	7041	CIA		3
4	761	000747	3071	DCA T2		4
5	762	000750	1067	TAD LINENO		5
6	763	000751	0104	AND P7600		6
7	764	000752	1071	TAD T2		7
8	765	000753	7650	SNA CLA		8
9	766	000754	2344	ISZ GRPTST		9
10	767	000755	5744	JMP I GRPTST		10
11	768					11
12	769			/ INPUT FROM TEXT OR KEYBOARD;		12
13	770			/ IF BACK-ARROW, RESTART INPUT		13
14	771					14
15	772	000756	0000	INPUT, 0	/INPUT A CHARACTER	15
16	773	000757	1036	TAD INSUB	/NON-ZERO FOR KEYBOARD	16
17	774	000760	7640	SZA CLA		17
18	775	000761	5364	JMP .+3		18
19	776	000762	4545	GETC		19
20	777	000763	5756	JMP I INPUT		20
21	778	000764	4552	READC		21
22	779	000765	4547	SORTJ		22
23	780	000766	6776	SPECIAL-1		23
24	781	000767	3402	INFIX-SPECIAL		24
25	782	000770	5756	JMP I INPUT		25
26	783					26
27	784	000771	1035	ILIST, IF1	/,	27
28	785	000772	0610	PROCESS	/;	28
29	786	000773	0614	PC1	/CR	29
30	787					30
31	788			/	ENGLISH-FRENCH	31
32	789	0774		COMLST= .	/COMMAND DECODING LIST	32
33	790					33
34	791	000774	0323	"S	/SET - ORGANIZE	34
35	792	000775	0306	"F	/FOR - QUAND	35
36	793	000776	0311	"I	/IF - SI	36
37	794	000777	0304	"D	/DO - FAIZ	37
38	795	001000	0307	"G	/GOTO - VA	38
39	796	001001	0303	"C	/COMMENT- COMMENTE	39
40	797	001002	0301	"A	/ASK - DEMANDE	40
41	798	001003	0324	"T	/TYPE - TAPE	41
42	799	001004	0314	"L	/LIBRARY- ENTREPOSE	42
43	800	001005	0305	"E	/ERASE - BIFFE	43
44	801	001006	0327	"W	/WRITE - INSCRIS	44
45	802	001007	0315	"M	/MODIFY - MODIFIE	45
46	803	001010	0321	"Q	/QUIT - ARRETE	46
47	804	001011	0322	"R	/RETURN - RETOURNE	47
48	805	001012	0212	"*-40	/(ASTERISK)=EXPANDABLE COMMAND	48
49	806					49
50	807			/ THIS COMMAND LIST IS SPEED OPTIMIZED.		50
51						51
52						52
53						53
54						54
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1	809		/	CONDITIONAL TRANSFER PROCESS.	1
2	810				2
3	811	001013	4564	IF, TESTC /IGNORE SPACES AND TEST	3
4	812	001014	4637	JMS I IECALL /T	4
5	813	001015	2013	ISZ PDLXR /N-DUMP THE (EFOP)	5
6	814	001016	4640	JMS I IPART /F-CHECK FOR PAREN MATCH	6
7	815	001017	1111	TAD M2 /A	7
8	816	001020	3032	DCA T1	8
9	817	001021	1045	TAD FLAC+1 /TEST -, 0, +	9
10	818	001022	7510	SPA	10
11	819	001023	2032	ISZ T1 /N-T0 -1, -2, -3	11
12	820	001024	7750	SPA SNA CLA	12
13	821	001025	2032	IF3, ISZ T1 /COUNT COMMAS	13
14	822	001026	7410	SKP	14
15	823	001027	5767	JMP I COMGO+4 /TRANSFER	15
16	824	001030	4547	SORTJ /SEARCH TEXT UNTIL ,;C.R.	16
17	825	001031	1377	TLIST-1	17
18	826	001032	7371	ILIST-TLIST	18
19	827	001033	4545	GETC	19
20	828	001034	5230	JMP .-4	20
21	829	001035	4545	IF1, GETC /MOVE PAST COMMA	21
22	830	001036	5225	JMP IF3	22
23	831				23
24	832	001037	1601	IECALL, ECALL	24
25	833	001040	2047	IPART, PARTEST	25
26					26
27					27
28					28
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1	835		/	LOOP CONTROL STATEMENT		1
2	836					2
3	837	1041	SET=	.	/SUBSET OF "FOR".	3
4	838					4
5	839	001041	4540	FOR,	PUSHJ	/LOOPS, ETC.
6	840	001042	1403		GETARG	/LOOK FOR "=" NEXT
7	841	001043	4560		SPNOR	/IGNORE SPACES
8	842	001044	1066		TAD CHAR	
9	843	001045	1335		TAD MEQ	
10	844	001046	7440		SZA	
11	845	001047	4566		ERROR4	/LEFT OF "=" IN ERROR: 'FOR' OR 'SET'
12	846	001050	1030		TAD PT1	
13	847	001051	4542		PUSHA	/SAVE POINTER TO VARIABLE
14	848	001052	4540		PUSHJ	
15	849	001053	1612		EVAL-1	/GET INITIAL VALUE EXPRESSION
16	850	001054	1413		POPA	
17	851	001055	3030		DCA PT1	
18	852	001056	4407		FINT	/INITIALIZE NOW.
19	853	001057	6430		FPUT I PT1	
20	854	001060	0000		FXIT	
21	855	001061	4547		SORTJ	/TEST LAST CHAR FROM "EVAL"
22	856	001062	1377		TLIST-1	
23	857	001063	7177		FLIST1-TLIST	
24	858	001064	4566		ERROR4	/EXCESS R-PAR
25	859					
26	860	001065	1030	FINCR,	TAD PT1	/SAVE VARIABLE ADDRESS *
27	861	001066	4542		PUSHA	
28	862	001067	4540		PUSHJ	/EVALUATE THE INCREMENT,IF ANY.
29	863	001070	1612		EVAL-1	
30	864	001071	4547		SORTJ	/TEST TERMINATORS
31	865	001072	1377		TLIST-1	
32	866	001073	7174		FLIST2-TLIST	
33	867	001074	4566		ERROR4	/ILLEGAL TERMINATOR IN 'FOR'
34	868					
35	869	001075	4543	FLIMIT,	PUSHF	/SAVE THE INCREMENT. *
36	870	001076	2030		FLARG	
37	871	001077	4540		PUSHJ	/GET THE LIMIT (NO ERROR DETECTION AFTER LIMIT)
38	872	001100	1612		EVAL-1	
39	873	001101	4543	FCONT,	PUSHF	/SAVE THE LIMIT *
40	874	001102	2030		FLARG	
41	875	001103	4543		PUSHF	/SAVE TEXT OF OBJECT STATEMENTS
42	876	001104	0017		TEXTP	
43	877	001105	4540		PUSHJ	/DO THE OBJECT STATEMENTS
44	878	001106	0610		PROCESS	
45	879	001107	4544		POPF	/RESTORE REMAINING TEXT.
46	880	001110	0017		TEXTP	
47	881	001111	4544		POPF	/GET LIMIT
48	882	001112	2030		FLARG	
49	883	001113	4544		POPF	/GET INCREMENT
50	884	001114	7470		ITER1	
51	885	001115	1413		POPA	/GET VARIABLE ADDRESS
52	886	001116	3030		DCA PT1	
53						
54						
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1	888	001117	4407		FINT		/INCREMENT AND TEST	1
2	889	001120	0430		FGET I	PT1	/LOAD THE VARIABLE	2
3	890	001121	1733		FADD I	FINKP	/INCREMENT IT	3
4	891	001122	6430		FPUT I	PT1	/CHANGE IT	4
5	892	001123	2525		FSUB I	FLARGP	/TEST IT	5
6	893	001124	0000		FXIT			6
7	894	001125	1045		TAD	FLAC+1		7
8	895	001126	7740		SMA SZA	CLA		8
9	896	001127	5541		POPJ		/END OF LOOP	9
10	897	001130	1030		TAD	PT1		10
11	898	001131	4542		PUSHA		/SAVE ADDRESS *	11
12	899	001132	4543		PUSHF		/SAVE INCREMENT AGAIN *	12
13	900	001133	7470	FINKP,	ITER1			13
14	901	001134	5301		JMP	FCONT		14
15	902							15
16	903	001135	7503	MEQ,	-"	=		16
17	904	001136	7524	MCOM,	-"	,		17
18	905							18
19	906	001137	4543	FINFIN,	PUSHF		/SET INCREMENT TO ONE.	19
20	907	001140	2405		FLTONE			20
21	908	001141	5301		JMP	FCONT		21
22	909							22
23	910			/	SET AND INTENSIFY THE POINT:	FDIS(X,Y)		23
24	911							24
25	912	001142	4453	XDYS,	JMS I	INTEGER	/RETURN=INTEGER VALUE OF Y.	25
26	913	001143	4542		PUSHA			26
27	914	001144	1066		TAD	CHAR		27
28	915	001145	1336		TAD	MCOM		28
29	916	001146	7640		SZA CLA			29
30	917	001147	4566		ERROR3		/ARG MISSING	30
31	918	001150	4540		PUSHJ			31
32	919	001151	1612		EVAL-1			32
33	920	001152	4453		JMS I	INTEGER		33
34	921	001153	6063		6063		/DYL	34
35	922	001154	7200		CLA			35
36	923	001155	1413		POPA			36
37	924	001156	6057		6057		/DXL DIX	37
38	925	001157	7410		SKP			38
39	926							39
40	927			/	TAKE THE INTEGER PART			40
41	928							41
42	929	001160	4453	XINT,	JMS I	INTEGER	/(FIX)	42
43	930	001161	7200		CLA			43
44	931	001162	5536		JMP I	EFUN3I		44
45								45
46								46
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1	933		1163		COMGO=	.	/COMMAND ROUTINE ADDRESSES	1
2	934							2
3	935	001163	1041		SET			3
4	936	001164	1041		FOR			4
5	937	001165	1013		IF			5
6	938	001166	0420		DO			6
7	939	001167	0603		GOTO		/(REFERENCED)	7
8	940	001170	0614		COMMENT			8
9	941	001171	1202		ASK			9
10	942	001172	1203		TYPE			10
11	943	001173	7503		LIBRARY			11
12	944	001174	2204		ERASE			12
13	945	001175	0635		WRITE			13
14	946	001176	1256		MODIFY			14
15	947	001177	0177		START		/RETURN TO COMMAND MODE VIA 'QUIT'	15
16	948	001200	1563		RETRN			16
17	949	001201	6361		HSPX		/ACTIVATE THE HIGH SPEED READER	17
18	950							18
19	951							19
20	952							20
21	953							21
22	954							22
23	955			/	INPUT-OUTPUT STATEMENTS			23
24	956							24
25	957	001202	7240	ASK,	CLA CMA		/REMEMBER WHICH CALL.	25
26	958	001203	3056	TYPE,	DCA ATSW			26
27	959	001204	3026	TASK,	DCA DEBSW		/RE-ENABLE THE TRACE	27
28	960	001205	4547		SORTJ		/SPECIAL CHARACTER?	28
29	961	001206	1371		ALIST-1			29
30	962	001207	0176		ATLIST-ALIST			30
31	963	001210	2056		ISZ ATSW		/TEST QUOTE SWITCH	31
32	964	001211	5226		JMP TYPE2			32
33	965	001212	4540		PUSHJ		/DO ASK; SETUP PT1	33
34	966	001213	1403		GETARG			34
35	967	001214	1066		TAD CHAR		/SAVE IN-LINE CHARACTER.	35
36	968	001215	4542		PUSHA			36
37	969	001216	1255		TAD COL		/TYPE COLON	37
38	970	001217	4551		PRINTC		/(CLA)- TO SUPRESS ":"	38
39	971	001220	2036		ISZ INSUB		/INDICATE 'READC'	39
40	972	001221	7001		IAC		/POINT PAST CHAR	40
41	973	001222	4531		JMS I FINPUT		/READ DATA AND SAVE	41
42	974	001223	1413		POPA		/RE-TEST LAST TERMINATOR	42
43	975	001224	3066		DCA CHAR			43
44	976	001225	5202		JMP ASK		/CONTINUE PROCESSING	44
45	977							45
46	978	001226	4540	TYPE2,	PUSHJ		/DO TYPE	46
47	979	001227	1613		EVAL			47
48	980	001230	4530		JMS I FOUTPUT		/PRINT	48
49	981	001231	5203		JMP TYPE			49
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1	983	001232	2026	TQUOT,	ISZ	DEBGSW	/DISABLE TRACE	1
2	984	001233	4545		GETC		/TYPE LITERALS	2
3	985	001234	4547		SORTJ			3
4	986	001235	1403		TLIST2-1			4
5	987	001236	0773		TLIST3-TLIST2			5
6	988	001237	4551		PRINTC			6
7	989	001240	5233		JMP	TQUOT+1		7
8	990							8
9	991	001241	4545	TINTR,	GETC		/PASS PERCENT SIGN	9
10	992	001242	4554		GETLN		/READ FORMAT CONTROL: "%7.03"	10
11	993	001243	1067		TAD	LINENO		11
12	994	001244	3052		DCA	FISW	/SAVE FORMAT CODE	12
13	995	001245	5204		JMP	TASK		13
14	996							14
15	997	001246	1077	TCRLF2,	TAD	CCR	/SPLAT=CR ALONE	15
16	998	001247	4463		JMS I	OUTDEV		16
17	999	001250	7040		CMA		/NON-PRINTING DELAY FOR CR	17
18	1000	001251	1077	TCRLF,	TAD	CCR	/EXCLAMATION POINT=CR, LF.	18
19	1001	001252	4551		PRINTC			19
20	1002	001253	4545	TASK4,	GETC		/MOVE TO NEXT CHARACTER	20
21	1003	001254	5204		JMP	TASK		21
22	1004							22
23	1005	001255	0272	COL,	" :		/" :	23
24	1006							24
25	1007							25
26	1008							26
27	1009			/	IF DEBGSW= 0:	ENABLE FLIP-FLOP "DMPSW"		27
28	1010			/		#0: DISABLE AND RETURN ALL "?" ' S.		28
29	1011			/	IF DMPSW= 0:	TRACE ON, IF ENABLED		29
30	1012			/		#0: TRACE OFF		30
31	1013			/	IF BOTH= 0:	PRINT TRACE.		31
32								32
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1	1015		/	SEARCH ROUTINES		1
2	1016					2
3	1017	001256	4554	MODIFY, GETLN	/READ LINE NO.	3
4	1018	001257	4555	FINDLN	/LOOK IT UP NOW.	4
5	1019	001260	4566	ERROR2	/NOT THERE = BAD COMMAND UNLESS ZERO.	5
6	1020	001261	1060	TAD BUFR	/SET POINTERS	6
7	1021	001262	3010	DCA AXIN	/FOR INPUT	7
8	1022	001263	3062	DCA XCTIN		8
9	1023	001264	1067	TAD LINENO	/COPY THE SAME LINE NUMBER.	9
10	1024	001265	3410	DCA I AXIN	/(X-MEM)	10
11	1025	001266	1010	TAD AXIN	/SAVE START OF NEW LINE	11
12	1026	001267	3027	DCA PACKST		12
13	1027	001270	4464	SCONT, JMS I INDEV	/READ THE TELETYPE INPUT SILENTLY.	13
14	1028	001271	3100	DCA LIST3+1	/SAVE SEARCH CHARACTER	14
15	1029	001272	2026	ISZ DEBGSW	/NO BREAKS.	15
16	1030	001273	4545	SCHAR, GETC	/TYPE+TEST-F.F.	16
17	1031	001274	4551	PRINTC	/PLAYBACK THE TEXT	17
18	1032	001275	4547	SORTJ	/LOOK FOR MATCH	18
19	1033	001276	0076	LIST3-1		19
20	1034	001277	1271	LISTG0-LIST3		20
21	1035	001300	4546	PACKC	/SAVE NEW LINE.	21
22	1036	001301	5273	JMP SCHAR		22
23	1037					23
24	1038	001302	1060	SBAR, TAD BUFR	/RESTART-B.A.	24
25	1039	001303	7001	IAC		25
26	1040	001304	3010	DCA AXIN	/SET POINTERS	26
27	1041	001305	3062	DCA XCTIN		27
28	1042	001306	4552	SFOUND, READC	/READ FROM KEYBOARD	28
29	1043	001307	4547	SORTJ	/TEST	29
30	1044	001310	0071	LIST6-1		30
31	1045	001311	1271	SRNLST-LIST6		31
32	1046	001312	4546	SGOT, PACKC	/PACK CHAR.	32
33	1047	001313	5306	JMP SFOUND	/MORE	33
34						34
35						35
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1	1049	001314	0000	SORTB,	0		/SORT AND BRANCH ROUTINE. - "SORTJ"	1
2	1050	001315	7450		SNA			2
3	1051	001316	1066		TAD	CHAR	/ASSUME CHAR IF AC=0	3
4	1052	001317	7041		CIA			4
5	1053	001320	3071		DCA	T2	/SAVE SORT ITEM	5
6	1054	001321	1714		TAD I	SORTB	/FIRST ARG IS LIST LESS ONE	6
7	1055	001322	2314		ISZ	SORTB	/2AND IS INTRA-LIST LENGTH	7
8	1056	001323	3012		DCA	XRT2		8
9	1057	001324	1412		TAD I	XRT2		9
10	1058	001325	7510		SPA		/**LISTS ENDED BY NEGATIVE NUMBERS**	10
11	1059	001326	5340		JMP	SEX	/READ EXIT	11
12	1060	001327	1071		TAD	T2	/FIND ADDRESS	12
13	1061	001330	7640		SZA	CLA		13
14	1062	001331	5324		JMP	.-5		14
15	1063	001332	1012		TAD	XRT2	/MATCH FOUND.	15
16	1064	001333	1714		TAD I	SORTB		16
17	1065	001334	3071		DCA	T2		17
18	1066	001335	1471		TAD I	T2		18
19	1067	001336	3071		DCA	T2	/DEBUG : AC = ADDRESS	19
20	1068	001337	5471		JMP I	T2		20
21	1069							21
22	1070	001340	2314	SEX,	ISZ	SORTB	/MATCH NOT FOUND.	22
23	1071	001341	7300		CLA	CLL		23
24	1072	001342	5714		JMP I	SORTB		24
25	1073							25
26	1074							26
27	1075							27
28	1076			/		ANALOGUE TO DIGITAL CONVERSION FOR AX08		28
29	1077							29
30	1078	001343	4453	XADC,	JMS I	INTEGER		30
31	1079	001344	7000		NOP		/(IOF)-FOR OTHER BRANDS	31
32	1080	001345	6375		6375		/ACMX ADCV	32
33	1081	001346	6332		6332		/SKAD	33
34	1082	001347	5346		JMP	.-1		34
35	1083	001350	6362		6362		/RADC	35
36	1084	001351	3046		DCA	FLAC+2		36
37	1085	001352	6001		ION			37
38	1086	001353	5536		JMP I	EFUN3I		38
39								39
40								40
41								41
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1	1088	001354	0000	OUTL,	0	/SLOW OUTPUT FOR ODT SYNCRONIZATION	1
2	1089	001355	6046		TLS	/AND FOR H.S. PUNCH	2
3	1090	001356	6026		PLS		3
4	1091	001357	6041		TSF	/IOT FOR SLOWEST DEVICE	4
5	1092	001360	5357		JMP	. -1	5
6	1093	001361	7200		CLA		6
7	1094	001362	5754		JMP I	OUTL	7
8	1095						8
9	1096						9
10	1097						10
11	1098						11
12	1099		1363		SRNLST=	/'MODIFY' CONTROL CHARACTER TABLE	12
13	1100						13
14	1101	001363	1273		SCHAR	/F.F. = CONTINUE	14
15	1102	001364	1270		SCONT	/BELL = CHANGE SEARCH CHARACTER	15
16	1103	001365	2740		RECOVR	/C.C. = BREAK	16
17	1104	001366	1302		SBAR	/B.A. = RESTART	17
18	1105	001367	1271		SCONT+1	/L.F. = FINISH THE LINE AS BEFORE.	18
19	1106						19
20	1107		1370		LISTG0=		20
21	1108						21
22	1109	001370	0261		SRETN	/C.R. = END THE LINE HERE AS IS.	22
23	1110	001371	1312		SGOT	/CHAR = SEARCH CHARACTER	23
24	1111						24
25	1112						25
26	1113						26
27	1114						27
28	1115						28
29	1116		1372		ALIST=	/ASK/TYPE LIST OF CONTROLS.	29
30	1117						30
31	1118	001372	0245		"%	/%	31
32	1119	001373	0242		" "	/"	32
33	1120	001374	0241		"!	/!	33
34	1121	001375	0243		"#	/#	34
35	1122	001376	0244		"\$	/\$	35
36	1123						36
37	1124		1377		GLIST=		37
38	1125						38
39	1126	001377	0240		"	/SPACE	39
40	1127		1400		TLIST=		40
41	1128						41
42	1129	001400	0254		"	/,	42
43	1130	001401	0273		" ;	/;	43
44	1131	001402	0215		"M&237	/C.R.	44
45	1132						45
46	1133			/		THIS LIST IS ENDED BY 'TESTC'.	46
47							47
48							48
49							49
50							50
51							51
52							52
53							53
54							54
55							55
56							56
57							57
58							58

1	1135		/	FIND OR ENTER A VARIABLE IN THE LIST.	1
2	1136				2
3	1137	001403	4564	GETARG, TESTC	3
4	1138	001404	0242	TLIST2, ""	4
5	1139	001405	0215	"M&237	5
6	1140	001406	4566	ERROR4	6
7	1141	001407	3062	GETVAR, DCA XCTIN	7
8	1142	001410	4546	PACKC	8
9	1143	001411	4545	GETC	9
10	1144	001412	4550	SORTC	10
11	1145	001413	1767	TERMS-1	11
12	1146	001414	5226	JMP GSERCH	12
13	1147	001415	1066	TAD CHAR	13
14	1148	001416	0122	AND P77	14
15	1149	001417	1061	TAD ADD	15
16	1150	001420	3061	DCA ADD	16
17	1151	001421	4545	GETC	17
18	1152	001422	4550	SORTC	18
19	1153	001423	1767	TERMS-1	19
20	1154	001424	5226	JMP GSERCH	20
21	1155	001425	5221	JMP .-4	21
22	1156				22
23	1157	001426	4562	GSERCH, TSTLPR	23
24	1158	001427	5237	JMP GS1	24
25	1159	001430	1061	TAD ADD	25
26	1160	001431	3056	DCA EFOP	26
27	1161	001432	4660	JMS I GECALL	27
28	1162	001433	1413	POPA	28
29	1163	001434	3061	DCA ADD	29
30	1164	001435	4657	JMS I PTEST	30
31	1165	001436	4453	JMS I INTEGER	31
32	1166	001437	3317	GS1, DCA SUBS	32
33	1167	001440	1060	TAD STARTV	33
34	1168	001441	3030	GS3, DCA PT1	34
35	1169	001442	1030	TAD PT1	35
36	1170	001443	7041	CIA	36
37	1171	001444	1031	TAD LASTV	37
38	1172	001445	7750	SPA SNA CLA	38
39	1173	001446	5261	JMP GS2	39
40	1174	001447	1430	TAD I PT1	40
41	1175	001450	7041	CIA	41
42	1176	001451	1061	TAD ADD	42
43	1177	001452	7650	SNA CLA	43
44	1178	001453	5305	JMP GFND1	44
45	1179	001454	1030	GS4, TAD PT1	45
46	1180	001455	1070	TAD GINC	46
47	1181	001456	5241	JMP GS3	47
48	1182				48
49	1183	001457	2047	PTEST, PARTEST	49
50	1184	001460	1601	GECALL, ECALL	50
51					51
52					52
53					53
54					54
55					55
56					56
57					57
58					58

1	1186	001461	1031	GS2,	TAD	LASTV	/ADD THE VARIABLE	1
2	1187	001462	1005		TAD	P13	/TEST STORAGE LIMITS	2
3	1188	001463	7141		CLL CIA			3
4	1189	001464	1013		TAD	PDLXR		4
5	1190	001465	7620		SNL CLA			5
6	1191	001466	4566		ERROR3			6
7	1192	001467	1031		TAD	LASTV	/UPDATE THE LIST.	7
8	1193	001470	1070		TAD	GINC		8
9	1194	001471	3031		DCA	LASTV		9
10	1195	001472	1061		TAD	ADD	/SAVE NAME	10
11	1196	001473	3430		DCA I	PT1		11
12	1197	001474	2030		ISZ	PT1	/SAVE SUBSCRIPT	12
13	1198	001475	1317		TAD	SUBS		13
14	1199	001476	3430		DCA I	PT1		14
15	1200	001477	2030		ISZ	PT1	/SET PT1	15
16	1201	001500	4407		FINT			16
17	1202	001501	0537		FGET I	CFRSX		17
18	1203	001502	6430		FPUT I	PT1		18
19	1204	001503	0000		FXIT			19
20	1205	001504	5541		POPJ		/EXIT	20
21	1206							21
22	1207	001505	1030	GFND1,	TAD	PT1	/FOUND SAME	22
23	1208	001506	3011		DCA	XRT	/TEST SUBSCRIPTS	23
24	1209	001507	1411		TAD I	XRT		24
25	1210	001510	7041		CIA			25
26	1211	001511	1317		TAD	SUBS		26
27	1212	001512	7640		SZA CLA			27
28	1213	001513	5254		JMP	GS4	/WRONG SUBSCRIPT	28
29	1214	001514	2030		ISZ	PT1	/SET POINTER TO DATA	29
30	1215	001515	2030		ISZ	PT1		30
31	1216	001516	5541		POPJ			31
32								32
33								33
34								34
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37								37
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1	1218	1517		SUBS=	.		1
2	1219						2
3	1220	001517	0000	XSPNOR,	0	/IGNORE LEADING SPACES - "SPNOR"	3
4	1221	001520	1066	TAD	CHAR		4
5	1222	001521	1114	TAD	M240		5
6	1223	001522	7640	SZA	CLA		6
7	1224	001523	5717	JMP I	XSPNOR		7
8	1225	001524	4545	GETC			8
9	1226	001525	5320	JMP	XSPNOR+1		9
10	1227						10
11	1228						11
12	1229	001526	7520	M260,	- "0		12
13	1230	001527	7507	M271,	- "9		13
14	1231						14
15	1232						15
16	1233						16
17	1234	001530	0000	RANO,	0000	/RANDOM NUMBER STORAGE!	17
18	1235	001531	2000		2000		18
19	1236	001532	0000		0000		19
20	1237						20
21	1238	001533	0000	XTESTN,	0	/RETURNS: .; OTHER; NUMBER - "TESTN"	21
22	1239	001534	1066	TAD	CHAR		22
23	1240	001535	1115	TAD	MPER		23
24	1241	001536	7640	SZA	CLA		24
25	1242	001537	2333	ISZ	XTESTN		25
26	1243	001540	1066	TAD	CHAR		26
27	1244	001541	1326	TAD	M260		27
28	1245	001542	3054	DCA	SORTCN	/SAVE VALUE OF THE NUMBER	28
29	1246	001543	1054	TAD	SORTCN	/TEST IF REALLY A DIGIT.	29
30	1247	001544	7710	SPA	CLA		30
31	1248	001545	5733	JMP I	XTESTN		31
32	1249	001546	1066	TAD	CHAR		32
33	1250	001547	1327	TAD	M271		33
34	1251	001550	7750	SPA	SNA CLA		34
35	1252	001551	2333	ISZ	XTESTN	/IF A NUMBER	35
36	1253	001552	5733	JMP I	XTESTN		36
37	1254						37
38	1255	001553	4407	XRAN,	FINT	/PSEUDO-RANDOM NUMBER GENERATOR.	38
39	1256	001554	1330	FADD	RANO	/ADD RUNNING RESULT TO THE ARGUMENT, IF ANY.	39
40	1257	001555	4350	FMUL	.-5	/BLAST THE ARGUMENT	40
41	1258	001556	6330	FPUT	RANO		41
42	1259	001557	0000	FXIT			42
43	1260	001560	3330	DCA	RANO	/CONVERT TO .5 THROUGH .999	43
44	1261	001561	3044	DCA	FLAC	/SAME AS RETURN	44
45	1262	001562	5536	JMP I	EFUN3I		45
46							46
47							47
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54							54
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57							57
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1	1264		/	EXIT FROM A "DO" SUBROUTINE		1
2	1265					2
3	1266	001563	1137	RETRN, TAD CFRSX	/(PC) => 0	3
4	1267	001564	3022	DCA PC		4
5	1268	001565	1413	XPOPJ, TAD I PDLXR	/RECURSIVE EXIT - "POPJ"	5
6	1269	001566	3071	DCA T2		6
7	1270	001567	5471	JMP I T2		7
8	1271					8
9	1272					9
10	1273					10
11	1274		1570	ATLIST= .	/ASK-TYPE CONTROL CHARACTER TABLE	11
12	1275					12
13	1276	001570	1241	TINTR	/% - FORMAT DELIMITER	13
14	1277	001571	1232	TQUOT	/ " - LITERAL DELIMITER	14
15	1278	001572	1251	TCRLF	/! - CARRIAGE RETURN AND LINE FEED	15
16	1279	001573	1246	TCRLF2	/# - CARRIAGE RETURN ONLY	16
17	1280	001574	3052	TDUMP	/\$ - DUMP THE SYMBOL TABLE CONTENTS	17
18	1281	001575	1253	TASK4	/SP- TERMINATOR FOR NAMES	18
19	1282	001576	1253	TASK4	/, - TERMINATOR FOR EXPRESSIONS	19
20	1283	001577	0610	PROCESS	/; - TERMINATOR FOR COMMANDS	20
21	1284	001600	0614	PC1	/C.R. - TERMINATOR FOR STRINGS	21
22	1285					22
23	1286		/	\$ - FOR 'TDUMP' TERMINATES THE COMMAND.		23
24						24
25						25
26						26
27						27
28						28
29						29
30						30
31						31
32						32
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1	1288		/			EVALUATE AN EXPRESSION WHICH	1
2	1289		/			TERMINATES WITH AN R-PAR,; OR C.R. AND	2
3	1290		/			LEAVE THE RESULT IN FLAC AND IN FLARG.	3
4	1291						4
5	1292	001601	0000	ECALL,	0	/RECURSIVE CALL TO "EVAL"	5
6	1293	001602	1054	TAD	SORTCN	/SAVE 'SORTCN', 'LASTOP', AND 'EFOP'	6
7	1294	001603	4542	PUSHA			7
8	1295	001604	1055	TAD	LASTOP		8
9	1296	001605	4542	PUSHA			9
10	1297	001606	1056	TAD	EFOP	/SAVE FUNCTION CODE.	10
11	1298	001607	4542	PUSHA			11
12	1299	001610	1201	TAD	ECALL	/RETURN TO CALLING	12
13	1300	001611	4542	PUSHA		/ADDRESS AFTER NEXT POPJ	13
14	1301						14
15	1302	001612	4545	GETC		/MOVE PAST EXTRA CHARACTER	15
16	1303	001613	3055	EVAL,	DCA	LASTOP	16
17	1304	001614	4564	TESTC		/EVALUATION CONTROLLER (CHECKPOINT?)	17
18	1305	001615	5227	JMP	ETERM1	/TEST CHARACTER AND IGNORE SPACES	18
19	1306	001616	5332	JMP	ENUM	/TERMINATOR	19
20	1307	001617	5343	JMP	EFUN	/NUMBER	20
21	1308	001620	4540	PUSHJ		/FUNCTION	21
22	1309	001621	1407	GETVAR		/LETTER OF VARIABLE	22
23	1310	001622	4564	OPNEXT,	TESTC	/FIND OR CREATE VARIABLE; ALSO SET PT1.	23
24	1311	001623	5244	JMP	ETERMN	/PT1=>ARG	24
25	1312	001624	0212	ECHOLST,	"J&237	/T	25
26	1313	001625	0377			/N-ERROR IN FORMAT	26
27	1314	001626	4566	ERROR4		/F	27
28	1315					/L - MISSING OPERATOR	28
29	1316	001627	1137	ETERM1,	TAD	CFRSX	29
30	1317	001630	3030	DCA	PT1	/SET PT1.	30
31	1318	001631	1111	TAD	M2	/TO POINT TO ZERO	31
32	1319	001632	1054	TAD	SORTCN	/TEST FOR UNARY OPERATIONS	32
33	1320	001633	7450	SNA			33
34	1321	001634	5247	JMP	ETERM	/CREATE DUMMY FOR UNARY MINUS	34
35	1322	001635	7001	IAC			35
36	1323	001636	7650	SNA	CLA		36
37	1324	001637	5323	JMP	ARGNXT	/IGNORE UNARY PLUS	37
38	1325	001640	1054	TAD	SORTCN	/TEST FOR NULL PARENS.	38
39	1326	001641	1121	TAD	M11		39
40	1327	001642	7710	SPA	CLA		40
41	1328	001643	5363	JMP	ELPAR	/MIGHT BE AN L-PAR.	41
42	1329	001644	4562	ETERMN,	TSTLPR		42
43	1330	001645	7410	SKP			43
44	1331	001646	4566	ERROR4		/OPERATOR MISSING BEFORE PAREN	44
45	1332	001647	1054	ETERM,	TAD	SORTCN	45
46	1333	001650	3024	DCA	THISOP	/SET FROM "TESTC" - "SORTC"	46
47	1334	001651	1024	TAD	THISOP		47
48	1335	001652	1121	TAD	M11		48
49	1336	001653	7700	SMA	CLA	/END?	49
50	1337	001654	3024	DCA	THISOP	/ "THISOP" EQUIV. TO END OF EXP.	50
51							51
52							52
53							53
54							54
55							55
56							56
57							57
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1	1339	001655	1024	ETERM2,	TAD	THISOP	/COMPARE PRIORITIES	1
2	1340	001656	7041		CIA			2
3	1341	001657	1055		TAD	LASTOP		3
4	1342	001660	7710		SPA	CLA		4
5	1343	001661	5310		JMP	EPAR	/CONTINUE	5
6	1344	001662	1055		TAD	LASTOP	/FIND OPERATION	6
7	1345	001663	7112		CLL	RTR		7
8	1346	001664	7012		RTR			8
9	1347	001665	1331		TAD	OPTABL		9
10	1348	001666	3274		DCA	FLOP		10
11	1349	001667	1055		TAD	LASTOP		11
12	1350	001670	7640		SZA	CLA	/TEST FOR END OF DATA INTO FLOATING AC.	12
13	1351	001671	4544		POPF		/GET LAST DATA	13
14	1352	001672	0044		FLAC			14
15	1353	001673	4407		FINT			15
16	1354	001674	0000	FLOP,	00		/((FLOPR I PT1)+-*/	16
17	1355	001675	6525		FPUT I	FLARGP	/SAVE RESULT	17
18	1356	001676	0000		FXIT			18
19	1357	001677	1125		TAD	FLARGP		19
20	1358	001700	3030		DCA	PT1		20
21	1359	001701	1024		TAD	THISOP		21
22	1360	001702	1055		TAD	LASTOP	/=0?	22
23	1361	001703	7650		SNA	CLA		23
24	1362	001704	5541		POPJ		/EXIT "EVAL"	24
25	1363	001705	1413		POPA		/GET PRIOR OP	25
26	1364	001706	3055		DCA	LASTOP		26
27	1365	001707	5255		JMP	ETERM2	/COMPARE THIS OP	27
28	1366							28
29	1367	001710	4562	EPAR,	TSTLPR		/TEST FOR SUB-EXPRESSION	29
30	1368	001711	7410		SKP			30
31	1369	001712	5365		JMP	EPAR2	/GO EVALUATE EXPRESSION	31
32	1370	001713	1055		TAD	LASTOP	/CONTINUE READING THE EXPRESSION	32
33	1371	001714	4542		PUSHA		/SAVE "LASTOP".	33
34	1372	001715	1030		TAD	PT1		34
35	1373	001716	3320		DCA	.+2		35
36	1374	001717	4543		PUSHF		/SAVE LAST ARGUMENT	36
37	1375	001720	0000		00			37
38	1376	001721	1024		TAD	THISOP	/MORE TO COME	38
39	1377	001722	3055		DCA	LASTOP		39
40	1378	001723	4545	ARGNXT,	GETC		/READ 1ST CHAR OF AN ARG.	40
41	1379	001724	4564		TESTC		/DO SPECIAL CHECK	41
42	1380	001725	5363		JMP	ELPAR	/COULD BE LEFT PAREN	42
43	1381	001726	5332		JMP	ENUM	/N	43
44	1382	001727	5343		JMP	EFUN	/F	44
45	1383	001730	5220		JMP	OPNEXT-2	/L	45
46	1384							46
47	1385	001731	0430	OPTABL,	FGET I	PT1	/BASE FOR OPERATION COMPUTATION	47
48								48
49								49
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1	1387	001732	4543	ENUM,	PUSHF		/TO PROCESS A NUMBER, SAVE AC	1
2	1388	001733	0044		FLAC			2
3	1389	001734	1125		TAD	FLARGP	/SET POINTER AS FOR A VARIABLE.	3
4	1390	001735	3030		DCA	PT1		4
5	1391	001736	3036		DCA	INSUB	/POINT TO 'GETC' AND USE CHAR	5
6	1392	001737	4531		JMS I	FINPUT	/READ TEXT NUMBER => (PT1)	6
7	1393	001740	4544		POPF		/RESTORE THE AC	7
8	1394	001741	0044		FLAC			8
9	1395	001742	5222		JMP	OPNEXT	/CONTINUE	9
10	1396							10
11	1397	001743	3056	EFUN,	DCA	EFOP	/SET CODE	11
12	1398	001744	4545		GETC		/READ FUNCTION NAME.(1, 2, OR 3 LETTERS)	12
13	1399	001745	4550		SORTC		/LOOK FOR TERMINATION CHARACTER.	13
14	1400	001746	1767		TERMS-1			14
15	1401	001747	5354		JMP	EFUN2	/YES	15
16	1402	001750	1056		TAD	EFOP	/NO	16
17	1403	001751	7104		CLL RAL		/MISH-MASH HASH CODE	17
18	1404	001752	1066		TAD	CHAR		18
19	1405	001753	5343		JMP	EFUN		19
20	1406							20
21	1407	001754	4562	EFUN2,	TSTLPR			21
22	1408	001755	4566		ERROR4		/MUST BE FOLLOWED BY PARENS TO SET ARGUMENT	22
23	1409	001756	4201		JMS	ECALL	/CALL "EVAL" TO COMPUTE ARGUMENT	23
24	1410	001757	1413		POPA		/BRANCH ON FUNCTION CODE; RETURN VIA EFUN3I.	24
25	1411	001760	4547		SORTJ			25
26	1412	001761	2164		FNTABL-1			26
27	1413	001762	6207		FNTABF-FNTABL			27
28	1414	001763	4562	ELPAR,	TSTLPR		/LEFT PAREN OR FELL THROUGH FUNCTION TABLE	28
29	1415	001764	4566		ERROR4		/DOUBLE OPERATORS OR ILLEGAL FUNCTION NAME.	29
30	1416	001765	4201	EPAR2,	JMS	ECALL	/EVALUATE NESTED EXPRESSION	30
31	1417	001766	2013		ISZ	PDLXR	/DUMP EXTRA ARG.	31
32	1418	001767	5536		JMP I	EFUN3I		32
33								33
34								34
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1	1420		1770		TERMS=	.	/TERMINATOR TABLE FOR 'EVAL' AND 'GETVAR'	1
2	1421							2
3	1422	001770	0240		"		/SPACE 0	3
4	1423	001771	0253		" +		/+ 1	4
5	1424	001772	0255		" -		/- 2	5
6	1425	001773	0257		" /		// 3	6
7	1426	001774	0252		" *		/* 4	7
8	1427	001775	0336		" ^		/UP ARR 5	8
9	1428	001776	0250		" (		/(6 L-PARS	9
10	1429	001777	0333		" [		/[7	10
11	1430	002000	0274		" <		/<< 10	11
12	1431	002001	0251		")		/) 11 R-PARS	12
13	1432	002002	0335		"]		/] 12	13
14	1433	002003	0276		" >		/> 13	14
15	1434	002004	0254		" ,		/, 14	15
16	1435	002005	0273		" ;		/; 15	16
17	1436	002006	0215		" M&237		/C.R. 16	17
18	1437	002007	0275		" =		/= TO END GETARG FROM 'SET'	18
19	1438							19
20	1439			/		TWO MINOR FUNCTIONS		20
21	1440							21
22	1441							22
23	1442	002010	4543	XSGN,	PUSHF		/TAKE SIGN*1 OF FLARG	23
24	1443	002011	2405		FLTONE			24
25	1444	002012	4544		POPF			25
26	1445	002013	0044		FLAC			26
27	1446	002014	1231	XABS,	TAD	FLARG+1	/TAKE ABSOLUTE VALUE OF FLAC	27
28	1447	002015	7710		SPA CLA		/SKIP TO CONTINUE	28
29	1448	002016	4451		JMS I	MINSKI	/NEGATE THE FLOATING AC	29
30	1449							30
31	1450			/		CONTINUATION OF FUNCTION CALLS.		31
32	1451							32
33	1452	002017	4407	EFUN3,	FINT			33
34	1453	002020	7000		FNOR		/NORMALIZE FUNCTION RETURN	34
35	1454	002021	6230		FPUT	FLARG	/SAVE FUNCTION VALUE	35
36	1455	002022	0000		EXIT			36
37	1456	002023	1125		TAD	FLARGP	/SET POINTER	37
38	1457	002024	3030		DCA	PT1		38
39	1458	002025	4247		JMS	PARTEST		39
40	1459	002026	5627		JMP I	.+1; OPNEXT	/FUNCTION RETURN IS OK	40
41	1460	002027	1622					41
42								42
43								43
44								44
45								45
46								46
47								47
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1	1462	002030	0000	FLARG,	0		/DATA TEMPORARY STORAGE	1
2	1463	002031	0000		0			2
3	1464	002032	0000		0			3
4	1465	002033	0000		0			4
5	1466							5
6	1467	002034	0003	P3,	3			6
7	1468							7
8	1469	002035	0000	LPRTST,	0		/SKIP IF LEFT PAREN. - 'TSTLPR'	8
9	1470	002036	1054	TAD		SORTCN		9
10	1471	002037	1121	TAD		M11		10
11	1472	002040	7700	SMA	CLA			11
12	1473	002041	5635	JMP	I	LPRTST		12
13	1474	002042	1054	TAD		SORTCN		13
14	1475	002043	1120	TAD		M5		14
15	1476	002044	7740	SMA	SZA	CLA		15
16	1477	002045	2235	ISZ		LPRTST		16
17	1478	002046	5635	JMP	I	LPRTST		17
18	1479							18
19	1480	002047	0000	PARTEST,	0		/TEST THE PAREN MATCHINGS	19
20	1481	002050	1413	POPA			/RESTORE LAST OPERATION	20
21	1482	002051	3055	DCA		LASTOP		21
22	1483	002052	1234	TAD		P3	/+3 TO COMPARE CODES	22
23	1484	002053	1413	POPA			/GET LAST PAREN CODE.	23
24	1485	002054	7041	CIA			/CHECK FOR PAREN MATCH.	24
25	1486	002055	1054	TAD		SORTCN	/(STILL SET FROM THE LAST "EVAL")	25
26	1487	002056	7640	SZA	CLA		/SKIP IF MATCH	26
27	1488	002057	4566	ERROR4			/PAREN ERROR	27
28	1489	002060	4545	GETC			/MOVE PAST R-PAR	28
29	1490	002061	5647	JMP	I	PARTEST		29
30								30
31								31
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1	1492		/	THE DELETE A LINE ROUTINE		1
2	1493					2
3	1494	002062	0000	XDELETE,0	/UNCHAIN A LINE AND RECOVER THE SPACE.	3
4	1495	002063	6002	IOF	/PROTECT POINTER CHANGES FROM INTERRUPTIONS	4
5	1496	002064	4555	FINDLN	/SETS "THISLN" AND "LASTLN".	5
6	1497	002065	5662	JMP I XDELETE	/ALREADY GONE	6
7	1498	002066	2026	ISZ DEBSW	/DISABLE TRACE	7
8	1499	002067	4545	GETC	/MEASURE LENGTH	8
9	1500	002070	1066	TAD CHAR		9
10	1501	002071	1116	TAD MCR		10
11	1502	002072	7640	SZA CLA		11
12	1503	002073	5267	JMP .-4		12
13	1504	002074	1017	TAD AXOUT	/SAVE LAST ADDRESS	13
14	1505	002075	7040	CMA		14
15	1506	002076	1023	TAD THISLN		15
16	1507	002077	3057	DCA CNTR	/LENGTH < 0	16
17	1508	002100	1133	TAD CFRS	/IT IS ILLEGAL TO DELETE THE FIRST LINE	17
18	1509	002101	7041	CIA		18
19	1510	002102	1023	TAD THISLN		19
20	1511	002103	7650	SNA CLA		20
21	1512	002104	5177	JMP START	/JUST IGNORE SUCH COMMANDS	21
22	1513	002105	7000	CDF T	/CHANGE DATA FIELD TO TEXT.(X-MEM)	22
23	1514	002106	1423	TAD I THISLN	/DISCONNECT	23
24	1515	002107	3425	DCA I LASTLN		24
25	1516	002110	1133	TAD CFRS	/START LIST AT TOP	25
26	1517	002111	3071	DCA T2	/EXAMINATION ADDRESS	26
27	1518	002112	1471	TAD I T2	/GET THE NEXT ADDR.	27
28	1519	002113	7450	SNA	/TEST FOR END	28
29	1520	002114	5327	JMP DONE	/YES-WRAP UP ALL.	29
30	1521	002115	3032	DCA T1	/SAVE NEXT ADDRESS.	30
31	1522	002116	1023	TAD THISLN	/COMPARE LINE POSITIONS	31
32	1523	002117	7141	CLL CIA		32
33	1524	002120	1032	TAD T1		33
34	1525	002121	7630	SZL CLA	/SKIP IF THISLN > X	34
35	1526	002122	1057	TAD CNTR	/CHANGE (X) TO ACCOUNT FOR	35
36	1527	002123	1032	TAD T1	/GARBAGE COLLECTION.	36
37	1528	002124	3471	DCA I T2		37
38	1529	002125	1032	TAD T1	/GET NEXT	38
39	1530	002126	5311	JMP DOK		39
40						40
41						41
42						42
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1	1532		/	GARBAGE COLLECTION		1
2	1533					2
3	1534	002127	7040	DONE,	CMA /BACKUP L FOR XR	3
4	1535	002130	1023	TAD	THISLN	4
5	1536	002131	3011	DCA	XRT	5
6	1537	002132	1057	TAD	CNTR /SETUP END OF HOSE	6
7	1538	002133	7040	CMA		7
8	1539	002134	1023	TAD	THISLN	8
9	1540	002135	3012	DCA	XRT2	9
10	1541	002136	1057	TAD	CNTR /CORRECT END OF BUFFER POINTER.	10
11	1542	002137	1060	TAD	BUFR	11
12	1543	002140	3060	DCA	BUFR	12
13	1544	002141	1010	TAD	AXIN /COMPUTE COUNT	13
14	1545	002142	7040	CMA		14
15	1546	002143	1012	TAD	XRT2	15
16	1547	002144	3032	DCA	T1	16
17	1548	002145	1010	TAD	AXIN	17
18	1549	002146	1057	TAD	CNTR	18
19	1550	002147	3010	DCA	AXIN	19
20	1551	002150	1412	TAD I	XRT2 /SIPHON LOWER PART.	20
21	1552	002151	3411	DCA I	XRT	21
22	1553	002152	2032	ISZ	T1	22
23	1554	002153	5350	JMP	.-3	23
24	1555	002154	5263	JMP	XDELETE+1 /RESET 'LASTLN', 'THISLN', AND DATA FIELD.	24
25	1556					25
26	1557					26
27	1558	002155	0000	CHIN,	0 /READ IN A CHARACTER SUBR. - "READC"	27
28	1559	002156	4464	JMS I	INDEV	28
29	1560	002157	3066	DCA	CHAR	29
30	1561	002160	4550	SORTC	/LINEFEED OR RUBOUT?	30
31	1562	002161	1623	ECHOLST-1		31
32	1563	002162	5755	JMP I	CHIN /YES	32
33	1564	002163	4551	PRINTC	/ECHO THE INPUT	33
34	1565	002164	5755	JMP I	CHIN	34
35	1566					35
36	1567					36
37	1568		2165	FNTABL=	.	37
38	1569					38
39	1570	002165	2533	"A^2+"B^2+"S	/ABS	39
40	1571	002166	2650	"S^2+"G^2+"N	/SGN	40
41	1572	002167	2636	"I^2+"T^2+"R	/ITR	41
42	1573	002170	2565	"D^2+"I^2+"S	/DIS	42
43	1574	002171	2630	"R^2+"A^2+"N	/RAN	43
44	1575	002172	2517	"A^2+"D^2+"C	/ADC	44
45	1576	002173	2572	"A^2+"T^2+"N	/ATN	45
46	1577	002174	2624	"E^2+"X^2+"P	/EXP	46
47	1578	002175	2625	"L^2+"O^2+"G	/LOG	47
48	1579	002176	2654	"S^2+"I^2+"N	/SIN LIST OF CODED FUNCTION NAMES	48
49	1580	002177	2575	"C^2+"O^2+"S	/COS	49
50	1581	002200	2702	"S^2+"Q^2+"T	/SQT	50
51	1582	002201	2631	"N^2+"E^2+"W	/NEW	51
52	1583	002202	2567	"C^2+"O^2+"M	/COM	52
53	1584	002203	0330	"X	/X	53
54						54
55						55
56						56
57						57
58						58

1	1586		/	ERASE SINGLE LINES, GROUPS, OR VARIABLES			1
2	1587						2
3	1588	002204	4564	ERASE,	TESTC	/TEST THE SECOND WORD, IF ANY.	3
4	1589	002205	5237		JMP ERVX	/ERASE VARIABLES	4
5	1590	002206	5222		JMP ERL	/LINES OR GROUPS	5
6	1591	002207	5213		JMP .+4	/ERROR	6
7	1592	002210	1066		TAD CHAR	/ALL TEXT	7
8	1593	002211	1112		TAD MINUSA		8
9	1594	002212	7440		SZA		9
10	1595	002213	4566		ERROR3	/BAD ARG FOR ERASE.	10
11	1596	002214	1135	ERT,	TAD ENDT	/ERASE ALL TEXT **	11
12	1597	002215	3060		DCA BUFR		12
13	1598	002216	3533		DCA I CFRS	/(X-MEM)	13
14	1599	002217	1060	ERV,	TAD STARTV	/ERASE VARIABLES **	14
15	1600	002220	3031		DCA LASTV		15
16	1601	002221	5177		JMP START	/POINTERS MAY BE DIFFERENT NOW.	16
17	1602						17
18	1603	002222	4554	ERL,	GETLN	/ERASE LINES.	18
19	1604	002223	1060		TAD BUFR	/PROTECT REST OF TEXT.	19
20	1605	002224	3010		DCA AXIN		20
21	1606	002225	4565	ERG,	DELETE	/EXTRACT ONE LINE	21
22	1607	002226	2023		ISZ THISLN		22
23	1608	002227	1065		TAD NAGSW		23
24	1609	002230	7700		SMA CLA		24
25	1610	002231	1423		TAD I THISLN	/(X-MEM)	25
26	1611	002232	4563		TSTGRP	/SKIP IF G(AC) = G(LINENO)	26
27	1612	002233	5217		JMP ERV		27
28	1613	002234	1423		TAD I THISLN	/(X-MEM)	28
29	1614	002235	3067		DCA LINENO		29
30	1615	002236	5225		JMP ERG		30
31	1616						31
32	1617	002237	1060	ERVX,	TAD STARTV	/INIT VARIABLES MAY BE INDIRECT COMMAND	32
33	1618	002240	3031		DCA LASTV		33
34	1619	002241	5541		POPJ		34
35							35
36							36
37							37
38							38
39							39
40							40
41							41
42							42
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57							57
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1	1621		/			ROUTINE CALLED VIA "FINDLN":	1
2	1622						2
3	1623		/			SEARCH FOR A GIVEN LINE I.D. =["LINENO"]	3
4	1624		/			1ST RETURN IF NOT FOUND,	4
5	1625		/			2AND IF FOUND.	5
6	1626		/			"THISLN" = FOUND LINE OR NEXT LARGER.	6
7	1627		/			"LASTLN" = LESSER AND/OR LAST.	7
8	1628		/			"TEXTP" IS SET.	8
9	1629						9
10	1630	002242	0000	XFIND,	0		10
11	1631	002243	1133	TAD	CFRS	/INITIALIZE POINTERS TO FIRST LINE	11
12	1632	002244	3025	DCA	LASTLN		12
13	1633	002245	1133	TAD	CFRS		13
14	1634	002246	3023	DCA	THISLN	/SAVE THIS ONE	14
15	1635	002247	1023	TAD	THISLN		15
16	1636	002250	3011	DCA	XRT		16
17	1637	002251	1067	TAD	LINENO		17
18	1638	002252	7141	CLL CMA	IAC	/CLEAR LINK AND NEGATE LINENO.	18
19	1639	002253	1411	TAD I	XRT	/LINENO=0 WILL ALSO BE FOUND(X-MEM)	19
20	1640	002254	7450	SNA			20
21	1641	002255	5266	JMP	FEND3-1	/FOUND IT.	21
22	1642	002256	7630	SZL CLA			22
23	1643	002257	5267	JMP	FEND3	/PAST IT.	23
24	1644	002260	1023	TAD	THISLN	/MOVE POINTERS	24
25	1645	002261	3025	DCA	LASTLN		25
26	1646	002262	1423	TAD I	THISLN	/END OF TEXT? (X-MEM)	26
27	1647	002263	7440	SZA			27
28	1648	002264	5246	JMP	FINDN	/NOT YET	28
29	1649	002265	7410	SKP			29
30	1650	002266	2242	ISZ	XFIND	/2ND EXIT = FOUND	30
31	1651	002267	1023	FEND3, TAD	THISLN	/1ST RETURN = NOT FOUND	31
32	1652	002270	7001	IAC			32
33	1653	002271	3017	DCA	AXOUT	/SET "TEXTP".	33
34	1654	002272	3020	DCA	XCT		34
35	1655	002273	5642	JMP I	XFIND		35
36	1656						36
37	1657	002274	0000	UTRA,	0	/UNPACK CHARACTER. - "GETC"	37
38	1658	002275	4330	JMS	GET1		38
39	1659	002276	7710	UTE,	SPA CLA	/NORM & EXTEND	39
40	1660	002277	1006	TAD	C100	/300-337 & 340-376	40
41	1661	002300	1357	TAD	M137	/240-276 & 200-236	41
42	1662	002301	1066	TAD	CHAR		42
43	1663	002302	7450	SNA			43
44	1664	002303	5316	JMP	UTX	/"?" FOUND	44
45	1665	002304	1075	TAD	P337		45
46	1666	002305	3066	UTQ,	DCA	CHAR	46
47	1667	002306	1026	TAD	DEBGSW		47
48	1668	002307	1100	TAD	DMPSW		48
49	1669	002310	7650	SNA CLA		/PRINT ONLY IF BOTH ARE ZERO.	49
50	1670	002311	4551	PRINTC			50
51	1671	002312	5674	JMP I	UTRA		51
52	1672						52
53	1673	002313	4330	EXTR,	JMS	GET1	53
54	1674	002314	7040	CMA			54
55	1675	002315	5276	JMP	UTE		55
56							56
57							57
58							58

1	1677	002316	1026	UTX,	TAD	DEBGSW	/TEST FOR TRACE-ENABLED	1
2	1678	002317	7640		SZA CLA			2
3	1679	002320	5326		JMP	.+6		3
4	1680	002321	1100		TAD	DMP SW	/FLIP THE TRACE FLOP	4
5	1681	002322	7650		SNA CLA			5
6	1682	002323	7001		IAC			6
7	1683	002324	3100		DCA	DMP SW		7
8	1684	002325	5275		JMP	UTRA+1	/GET NEXT CHARACTER INSTEAD.	8
9	1685							9
10	1686	002326	1110		TAD	P277	/TRACE DISABLED = RETURN "?"	10
11	1687	002327	5305		JMP	UTQ		11
12	1688							12
13	1689	002330	0000	GET1,	0		/UNPACK 6-BITS	13
14	1690	002331	2020		ISZ	XCT	/STARTS=0	14
15	1691	002332	5345		JMP	GET3		15
16	1692	002333	1021		TAD	GTEM		16
17	1693	002334	0122	GEND,	AND	P77		17
18	1694	002335	3066		DCA	CHAR	/SAVE	18
19	1695	002336	1066		TAD	CHAR		19
20	1696	002337	1103		TAD	M77		20
21	1697	002340	7650		SNA CLA			21
22	1698	002341	5313		JMP	EXTR	/EXTENDED	22
23	1699	002342	1066		TAD	CHAR		23
24	1700	002343	1356		TAD	M40		24
25	1701	002344	5730		JMP I	GET1		25
26	1702							26
27	1703	002345	1417	GET3,	TAD I	AXOUT	/(X-MEM)	27
28	1704	002346	3021		DCA	GTEM		28
29	1705	002347	7040		CMA			29
30	1706	002350	3020		DCA	XCT		30
31	1707	002351	1021		TAD	GTEM		31
32	1708	002352	7112		CLL RTR			32
33	1709	002353	7012		RTR			33
34	1710	002354	7012		RTR			34
35	1711	002355	5334		JMP	GEND		35
36	1712							36
37	1713	002356	7740	M40,	-40			37
38	1714	002357	7641	M137,	-137			38
39								39
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1	1716	002360	0000	XENDLN, 0		/TERMINATE THE BUFFERED LINE - "ENDLN"	1
2	1717	002361	7000	CDF	T	/(X-MEM)	2
3	1718	002362	1425	TAD I	LASTLN	/SAVE OLD POINTER	3
4	1719	002363	3460	DCA I	BUFR		4
5	1720	002364	1060	TAD	BUFR	/POINT TO NEW LAST LINE	5
6	1721	002365	3425	DCA I	LASTLN		6
7	1722	002366	1061	TAD	ADD	/CHECK FOR EXTRA INFO	7
8	1723	002367	7440	SZA			8
9	1724	002370	3410	DCA I	AXIN		9
10	1725	002371	1010	TAD	AXIN	/COMPUTE NEW END OF BUFFER	10
11	1726	002372	7001	IAC			11
12	1727	002373	3060	DCA	BUFR		12
13	1728	002374	1060	TAD	STARTV	/RESET VARIABLE LIST (X-MEM)	13
14	1729	002375	3031	DCA	LASTV		14
15	1730	002376	5760	JMP I	XENDLN		15
16	1731						16
17	1732						17
18	1733						18
19	1734		2377	TLIST3=	.	/LITERAL TERMINATORS	19
20	1735						20
21	1736	002377	1253	TASK4		/"	21
22	1737	002400	0614	PC1		/C.R. = AUTOMATIC QUOTE MATCH	22
23	1738						23
24	1739						24
25	1740		2401	INFIX=	.	/DATA CONTROL CHARACTERS	25
26	1741						26
27	1742	002401	6202	FLINTP+2		/LEFT ARROW = KILL	27
28	1743	002402	0757	INPUT+1		/RUBOUT = IGNORE	28
29	1744	002403	0757	INPUT+1		/L.F. = IGNORE	29
30	1745	002404	6250	ENDFI+5		/ALT MODE = EXIT	30
31	1746						31
32	1747	002405	0001	FLTONE, 0001		/(NO RELATIVE REFERENCES)	32
33	1748	002406	2000		2000		33
34	1749	002407	0000	FLTZER, 0000			34
35	1750	002410	0000		0000		35
36	1751	002411	0000		0000		36
37	1752	002412	0000		0000		37
38	1753	002413	7766	M12, -12		/DECIMAL CONVERSION FACTOR FOR "PRNT"	38
39	1754						39
40	1755						40
41	1756	002414	0000	I33, 0		/NO-INTERRUPT INPUT ROUTINE	41
42	1757	002415	6031	KSF			42
43	1758	002416	5215	JMP	.-1		43
44	1759	002417	6036	KRB			44
45	1760	002420	0106	AND	P177	/IGNORE PARITY BIT	45
46	1761	002421	7450	SNA			46
47	1762	002422	5215	JMP	.-5		47
48	1763	002423	1123	TAD	C200		48
49	1764	002424	5614	JMP I	I33		49
50							50
51							51
52							52
53							53
54							54
55							55
56							56
57							57
58							58

1	1766	002425	0000	XPRNT,	0		/PRINT A LINE NUMBER - "PRNTLN"	1
2	1767	002426	1067		TAD	LINENO		2
3	1768	002427	4557		RTL6			3
4	1769	002430	0122		AND	P77		4
5	1770	002431	4242		JMS	PRNT	/TWO DIGIT "PART" NUMBER	5
6	1771	002432	1102		TAD	PER		6
7	1772	002433	4551		PRINTC		/PERIOD FOR SEPARATION	7
8	1773	002434	1067		TAD	LINENO		8
9	1774	002435	4242		JMS	PRNT	/TWO DIGIT "STEP" NUMBER.	9
10	1775	002436	1356		TAD	M140		10
11	1776	002437	3066		DCA	CHAR	/SAVE SPACE IN CHAR.	11
12	1777	002440	4551		PRINTC		/PRINT TRAILING SPACE	12
13	1778	002441	5625		JMP I	XPRNT		13
14	1779							14
15	1780		0032		VAL=	T1		15
16	1781							16
17	1782	002442	0000	PRNT,	0		/PRINT TWO DECIMAL DIGITS	17
18	1783	002443	0106		AND	P177		18
19	1784	002444	3032		DCA	VAL		19
20	1785	002445	1113		TAD	C260		20
21	1786	002446	3033		DCA	T3		21
22	1787	002447	5252		JMP	.+3		22
23	1788	002450	2033		ISZ	T3		23
24	1789	002451	3032	XYZ,	DCA	VAL		24
25	1790	002452	1032		TAD	VAL		25
26	1791	002453	1213		TAD	M12		26
27	1792	002454	7500		SMA			27
28	1793	002455	5250		JMP	XYZ-1		28
29	1794	002456	7200		CLA			29
30	1795	002457	1033		TAD	T3		30
31	1796	002460	4551		PRINTC			31
32	1797	002461	1032		TAD	VAL		32
33	1798	002462	1113		TAD	C260		33
34	1799	002463	4551		PRINTC			34
35	1800	002464	5642		JMP I	PRNT		35
36	1801							36
37	1802	002465	0000	OUT,	0		/OUTPUT A CHARACTER - "PRINTC"	37
38	1803	002466	7450		SNA		/USE (AC) OR (CHAR)	38
39	1804	002467	1066		TAD	CHAR		39
40	1805	002470	1116		TAD	MCR		40
41	1806	002471	7450		SNA			41
42	1807	002472	5276		JMP	OUTCR		42
43	1808	002473	1077		TAD	CCR		43
44	1809	002474	4463		JMS I	OUTDEV		44
45	1810	002475	5665	OUTX,	JMP I	OUT		45
46	1811							46
47	1812	002476	1077	OUTCR,	TAD	CCR		47
48	1813	002477	4463		JMS I	OUTDEV		48
49	1814	002500	1076		TAD	CLF		49
50	1815	002501	5274		JMP	OUTX-1		50
51								51
52								52
53								53
54								54
55								55
56								56
57								57
58								58

1	1817	002502	0000	PACBUF,	0		/PACK A CHARACTER - "PACKC"	1
2	1818	002503	1110		TAD	P277		2
3	1819	002504	7041		CIA			3
4	1820	002505	1066		TAD	CHAR		4
5	1821	002506	7450		SNA		/CHANGE 277 TO 337	5
6	1822	002507	1352		TAD	P40		6
7	1823	002510	1101		TAD	M100		7
8	1824	002511	7450		SNA		/TEST FOR RUBOUT.	8
9	1825	002512	5755		JMP I	RUBIT		9
10	1826	002513	1353		TAD	P377		10
11	1827	002514	3071		DCA	T2	/SAVE INPUT ITEM	11
12	1828	002515	1071		TAD	T2	/SO THAT QUESTION DOESN'T MAKE	12
13	1829	002516	0354		AND	C140	/CHAR LOOK LIKE A LEFT-ARROW	13
14	1830	002517	1356		TAD	M140		14
15	1831	002520	7440		SZA		/DATA WORD.	15
16	1832	002521	1354		TAD	C140		16
17	1833	002522	7650		SNA	CLA		17
18	1834	002523	5332		JMP	ESCA	/340-377 AND 200-237	18
19	1835	002524	1071	PA1,	TAD	T2	/240-337	19
20	1836	002525	0122		AND	P77		20
21	1837	002526	7440		SZA		/IGNORE 300	21
22	1838	002527	4335		JMS	PCK1		22
23	1839	002530	7000	PACX,	CDF	P	/(X-MEM)	23
24	1840	002531	5702		JMP I	PACBUF		24
25	1841							25
26	1842	002532	1122	ESCA,	TAD	P77		26
27	1843	002533	4335		JMS	PCK1		27
28	1844	002534	5324		JMP	PA1		28
29	1845							29
30	1846	002535	0000	PCK1,	0			30
31	1847	002536	2062		ISZ	XCTIN	/=0 TO START	31
32	1848	002537	5357		JMP	ROT		32
33	1849	002540	1061		TAD	ADD		33
34	1850	002541	3410		DCA I	AXIN	/(X-MEM)	34
35	1851	002542	3061		DCA	ADD	/CLEAR PACKING WORD	35
36	1852	002543	1013		TAD	PDLXR	/CHECK FOR OVERFLOW	36
37	1853	002544	7141		CLL CMA	IAC		37
38	1854	002545	1005		TAD	P13	/RESERVATIONS FOR PUSH-DOWN LIST	38
39	1855	002546	1010		TAD	AXIN		39
40	1856	002547	7620		SNL CLA			40
41	1857	002550	5735		JMP I	PCK1		41
42	1858	002551	4566		ERROR2		/FULL BUFFER	42
43	1859							43
44	1860	002552	0040	P40,	40			44
45	1861	002553	0377	P377,	377			45
46	1862	002554	0140	C140,	140			46
47	1863	002555	3004	RUBIT,	RUB1			47
48	1864	002556	7640	M140,	-140			48
49	1865							49
50	1866	002557	4557	ROT,	RTL6		/(EAE)	50
51	1867	002560	3061		DCA	ADD		51
52	1868	002561	7040		CMA			52
53	1869	002562	3062		DCA	XCTIN		53
54	1870	002563	5735		JMP I	PCK1	/REST OF PAGE USED BY 8K	54
55								55
56								56
57								57
58								58

1	1872	*2600		PAGE			1
2	1873						2
3	1874	/		INTERRUPT PROCESSOR.			3
4	1875						4
5	1876	002600	0000	SAVAC,	0	/CONTENTS OF AC	5
6	1877	002601	0000	SAVLK,	0	/CONTENTS OF LINK	6
7	1878	002602	7575	MBREAK,	-203	/CONTROL-C	7
8	1879						8
9	1880	002603	3200	INTRPT,	DCA	SAVAC	9
10	1881	002604	7010		RAR		10
11	1882	002605	3201		DCA	SAVLK	11
12	1883	002606	6041		TSF		12
13	1884	002607	5225		JMP	KINT	13
14	1885	002610	6042		TCF		14
15	1886	002611	3016		DCA	TELSW	15
16	1887	002612	1665		TAD I	OPTRI	16
17	1888	002613	7450		SNA		17
18	1889	002614	5225		JMP	KINT	18
19	1890	002615	6044		TPC		19
20	1891	002616	3016		DCA	TELSW	20
21	1892	002617	3665		DCA I	OPTRI	21
22	1893	002620	1265		TAD	OPTRI	22
23	1894	002621	7001		IAC		23
24	1895	002622	0107		AND	P17	24
25	1896	002623	1263		TAD	OPTR0	25
26	1897	002624	3265		DCA	OPTRI	26
27	1898	002625	6031	KINT,	KSF		27
28	1899	002626	5246		JMP	EXIT	28
29	1900	002627	6036		KRB		29
30	1901	002630	0106		AND	P177	30
31	1902	002631	7450		SNA		31
32	1903	002632	5246		JMP	EXIT	32
33	1904	002633	1123		TAD	C200	33
34	1905	002634	3262		DCA	SIN	34
35	1906	002635	1262		TAD	SIN	35
36	1907	002636	1202		TAD	MBREAK	36
37	1908	002637	7650		SNA CLA		37
38	1909	002640	5340		JMP	RECOVR	38
39	1910	002641	1034		TAD	INBUF	39
40	1911	002642	7640		SZA CLA		40
41	1912	002643	4566		ERROR2		41
42	1913	002644	1262		TAD	SIN	42
43	1914	002645	3034		DCA	INBUF	43
44	1915	002646	6011	EXIT,	RSF		44
45	1916	002647	5252		JMP	+.3	45
46	1917	002650	6012		RRB		46
47	1918	002651	3037		DCA	HINBUF	47
48	1919	002652	6244		RMF		48
49	1920	002653	6101		SMP		49
50	1921	002654	7000		NOP		50
51	1922	002655	1201		TAD	SAVLK	51
52	1923	002656	7104		CLL RAL		52
53	1924	002657	1200		TAD	SAVAC	53
54	1925	002660	6001		ION		54
55	1926	002661	5400	EXITJ,	JMP I	0	55
56							56
57							57
58							58

[illegible]

1	1966		/	ERROR RECOVERY PROCEEDURE			1
2	1967						2
3	1968	002725	3326	ERROR5,	DCA	+.1	3
4	1969	002726	0000	ERR2,	0		4
5	1970	002727	7240		CLA	CMA	5
6	1971	002730	1326		TAD	ERR2	6
7	1972	002731	3067		DCA	LINENO	7
8	1973	002732	6001		ION		8
9	1974	002733	1016		TAD	TELSW	9
10	1975	002734	7640		SZA	CLA	10
11	1976	002735	5333		JMP	.-2	11
12	1977	002736	6002		IOF		12
13	1978	002737	5342		JMP	+.3	13
14	1979	002740	1123	RECOVR,	TAD	C200	14
15	1980	002741	3067		DCA	LINENO	15
16	1981	002742	2016		ISZ	TELSW	16
17	1982	002743	1105		TAD	M20	17
18	1983	002744	3057		DCA	CNTR	18
19	1984	002745	7040		CMA		19
20	1985	002746	1263		TAD	OPTR0	20
21	1986	002747	3010		DCA	AXIN	21
22	1987	002750	7000		CDF	P	22
23	1988	002751	3410		DCA I	AXIN	23
24	1989	002752	2057		ISZ	CNTR	24
25	1990	002753	5351		JMP	.-2	25
26	1991	002754	3034		DCA	INBUF	26
27	1992	002755	1263		TAD	OPTR0	27
28	1993	002756	3265		DCA	OPTRI	28
29	1994	002757	1263		TAD	OPTR0	29
30	1995	002760	3264		DCA	OPTR0	30
31	1996	002761	7040	RECOVX,	CMA		31
32	1997	002762	6046		TLS		32
33	1998	002763	1101		TAD	P7700	33
34	1999	002764	4551		PRINTC		34
35	2000	002765	4553		PRNTLN		35
36	2001	002766	2022		ISZ	PC	36
37	2002	002767	1422		TAD I	PC	37
38	2003	002770	7450		SNA		38
39	2004	002771	5377		JMP	+.6	39
40	2005	002772	3067		DCA	LINENO	40
41	2006	002773	1101		TAD	P7700	41
42	2007	002774	4551		PRINTC		42
43	2008	002775	4551		PRINTC		43
44	2009	002776	4553		PRNTLN		44
45	2010	002777	1077		TAD	CCR	45
46	2011	003000	4551		PRINTC		46
47	2012	003001	1126		TAD	PTCH	47
48	2013	003002	3152		DCA	RDIV	48
49	2014	003003	5177		JMP	START	49
50							50
51							51
52							52
53							53
54							54
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56							56
57							57
58							58

1	2016		/	CHRACTER REMOVAL ROUTINE				1
2	2017							2
3	2018	003004	1062	RUB1,	TAD	XCTIN	/RUBOUT ONE LETTER	3
4	2019	003005	7640		SZA	CLA		4
5	2020	003006	5214		JMP	.+6		5
6	2021	003007	1010		TAD	AXIN		6
7	2022	003010	7041		CIA			7
8	2023	003011	1027		TAD	PACKST		8
9	2024	003012	7700		SMA	CLA	/TEST NULL LINE	9
10	2025	003013	5641		JMP	I	RUB5	10
11	2026	003014	1251		TAD	SPLAT	/FOR A RUBOUT ACKNOWLEDGEMENT	11
12	2027	003015	4551		PRINTC			12
13	2028	003016	1010		TAD	AXIN		13
14	2029	003017	3071		DCA	T2		14
15	2030	003020	7000		CDF	T	/(X-MEM)	15
16	2031	003021	2062		ISZ	XCTIN	/TEST HALF	16
17	2032	003022	5242		JMP	RUB2		17
18	2033	003023	1471		TAD	I	T2	18
19	2034	003024	0122		AND	P77		19
20	2035	003025	1103		TAD	M77		20
21	2036	003026	7640		SZA	CLA	/TEST FOR EXTEND	21
22	2037	003027	5237		JMP		RUB4	22
23	2038	003030	7040	RUB3,	CMA		/SET SWITCH	23
24	2039	003031	3062		DCA	XCTIN		24
25	2040	003032	7040		CMA		/BACKUP POINTER	25
26	2041	003033	1010		TAD	AXIN		26
27	2042	003034	3010		DCA	AXIN		27
28	2043	003035	1471		TAD	I	T2	28
29	2044	003036	0101		AND	P7700	/RESET ADD	29
30	2045	003037	3061	RUB4,	DCA	ADD		30
31	2046	003040	5641		JMP	I	RUB5	31
32	2047							32
33	2048	003041	2530	RUB5,	PACX			33
34	2049							34
35	2050	003042	1471	RUB2,	TAD	I	T2	35
36	2051	003043	0101		AND	P7700	/CHECK FOR EXTENDED	36
37	2052	003044	1006		TAD	C100		37
38	2053	003045	7640		SZA	CLA		38
39	2054	003046	5230		JMP		RUB3	39
40	2055	003047	3471		DCA	I	T2	40
41	2056	003050	5231		JMP		RUB3+1	41
42	2057							42
43	2058	003051	0334	SPLAT,	334			43
44								44
45								45
46								46
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1	2060		/	SYMBOL TABLE TYPEOUT ROUTINE				1
2	2061							2
3	2062	003052	1060	TDUMP,	TAD	STARTV	/INIT POINTER FOR SYMBOL DUMP.(X-MEM)	3
4	2063	003053	3030		DCA	PT1		4
5	2064	003054	1031		TAD	LASTV	/TEST FOR END OF LIST	5
6	2065	003055	7041		CIA			6
7	2066	003056	1030		TAD	PT1		7
8	2067	003057	7650		SNA	CLA		8
9	2068	003060	5541		POPJ			9
10	2069	003061	1430		TAD I	PT1	/GET THE VARIABLE	10
11	2070	003062	3316		DCA	OP+1	/((DCA I (4)-FOR(X-MEM)):SAVE NAME	11
12	2071	003063	1315		TAD	OP	/SETUP UNPACK POINTERS	12
13	2072	003064	3017		DCA	AXOUT		13
14	2073	003065	3020		DCA	XCT		14
15	2074	003066	4545		GETC		/READ AND PRINT "XX("	15
16	2075	003067	4551		PRINTC			16
17	2076	003070	4545		GETC			17
18	2077	003071	4551		PRINTC			18
19	2078	003072	4545		GETC			19
20	2079	003073	4551		PRINTC			20
21	2080	003074	2030		ISZ	PT1		21
22	2081	003075	1430		TAD I	PT1	/PRINT SUBSCRIPT TO 99	22
23	2082	003076	4714		JMS I	PRNT2		23
24	2083	003077	4545		GETC		/PRINT ")"	24
25	2084	003100	4551		PRINTC			25
26	2085	003101	2030		ISZ	PT1		26
27	2086	003102	4407		FINT		/PICK UP VALUE	27
28	2087	003103	0430		FGET I	PT1		28
29	2088	003104	0000		FXIT			29
30	2089	003105	4530		JMS I	FOUTPUT	/PRINT VALUE	30
31	2090	003106	1077		TAD	CCR		31
32	2091	003107	4551		PRINTC			32
33	2092	003110	1070		TAD	GINC		33
34	2093	003111	1111		TAD	M2		34
35	2094	003112	1030		TAD	PT1		35
36	2095	003113	5253		JMP	TDUMP+1		36
37	2096							37
38	2097	003114	2442	PRNT2,	PRNT			38
39	2098	003115	3115	OP,	.		/ (X-MEM)	39
40	2099	003116	0000		0000		/ (X-MEM)	40
41	2100	003117	5051		"(^100+)"	-200	/(THESE GO IN 10005 FOR X-MEM)	41
42								42
43								43
44								44
45								45
46								46
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1	2102		/	OUTPUT CHARACTER BUFFER (ADDRESS IS A MULTIPLE OF 20)	1
2	2103				2
3	2104	3120		IOBUF= 3120	3
4	2105				4
5	2106	3140		COMEIN= IOBUF+20 /COMMAND - INPUT BUFFER	5
6	2107				6
7	2108	3206		COMEOUT=COMEIN+46	7
8	2109				8
9	2110				9
10	2111	*3206		*COMEOUT	10
11	2112				11
12	2113				12
13	2114				13
14	2115	003206	0000	FRST, 0 /TEXT POINTER	14
15	2116	003207	0000	0000 /DUMMY LINE NO.	15
16	2117	003210	0355	"C^100+" -200 /TITLE	16
17	2118	003211	0617	"F^100+"0-300 /FO	17
18	2119	003212	0301	"C^100+"A-300 /CA	18
19	2120	003213	1454	"L^100+" -200 /L,	19
20	2121	003214	6171	"1^100+"9-200 /19	20
21	2122	003215	6671	FRSTX, "6^100+"9-200 /69	21
22	2123	003216	7715	"?^100+"M-300 /DUMMY C.R.	22
23	2124				23
24	2125		/	TO SAVE TEXT, SAVE C(BUFR), C(LASTV), AND C(FRST TO C(BUFR))	24
25	2126		/	WITH ODT-JR46. THE TAPES MAY BE TOGETHER WITH	25
26	2127		/	THE SYMBOLIC DUMP LAST : FOCAL + FLOAT + DIALOG .	26
27	2128		/	LOADING THE LAST SECTION MAY BE CONSIDERED OPTIONAL.	27
28	2129				28
29	2130				29
30	2131				30
31	2132	3217		BUFBEQ= . /TEXT BUFFER STARTS HERE.	31
32					32
33					33
34					34
35					35
36					36
37					37
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39					39
40					40
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58					58

1	2171	004426	6141	T12,	6141	/BECOME A LINC	1
2	2172	004427	0017		0017	/COMPLEMENT AC	2
3	2173	004430	0002		0002	/BACK TO 8 MODE	3
4	2174	004431	7001		IAC	/SET TO ZERO IF PDP-12	4
5	2175	004432	7650		SNA CLA		5
6	2176	004433	5306		JMP	ATES-5	6
7	2177						7
8	2178	004434	7101		CLL IAC	/LAB-8?	8
9	2179	004435	6344		6344	/"OTEN"	9
10	2180	004436	6331		6331	/"XRIN"	10
11	2181	004437	7700		SMA CLA		11
12	2182	004440	5246		JMP	.+6	12
13	2183	004441	1350		TAD	L8A	13
14	2184	004442	3752		DCA I	L8AY	14
15	2185	004443	1351		TAD	L8B	15
16	2186	004444	3753		DCA I	L8AX	16
17	2187	004445	5307		JMP	ATES-4	17
18	2188						18
19	2189	004446	7354		7354	/NL3776	19
20	2190	004447	1367		TAD	PDP8I	20
21	2191	004450	7650		SNA CLA	/IS THIS A PDP-8/I OR 8/L?	21
22	2192	004451	5265		JMP	ATEI	22
23	2193	004452	7344		7344	/NL7776	23
24	2194	004453	1366		TAD	P2	24
25	2195	004454	7650		SNA CLA		25
26	2196	004455	5312		JMP	ATES-1	26
27	2197	004456	1100		TAD	CCR+1	27
28	2198	004457	3764		DCA I	06	28
29	2199	004460	1212		TAD	04	29
30	2200	004461	3763		DCA I	05	30
31	2201	004462	5313		JMP	ATES	31
32	2202						32
33	2203	004463	2761	PDP5X,	ISZ I	02	33
34	2204	004464	5314		JMP	ATES+1	34
35							35
36							36
37							37
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[illegible]

1	2231		/	INITIALIZE THE DIALOGUE	1
2	2232				2
3	2233	004514 6046		TLS	3
4	2234	004515 6001		ION /ENABLE INTERRUPT	4
5	2235	004516 4540		PUSHJ	5
6	2236	004517 0421		DO+1	6
7	2237	004520 6002		IOF	7
8	2238				8
9	2239		/	SHALL I KEEP EXP, LOG, ATN? (256)	9
10	2240		/	SHALL I KEEP SIN, COS? (128)	10
11	2241		/	XF = +1(NO) -1(YES) 0(YES)	11
12	2242				12
13	2243	004521 1360		TAD XF	13
14	2244	004522 4371		JMS LOOKUP	14
15	2245	004523 7450		SNA	15
16	2246	004524 5344		JMP 00UT /NO DIALOGUE EXECUTED	16
17	2247	004525 7710		SPA CLA	17
18	2248	004526 1366		TAD P2	18
19	2249	004527 1120		TAD M5 /DELETE X-FUNCTIONS	19
20	2250	004530 3057		DCA CNTR	20
21	2251	004531 1354		TAD FNPT	21
22	2252	004532 3011		DCA XRT	22
23	2253	004533 1355		TAD ER5	23
24	2254	004534 3411		DCA I XRT /SET THE TABLE	24
25	2255	004535 2057		ISZ CNTR	25
26	2256	004536 5333		JMP .-3	26
27	2257	004537 1360		TAD XF /CORRECT BUFFER PROTECT	27
28	2258	004540 4371		JMS LOOKUP	28
29	2259	004541 7710		SPA CLA	29
30	2260	004542 1104		TAD P7600 /(-200)	30
31	2261	004543 1356		TAD BFX	31
32	2262	004544 1357	00UT,	TAD BFX	32
33	2263	004545 3035		DCA BOTTOM	33
34	2264	004546 5747		JMP I .+1; ERT /GOES TO START AFTER ERASING ALL	34
35	2265	004547 2214			35
36					36
37					37
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[illegible]

1	2293		*6321		*6321		/STUCK INTO THE FLOATING POINT PACKAGE.	1
2	2294							2
3	2295	006321	0000	HREAD,	0		/HAMILTON STANDARD READER SUBROUTINE.	3
4	2296	006322	1105		TAD	M20	/(CMA)- FOR 8/S.	4
5	2297	006323	3343		DCA	HSWITC		5
6	2298	006324	1037	HREAD2,	TAD	HINBUF	/(RSF) -WHEN DEBUGGING	6
7	2299	006325	7700		SMA	CLA	/(SKP)	7
8	2300	006326	5364		JMP	HSGO		8
9	2301	006327	2032		ISZ	T1	/SKIP IF OUT OF TAPE	9
10	2302	006330	5324		JMP	HREAD2		10
11	2303	006331	2343		ISZ	HSWITC		11
12	2304	006332	5324		JMP	HREAD2		12
13	2305	006333	4343		JMS	HSWITC	/LEAVES LINK ZERO	13
14	2306	006334	1013		TAD	PDLXR	/ < FRST?	14
15	2307	006335	1376		TAD	HTST		15
16	2308	006336	7620		SNL	CLA		16
17	2309	006337	5742		JMP	I	.+3	17
18	2310	006340	2013		ISZ	PDLXR	/YES=DIRECT COMMAND /NO=OK TO DUMP CALL	18
19	2311	006341	5541		POPJ			19
20	2312							20
21	2313	006342	0212		IBAR			21
22	2314							22
23	2315	006343	0000	HSWITC,	0			23
24	2316	006344	1375		TAD	HSPSW	/INITIALIZE H.S. READER	24
25	2317	006345	7040		CMA			25
26	2318	006346	3375		DCA	HSPSW	/CHANGE STATUS	26
27	2319	006347	7140		CLL	CMA	/CLEAR LINK	27
28	2320	006350	3037		DCA	HINBUF	/CLEAR BUFFER	28
29	2321	006351	1375		TAD	HSPSW		29
30	2322	006352	7440		SZA			30
31	2323	006353	6014		RFC		/START HARDWARE	31
32	2324	006354	7640		SZA	CLA		32
33	2325	006355	1377		TAD	RESTR	/(HREAD)	33
34	2326	006356	1126		TAD	PTCH		34
35	2327	006357	3152		DCA	RDIV	/"READC"	35
36	2328	006360	5743		JMP	I	HSWITC	36
37	2329							37
38	2330	006361	4343	HSPX,	JMS	HSWITC	/COMMAND "*" - SWAP	38
39	2331	006362	5763		JMP	I	.+1; PROC	39
40	2332	006363	0611					40
41	2333							41
42	2334	006364	7040	HSGO,	CMA		/FETCH NEXT CHARACTER	42
43	2335	006365	3037		DCA	HINBUF		43
44	2336	006366	6016		RFC	RRB	/PICK UP NEXT CHARACTER	44
45	2337	006367	0106		AND	P177	/CHECK FOR LEADER-TRAILER,ETC.	45
46	2338	006370	7450		SNA			46
47	2339	006371	5322		JMP	HREAD+1		47
48	2340	006372	1123		TAD	C200		48
49	2341	006373	3066		DCA	CHAR	/SAVE INPUT	49
50	2342	006374	5721		JMP	I	HREAD	50
51	2343							51
52	2344	006375	0000	HSPSW,	0			52
53	2345	006376	4557	HTST,	-COMEOUT-13			53
54	2346	006377	4144	RESTR,	HREAD-CHIN			54
55								55
56								56
57								57
58								58

1	2348		/	DISK MONITOR INTERACTIVE COMMAND OPERATES VIA THE KEYBOARD.			1
2	2349		/	THIS FITS UNDER THE 10 DIGIT FLOATING POINT OUTPUT BUFFER.			2
3	2350						3
4	2351	*7503		*7503			4
5	2352						5
6	2353	007503	1133	LIBRARY,	TAD	CFRS	6
7	2354	007504	4327		JMS	PRNT8	7
8	2355	007505	1060		TAD	BUFR	8
9	2356	007506	4327		JMS	PRNT8	9
10	2357	007507	1031		TAD	LASTV	10
11	2358	007510	4327		JMS	PRNT8	11
12	2359	007511	1035		TAD	BOTTOM	12
13	2360	007512	4327		JMS	PRNT8	13
14	2361	007513	5316		JMP	.+3	14
15	2362						15
16	2363	007514	4545		GETC		16
17	2364	007515	4551		PRINTC		17
18	2365	007516	1066		TAD	CHAR	18
19	2366	007517	1116		TAD	MCR	19
20	2367	007520	7640		SZA	CLA	20
21	2368	007521	5314		JMP	.-5	21
22	2369	007522	1016		TAD	TELSW	22
23	2370	007523	7640		SZA	CLA	23
24	2371	007524	5322		JMP	.-2	24
25	2372	007525	6002		IOF		25
26	2373	007526	5504		JMP I	P7600	26
27	2374						27
28	2375	007527	0000	PRNT8,	0		28
29	2376	007530	3032		DCA	T1	29
30	2377	007531	1032		TAD	T1	30
31	2378	007532	7006		RTL		31
32	2379	007533	7006		RTL		32
33	2380	007534	4350		JMS	PRINTD	33
34	2381	007535	4557		RTL6		34
35	2382	007536	7004		RAL		35
36	2383	007537	4350		JMS	PRINTD	36
37	2384	007540	7012		RTR		37
38	2385	007541	7010		RAR		38
39	2386	007542	4350		JMS	PRINTD	39
40	2387	007543	4350		JMS	PRINTD	40
41	2388	007544	7200		CLA		41
42	2389	007545	1077		TAD	CCR	42
43	2390	007546	4551		PRINTC		43
44	2391	007547	5727		JMP I	PRNT8	44
45	2392						45
46	2393	007550	0000	PRINTD,	0		46
47	2394	007551	0356		AND	LP7	47
48	2395	007552	1113		TAD	C260	48
49	2396	007553	4551		PRINTC		49
50	2397	007554	1032		TAD	T1	50
51	2398	007555	5750		JMP I	PRINTD	51
52	2399						52
53	2400	007556	0007	LP7,	7		53
54	2401						54
55	2402		/	USED BY 8K			55
56							56
57							57
58							58

1	2448		/	FLOATING POINT PACKAGE - EXPONENTIAL	1
2	2449				2
3	2450	1045		GETSGN= TAD FLAC+1	3
4	2451	5536		RETURN= JMP I EFUN3I	4
5	2452				5
6	2453	*4620		*4600+20	6
7	2454				7
8	2455	004620	1045	FEXP, GETSGN /TAKE ABSOLUTE VALUE	8
9	2456	004621	7710	SPA CLA	9
10	2457	004622	4724	JMS I NEGP	10
11	2458	004623	3033	DCA T3 /C(SIGN)=-1 IF I X2<0	11
12	2459	004624	4407	FINT	12
13	2460	004625	4313	FMUL LG2E	13
14	2461	004626	6675	FPUT I X2	14
15	2462	004627	0000	FEXT	15
16	2463	004630	4453	JMS I INTEGER /TAKE INTEGER PART	16
17	2464	004631	3325	DCA FLAG2 /SAVE LOW ORDER DATA	17
18	2465	004632	4407	FINT	18
19	2466	004633	7000	FNOR	19
20	2467	004634	6676	FPUT I XSQ2	20
21	2468	004635	0675	FGET I X2	21
22	2469	004636	2676	FSUB I XSQ2	22
23	2470	004637	6675	FPUT I X2	23
24	2471	004640	4675	FMUL I X2	24
25	2472	004641	6676	FPUT I XSQ2	25
26	2473	004642	1310	FADD DF	26
27	2474	004643	6326	FPUT TEMP	27
28	2475	004644	0305	FGET CF	28
29	2476	004645	3326	FDIV TEMP	29
30	2477	004646	2675	FSUB I X2	30
31	2478	004647	1277	FADD AF	31
32	2479	004650	6326	FPUT TEMP	32
33	2480	004651	0302	FGET BF	33
34	2481	004652	4676	FMUL I XSQ2	34
35	2482	004653	1326	FADD TEMP	35
36	2483	004654	6326	FPUT TEMP	36
37	2484	004655	0675	FGET I X2	37
38	2485	004656	3326	FDIV TEMP	38
39	2486	004657	4321	FMUL TWO	39
40	2487	004660	1316	FADD ONE	40
41	2488	004661	0000	FEXT	41
42	2489	004662	1325	TAD FLAG2	42
43	2490	004663	1044	TAD FLAC	43
44	2491	004664	3044	DCA FLAC	44
45	2492	004665	2033	ISZ T3	45
46	2493	004666	5536	RETURN	46
47	2494	004667	4407	FINT	47
48	2495	004670	6675	FPUT I X2	48
49	2496	004671	0316	FGET ONE	49
50	2497	004672	3675	FDIV I X2	50
51	2498	004673	0000	FEXT	51
52	2499	004674	5536	RETURN	52
53					53
54					54
55					55
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1	2501			/	CONSTANTS FOR FEXP	1
2	2502					2
3	2503	004675	5322	X2,	X	3
4	2504	004676	5326	XSQ2,	XSQR	4
5	2505	004677	0004	AF,	0004	5
6	2506	004700	2372		2372	6
7	2507	004701	1402		1402	7
8	2508	004702	7774	BF,	7774	8
9	2509	004703	2157		2157	9
10	2510	004704	5157		5157	10
11	2511	004705	0012	CF,	0012	11
12	2512	004706	5454		5454	12
13	2513	004707	0343		0343	13
14	2514	004710	0007	DF,	0007	14
15	2515	004711	2566		2566	15
16	2516	004712	5341		5341	16
17	2517	004713	0001	LG2E,	0001	17
18	2518	004714	2705		2705	18
19	2519	004715	2435		2435	19
20	2520	004716	0001	ONE,	0001	20
21	2521	004717	2000		2000	21
22	2522	004720	0000		0000	22
23	2523	004721	0002	TWO,	0002	23
24	2524	004722	2000		2000	24
25	2525	004723	0000		0000	25
26	2526	004724	5163	NEGP,	FNEG	26
27	2527					27
28	2528	004725	0000	FLAG2,	0	28
29	2529	004726	0000	TEMP,	0	29
30	2530	004727	0000		0	30
31	2531	004730	0000		0	31
32	2532	004731	0000		0	32
33						33
34						34
35						35
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1	2534		/	MAIN ALGORITHM FOR ARCTANGENT	1
2	2535				2
3	2536	004732	4407	ARCALG, FINT	3
4	2537	004733	0675	FGET I X2	4
5	2538	004734	4675	FMUL I X2	5
6	2539	004735	6676	FPUT I XSQ2	6
7	2540	004736	4374	FMUL BET2	7
8	2541	004737	1371	FADD BET1	8
9	2542	004740	4676	FMUL I XSQ2	9
10	2543	004741	1366	FADD BETZ	10
11	2544	004742	6326	FPUT TEMP	11
12	2545	004743	0363	FGET ALF2	12
13	2546	004744	4676	FMUL I XSQ2	13
14	2547	004745	1360	FADD ALF1	14
15	2548	004746	4676	FMUL I XSQ2	15
16	2549	004747	1355	FADD ALFZ	16
17	2550	004750	4675	FMUL I X2	17
18	2551	004751	3326	FDIV TEMP	18
19	2552	004752	0000	FEXT	19
20	2553	004753	5754	JMP I .+1; ARCRTN	20
21	2554	004754	5024		21
22	2555				22
23	2556				23
24	2557				24
25	2558		/	CONSTANTS - FLOATING ARC TANGENT	25
26	2559				26
27	2560	004755	0000	ALFZ, 0000	27
28	2561	004756	2437	2437	28
29	2562	004757	1643	1643	29
30	2563	004760	7777	ALF1, 7777	30
31	2564	004761	3304	3304	31
32	2565	004762	4434	4434	32
33	2566	004763	7773	ALF2, 7773	33
34	2567	004764	3306	3306	34
35	2568	004765	5454	5454	35
36	2569	004766	0000	BETZ, 0000	36
37	2570	004767	2437	2437	37
38	2571	004770	1646	1646	38
39	2572	004771	0000	BET1, 0000	39
40	2573	004772	2427	2427	40
41	2574	004773	2323	2323	41
42	2575	004774	7775	BET2, 7775	42
43	2576	004775	3427	3427	43
44	2577	004776	7052	7052	44
45					45
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1	2579	/			FLOATING POINT ARC TANGENT			1	
2	2580								2
3	2581	*5000			PAGE			3	
4	2582								4
5	2583	005000	1045	ARTN,	GETSGN	/TAKE ABSOLUTE VALUE		5	
6	2584	005001	7710		SPA CLA			6	
7	2585	005002	4363		JMS	FNEG		7	
8	2586	005003	3033		DCA	T3		8	
9	2587	005004	4407		FINT			9	
10	2588	005005	6635		FPUT I	X1		10	
11	2589	005006	2637		FSUB I	CON1		11	
12	2590	005007	0000		FEXT			12	
13	2591	005010	1045		GETSGN			13	
14	2592	005011	7710		SPA CLA			14	
15	2593	005012	5221		JMP	GO	/LESS THAN ONE	15	
16	2594	005013	4407		FINT			16	
17	2595	005014	0637		FGET I	CON1		17	
18	2596	005015	3635		FDIV I	X1		18	
19	2597	005016	6635		FPUT I	X1		19	
20	2598	005017	0000		FEXT			20	
21	2599	005020	7240		CLA CMA			21	
22	2600	005021	3362	GO,	DCA	FLAG1	/SIGN FLAG OF RESULT	22	
23	2601	005022	5623		JMP I	.+1;	ARCALG /CALL ALGORITHM	23	
24	2602	005023	4732					24	
25	2603								25
26	2604	005024	2362	ARCRTN,	ISZ	FLAG1	/RETURN HERE	26	
27	2605	005025	5634		JMP I	EXIT1		27	
28	2606	005026	4407		FINT			28	
29	2607	005027	6635		FPUT I	X1		29	
30	2608	005030	0636		FGET I	PI2		30	
31	2609	005031	2635		FSUB I	X1		31	
32	2610	005032	0000		FEXT			32	
33	2611	005033	5634		JMP I	.+1		33	
34	2612	005034	5302	EXIT1,	EXIT2			34	
35	2613								35
36	2614	/			CONSTANTS FOR ARCTANGENT			36	
37	2615								37
38	2616	005035	5322	X1,	X			38	
39	2617	005036	5316	PI2,	PI0T			39	
40	2618	005037	4716	CON1,	ONE			40	
41								41	
42								42	
43								43	
44								44	
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1	2620	005040	1045	FLOG,	GETSGN	/FLOATING LOGARITHM	1
2	2621	005041	7450		SNA		2
3	2622	005042	4566		ERROR3	/ZERO ARGUMENT FOR LOG	3
4	2623	005043	7710		SPA CLA		4
5	2624	005044	4451		JMS I MINSKI	/NEGATIVE ARGUMENT	5
6	2625	005045	4407		FINT		6
7	2626	005046	6756		FPUT I TEM		7
8	2627	005047	2637		FSUB I CON1		8
9	2628	005050	0000		FEXT		9
10	2629	005051	1045		GETSGN		10
11	2630	005052	7450		SNA		11
12	2631	005053	5536		RETURN		12
13	2632	005054	7700		SMA CLA		13
14	2633	005055	5264		JMP STARTL		14
15	2634	005056	4407		FINT		15
16	2635	005057	0637		FGET I CON1		16
17	2636	005060	3756		FDIV I TEM		17
18	2637	005061	6756		FPUT I TEM		18
19	2638	005062	0000		FEXT		19
20	2639	005063	7240		CLA CMA		20
21	2640	005064	3033	STARTL,	DCA T3		21
22	2641	005065	1005		TAD P13		22
23	2642	005066	3044		DCA FLAC		23
24	2643	005067	7040		CMA		24
25	2644	005070	1756		TAD I TEM		25
26	2645	005071	3045		DCA FLAC+1		26
27	2646	005072	3046		DCA FLAC+2		27
28	2647	005073	3047		DCA FLAC+3		28
29	2648	005074	7001		IAC		29
30	2649	005075	3756		DCA I TEM		30
31	2650	005076	4407		FINT		31
32	2651	005077	4357		FMUL LOG2		32
33	2652	005100	6635		FPUT I X1		33
34	2653	005101	0756		FGET I TEM		34
35	2654	005102	2637		FSUB I CON1		35
36	2655	005103	6756		FPUT I TEM		36
37	2656	005104	4353		FMUL LOG8		37
38	2657	005105	1350		FADD LOG7		38
39	2658	005106	4756		FMUL I TEM		39
40	2659	005107	1345		FADD LOG6		40
41	2660	005110	4756		FMUL I TEM		41
42	2661	005111	1342		FADD LOG5		42
43	2662	005112	4756		FMUL I TEM		43
44	2663	005113	1337		FADD L4		44
45	2664	005114	4756		FMUL I TEM		45
46	2665	005115	1334		FADD L3		46
47	2666	005116	4756		FMUL I TEM		47
48	2667	005117	1331		FADD L2		48
49	2668	005120	4756		FMUL I TEM		49
50	2669	005121	1326		FADD L1		50
51	2670	005122	4756		FMUL I TEM		51
52	2671	005123	1635		FADD I X1		52
53	2672	005124	0000		FEXT		53
54	2673	005125	5634		JMP I EXIT1		54
55							55
56							56
57							57
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1	2717		/			FLOATING POINT SINE AND COSINE	1
2	2718						2
3	2719	*5200				PAGE	3
4	2720						4
5	2721	005200	4407	FCOS,	FINT	/COS(X)=SIN(PI/2-X)	5
6	2722	005201	6322		FPUT	X	6
7	2723	005202	0316		FGET	PIOT	7
8	2724	005203	2322		FSUB	X	8
9	2725	005204	0000		FEXT		9
10	2726	005205	1045	FSIN,	GETSGN		10
11	2727	005206	7740		SMA SZA	CLA	11
12	2728	005207	5215		JMP	MOD	12
13	2729	005210	1045		GETSGN		13
14	2730	005211	7700		SMA CLA		14
15	2731	005212	5536		RETURN	/YES SIN(0)=0	15
16	2732	005213	4451		JMS I	MINSKI	16
17	2733	005214	7040		CMA	/NO: SIN(-X)=-SIN(X)	17
18	2734	005215	3033	MOD,	DCA	T3	18
19	2735			/		REDUCE X MODULO 2 PI	19
20	2736						20
21	2737						21
22	2738	005216	4407		FINT		22
23	2739	005217	3306		FDIV	TWOPI	23
24	2740	005220	6326		FPUT	XSQR	24
25	2741	005221	0000		FEXT		25
26	2742	005222	4453		JMS I	INTEGER	26
27	2743	005223	4407		FINT		27
28	2744	005224	7000		FNOR		28
29	2745	005225	6322		FPUT	X	29
30	2746	005226	0326		FGET	XSQR	30
31	2747	005227	2322		FSUB	X	31
32	2748	005230	4306		FMUL	TWOPI	32
33	2749	005231	6322		FPUT	X	33
34	2750	005232	2312		FSUB	PI	34
35	2751	005233	0000		FEXT	/X<PI?	35
36	2752	005234	1045		GETSGN		36
37	2753	005235	7710		SPA CLA		37
38	2754	005236	5245		JMP	PCHECK	38
39	2755	005237	4407		FINT	/NO, SIN(X-PI)=-SIN(X)	39
40	2756	005240	6322		FPUT	X	40
41	2757	005241	0000		FEXT		41
42	2758	005242	1033		TAD	T3	42
43	2759	005243	7040		CMA	/INVERT THE SIGN	43
44	2760	005244	3033		DCA	T3	44
45							45
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1	2762	005245	4407	PCHECK,	FINT		/X<PI/2?	1
2	2763	005246	0322		FGET	X		2
3	2764	005247	2316		FSUB	PIOT		3
4	2765	005250	0000		FEXT			4
5	2766	005251	1045		GETSGN			5
6	2767	005252	7710		SPA CLA			6
7	2768	005253	5261		JMP	PALG	/YES	7
8	2769	005254	4407		FINT		/NO	8
9	2770	005255	0312		FGET	PI	/SIN(X)=SIN(PI-X)	9
10	2771	005256	2322		FSUB	X		10
11	2772	005257	6322		FPUT	X		11
12	2773	005260	0000		FEXT			12
13	2774	005261	4407	PALG,	FINT			13
14	2775	005262	0322		FGET	X		14
15	2776	005263	3316		FDIV	PIOT		15
16	2777	005264	6322		FPUT	X		16
17	2778	005265	4322		FMUL	X		17
18	2779	005266	6326		FPUT	XSQR		18
19	2780	005267	0332		FGET	C9		19
20	2781	005270	4326		FMUL	XSQR		20
21	2782	005271	1336		FADD	C7		21
22	2783	005272	4326		FMUL	XSQR		22
23	2784	005273	1342		FADD	C5		23
24	2785	005274	4326		FMUL	XSQR		24
25	2786	005275	1346		FADD	C3		25
26	2787	005276	4326		FMUL	XSQR		26
27	2788	005277	1316		FADD	PIOT		27
28	2789	005300	4322		FMUL	X		28
29	2790	005301	0000		FEXT			29
30	2791	005302	2033	EXIT2,	ISZ	T3		30
31	2792	005303	5536		RETURN			31
32	2793	005304	4451		JMS I	MINSKI		32
33	2794	005305	5536		RETURN			33
34								34
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1	2796		/	CONSTANTS AND POINTERS	1
2	2797				2
3	2798				3
4	2799				4
5	2800	005306	0003	TWOPI,	0003
6	2801	005307	3110		3110
7	2802	005310	3756		3756
8	2803	005311	3235		3235
9	2804				
10	2805	005312	0002	PI,	0002
11	2806	005313	3110		3110
12	2807	005314	3756		3756
13	2808	005315	3235		3235
14	2809				
15	2810	005316	0001	PIOT,	0001
16	2811	005317	3110		3110
17	2812	005320	3756		3756
18	2813	005321	3235		3235
19	2814				
20	2815	005322	0000	X,	0000
21	2816	005323	0000		0000
22	2817	005324	0000		0000
23	2818	005325	0000		0000
24	2819				
25	2820	005326	0000	XSQR,	0000
26	2821	005327	0000		0000
27	2822	005330	0000		0000
28	2823	005331	0000		0000
29	2824				
30	2825		/	SINE CONSTANTS	
31	2826				
32	2827	005332	7764	C9,	7764
33	2828	005333	2401		2401
34	2829	005334	7015		7015
35	2830	005335	1042		1042
36	2831	005336	7771	C7,	7771
37	2832	005337	5464		5464
38	2833	005340	5514		5514
39	2834	005341	6150		6150
40	2835	005342	7775	C5,	7775
41	2836	005343	2431		2431
42	2837	005344	5361		5361
43	2838	005345	4736		4736
44	2839	005346	0000	C3,	0000
45	2840	005347	5325		5325
46	2841	005350	0414		0414
47	2842	005351	3167		3167
48	2843				
49	2844		/	END OF EXTENDED FUNCTIONS.	
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1	2846		/			PAGE 1 - INPUT/OUTPUT ROUTINES FOR THE FOCAL	1
2	2847		/			FLOATING POINT PACKAGE.	2
3	2848						3
4	2849						4
5	2850		/			IN THE COMMENTS BELOW:	5
6	2851		/			F = NUMBER OF DIGITS TO BE OUTPUT =FISW	6
7	2852		/			D = NUMBER OF DECIMAL PLACES =DECP	7
8	2853		/			E = DECIMAL EXPONENT =BEXP	8
9	2854		/			P = NUMBER OF PLACES REMAINING TO BE	9
10	2855		/			PRINTED BEFORE DECIMAL POINT	10
11	2856						11
12	2857	*5400			PAGE		12
13	2858						13
14	2859	0006			DIGITS= 6	/NUMBER OF DECIMAL DIGITS OUT	14
15	2860						15
16	2861	005400	0000	TGO,	0		16
17	2862	005401	3334	DCA	SCOUNT	/SAVE MAX. NUMBER OF DIGITS AVAILABLE - *SET COUNTS*	17
18	2863	005402	1052	TAD	FISW		18
19	2864	005403	4557	RTL6			19
20	2865	005404	0122	AND	P77		20
21	2866	005405	3032	DCA	T1		21
22	2867	005406	1032	TAD	T1		22
23	2868	005407	7041	CIA		/NO, COMPUTE FIELD SIZES	23
24	2869	005410	7450	SNA			24
25	2870	005411	1326	TAD	MD		25
26	2871	005412	3335	DCA	FCOUNT		26
27	2872	005413	1052	TAD	FISW	/(JMP FPRNT) - FOR NO ROUNDING.	27
28	2873	005414	7450	SNA		/FLOATING OUTPUT?	28
29	2874	005415	5241	JMP	R6	/YES, ROUND OFF TO MAX. NO. PLACES	29
30	2875	005416	0122	AND	P77		30
31	2876	005417	3333	DCA	DECP		31
32	2877	005420	1335	TAD	FCOUNT		32
33	2878	005421	1333	TAD	DECP		33
34	2879	005422	7510	SPA		/ F-D > 0?	34
35	2880	005423	5230	JMP	.+5	/YES	35
36	2881	005424	7240	CLA	CMA	/NO,	36
37	2882	005425	1032	TAD	T1		37
38	2883	005426	3333	DCA	DECP	/MAKE D = F-1	38
39	2884	005427	7040	CMA			39
40	2885	005430	1033	TAD	T3	/COMPARE DECIMAL EXPONENT	40
41	2886	005431	7500	SMA		/ F-D > E?	41
42	2887	005432	7200	CLA		/NO, ROUND OFF TO .F PLACES	42
43	2888	005433	1032	TAD	T1	/YES	43
44	2889	005434	7510	SPA		/ D+E < 0?	44
45	2890	005435	5263	JMP	FPRNT-2	/YES, NO ROUNDING NEEDED, GO TO PRINT	45
46	2891	005436	1326	TAD	MD	/NO, ROUND TO D+E PLACES,	46
47	2892	005437	7500	SMA		/TO A MAXIMUM OF D PLACES	47
48	2893	005440	7200	CLA			48
49							49
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1	2895	005441	1327	R6,	TAD	RND2	/ *ROUND UP *	1
2	2896	005442	3071		DCA	T2	/SAVE NUMBER+1 OF PLACES TO ROUND TO.	2
3	2897	005443	1731		TAD I	BUFST		3
4	2898	005444	1071		TAD	T2	/SET UP BUFFER ADDRESS AT WHICH	4
5	2899	005445	3336		DCA	PLCE	/ROUNDING OFF SHOULD START	5
6	2900	005446	1071		TAD	T2		6
7	2901	005447	7041		CIA		/SET UP COUNT OF MAXIMUM NUMBER	7
8	2902	005450	3071		DCA	T2	/OF CARRIES ALLOWABLE	8
9	2903	005451	1325		TAD	K5	/LITTLE EXTRA ON FIRST DIGIT.	9
10	2904	005452	2736	RET,	ISZ I	PLCE	/ADD 1 TO DIGIT AT CURRENT POSITION	10
11	2905	005453	1736		TAD I	PLCE		11
12	2906	005454	1330		TAD	OM12		12
13	2907	005455	7710		SPA CLA		/CARRY REQUIRED?	13
14	2908	005456	5265		JMP	FPRNT	/NO, GO TO OUTPUT	14
15	2909	005457	3736		DCA I	PLCE	/YES, MAKE CURRENT DIGIT ZERO	15
16	2910	005460	2071		ISZ	T2	/BEGINNING OF BUFFER REACHED?	16
17	2911	005461	5321		JMP	DECR	/NO, DECREMENT BUFFER ADDRESS AND REPEAT	17
18	2912	005462	2736		ISZ I	PLCE	/YES, SET MANTISSA TO 0.1	18
19	2913	005463	2033		ISZ	T3	/COMPENSATE BY INCREMENTING EXPONENT	19
20	2914	005464	7200		CLA			20
21	2915	005465	1052	FPRNT,	TAD	FISW	/AUTO-INDEX REGISTER ALREADY SET. - *PRINT*	21
22	2916	005466	7650		SNA CLA		/ F = 0?	22
23	2917	005467	5356		JMP	FLOUT	/YES, OUTPUT AS FLOATING NUMBER	23
24	2918	005470	1335		TAD	FCOUNT		24
25	2919	005471	1033		TAD	T3		25
26	2920	005472	7540		SMA SZA		/ E > F?	26
27	2921	005473	5355		JMP	FLOUT-1	/YES, CONVERT TO E FORMAT	27
28	2922	005474	1333		TAD	DECP		28
29	2923	005475	7500		SMA		/ E < F-D?	29
30	2924	005476	7200		CLA		/NO, TAKE P = E	30
31	2925	005477	7041		CIA		/YES, TAKE P = F-D	31
32	2926	005500	1033		TAD	T3		32
33	2927	005501	7041		CIA			33
34	2928	005502	3032		DCA	T1	/SET UP MINUS P	34
35	2929	005503	1033	BACK,	TAD	T3	/PRINT DD.DDD	35
36	2930	005504	1032		TAD	T1		36
37	2931	005505	7650		SNA CLA		/ P = E?	37
38	2932	005506	5343		JMP	DIG	/YES, PRINT DIGIT	38
39	2933	005507	1032		TAD	T1	/NO,	39
40	2934	005510	7001		IAC			40
41	2935	005511	7710		SPA CLA		/ P > 1?	41
42	2936	005512	1105		TAD	M20	/YES, TAKE SPACE (240-260); OTHERWISE ZERO	42
43	2937	005513	4336	IN,	JMS	OUTA	/PRINT CHARACTER	43
44	2938	005514	2032		ISZ	T1	/P CHARACTERS PRINTED?	44
45	2939	005515	5303		JMP	BACK	/NO	45
46	2940	005516	1102		TAD	PER	/YES,	46
47	2941	005517	4551		PRINTC		/PRINT DECIMAL POINT	47
48	2942	005520	5303		JMP	BACK		48
49	2943							49
50	2944	005521	7040	DECR,	CMA		/BACKUP TO TOP OF BUFFER.	50
51	2945	005522	1336		TAD	PLCE		51
52	2946	005523	3336		DCA	PLCE		52
53	2947	005524	5252		JMP	RET		53
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1	3001		/		DOUBLE PRECISION DECIMAL-BINARY	1
2	3002		/		INPUT AND CONVERSION FOR + OR - XXX...	2
3	3003					3
4	3004	*5600			PAGE	4
5	3005					5
6	3006	005600	0000	DECONV, 0		6
7	3007	005601	3046	DCA	LORD	7
8	3008	005602	3044	DCA	EXP	8
9	3009	005603	3045	DCA	HORD	9
10	3010	005604	3047	DCA	OVER2	10
11	3011	005605	3314	DCA	DNUMBR	11
12	3012	005606	3050	DCA	SIGNF	12
13	3013	005607	1066	TAD	CHAR	13
14	3014	005610	1264	TAD	MPLUS	14
15	3015	005611	7450	SNA		15
16	3016	005612	5220	JMP	.+6	16
17	3017	005613	1111	TAD	M2	17
18	3018	005614	7640	SZA	CLA	18
19	3019	005615	5221	JMP	.+4	19
20	3020	005616	7040	CMA		20
21	3021	005617	3050	DCA	SIGNF	21
22	3022	005620	4666	JMS I	XINPUT	22
23	3023	005621	1066	TAD	CHAR	23
24	3024	005622	1265	TAD	MSPACE	24
25	3025	005623	7650	SNA	CLA	25
26	3026	005624	5220	JMP	. -4	26
27	3027	005625	4227	JMS	DECON	27
28	3028	005626	5600	JMP I	DECONV	28
29						29
30						30
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1	3065	005667	0000	MULT10, 0			/ROUTINE TO MULTIPLY FLAC BY TEN (10)	1
2	3066	005670	1047	TAD	OVER2			2
3	3067	005671	3043	DCA	OVER1			3
4	3068	005672	1046	TAD	LORD		/DOUBLE PRECISION WORD	4
5	3069	005673	3042	DCA	AC1L		/BY TEN (DECIMAL)	5
6	3070	005674	1045	TAD	HORD		/REMAIN=REMAINDER	6
7	3071	005675	3041	DCA	AC1H			7
8	3072	005676	3312	DCA	REMAIN		/CLEAR OVERFLOW WORD	8
9	3073	005677	4315	JMS	MULT2		/CALL SUBROUTINE TO	9
10	3074	005700	4315	JMS	MULT2		/MULTIPLY BY TWO	10
11	3075	005701	4333	JMS	DUBLAD		/CALL DOUBLE ADD	11
12	3076	005702	4315	JMS	MULT2			12
13	3077	005703	1313	TAD	DIGIT		/ADD LAST DIGIT RECEIVED	13
14	3078	005704	3043	DCA	OVER1			14
15	3079	005705	3042	DCA	AC1L			15
16	3080	005706	3041	DCA	AC1H			16
17	3081	005707	4333	JMS	DUBLAD			17
18	3082	005710	1312	TAD	REMAIN		/EXIT WITH REMAINDER	18
19	3083	005711	5667	JMP I	MULT10		/IN AC	19
20	3084							20
21	3085	005712	0000	REMAIN, 0				21
22	3086	005713	0000	DIGIT, 0			/STORAGE FOR DIGIT	22
23	3087	005714	0000	DNUMBR, 0			/=NUMBER OF DIGITS	23
24	3088							24
25	3089	005715	0000	MULT2, 0			/MULTIPLY OVER2, LORD, HORD BY 2	25
26	3090	005716	1047	TAD	OVER2			26
27	3091	005717	7104	CLL RAL			/CARRY INSERT BIT IS IN LINK	27
28	3092	005720	3047	DCA	OVER2			28
29	3093	005721	1046	TAD	LORD			29
30	3094	005722	7004	RAL				30
31	3095	005723	3046	DCA	LORD			31
32	3096	005724	1045	TAD	HORD			32
33	3097	005725	7004	RAL				33
34	3098	005726	3045	DCA	HORD			34
35	3099	005727	1312	TAD	REMAIN			35
36	3100	005730	7004	RAL				36
37	3101	005731	3312	DCA	REMAIN			37
38	3102	005732	5715	JMP I	MULT2			38
39								39
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1	3104	005733	0000	DUBLAD, 0			/TRIPLE PRECISION ADDITION	1
2	3105	005734	7300	CLA CLL				2
3	3106	005735	1047	TAD	OVER2			3
4	3107	005736	1043	TAD	OVER1			4
5	3108	005737	3047	DCA	OVER2			5
6	3109	005740	7004	RAL				6
7	3110	005741	1046	TAD	LORD			7
8	3111	005742	1042	TAD	AC1L			8
9	3112	005743	3046	DCA	LORD			9
10	3113	005744	7004	RAL				10
11	3114	005745	1045	TAD	HORD			11
12	3115	005746	1041	TAD	AC1H			12
13	3116	005747	3045	DCA	HORD			13
14	3117	005750	7004	RAL				14
15	3118	005751	1312	TAD	REMAIN		/WITH OVERFLOW	15
16	3119	005752	3312	DCA	REMAIN			16
17	3120	005753	5733	JMP I	DUBLAD			17
18	3121							18
19	3122	005754	0000	DIV1, 0			/SHIFT OPERAND RIGHT	19
20	3123	005755	7300	CLA CLL			/TRIPLE PRECISION	20
21	3124	005756	1041	TAD	AC1H			21
22	3125	005757	7510	SPA				22
23	3126	005760	7120	CLL CML				23
24	3127	005761	7010	RAR				24
25	3128	005762	3041	DCA	AC1H			25
26	3129	005763	1042	TAD	AC1L			26
27	3130	005764	7010	RAR				27
28	3131	005765	3042	DCA	AC1L			28
29	3132	005766	1043	TAD	OVER1			29
30	3133	005767	7010	RAR				30
31	3134	005770	3043	DCA	OVER1			31
32	3135	005771	2040	ISZ	EX1			32
33	3136	005772	5754	JMP I	DIV1			33
34	3137	005773	5754	JMP I	DIV1			34
35								35
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1	3139		*6000	PAGE			1
2	3140						2
3	3141		/	FLOATING OUTPUT CONVERSION ROUTINE			3
4	3142						4
5	3143	006000	0000	FLOUTP,	0		5
6	3144	006001	1335	TAD	PEQ		6
7	3145	006002	4551	PRINTC		/(CLA)- TO SUPPRESS "="	7
8	3146	006003	1045	TAD	HORD	/NUMBER>0??	8
9	3147	006004	7700	SMA	CLA		9
10	3148	006005	1334	TAD	SMSP	/PRINT "-" OR A SPACE.	10
11	3149	006006	1336	TAD	SMIN		11
12	3150	006007	4551	PRINTC			12
13	3151	006010	4753	JMS I	ABSOL2		13
14	3152	006011	3033	DCA	T3	/INITIALIZE DECIMAL EXPONENT	14
15	3153	006012	1044	TAD	EXP	/IS EXP 0 TO 4?	15
16	3154	006013	7510	SPA			16
17	3155	006014	5227	JMP	FG03	/TOO LARGE: MULTIPLY BY 1/10	17
18	3156	006015	7440	SZA			18
19	3157	006016	1341	TAD	M4		19
20	3158	006017	7750	SPA SNA	CLA		20
21	3159	006020	5234	JMP	FG04		21
22	3160	006021	4407	FINT			22
23	3161	006022	4744	FMUL I	PPTEN		23
24	3162	006023	0000	FEXT			24
25	3163	006024	7001	IAC			25
26	3164	006025	1033	TAD	T3		26
27	3165	006026	5211	JMP	FG02		27
28	3166						28
29	3167	006027	4407	FG03,	FINT		29
30	3168	006030	4752	FMUL I	TENPT		30
31	3169	006031	0000	FEXT			31
32	3170	006032	7040	CMA			32
33	3171	006033	5225	JMP	. -6		33
34							34
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1	3173	006034	3745	FG04,	DCA I	DPT	/MULTIPLY BY TWO TO POSITION BIT0	1
2	3174	006035	3746		DCA I	REPT	/CLEAR OVERFLOW WORD	2
3	3175	006036	1350		TAD	SADR	/INIT BUFFER POINTER	3
4	3176	006037	3014		DCA	FLT XR		4
5	3177	006040	1044		TAD	EXP	/COMPUTE BITS IN 1ST DIGIT	5
6	3178	006041	7140		CLL CMA			6
7	3179	006042	3354		DCA	OUTDG	/TEMP COUNT	7
8	3180	006043	1343		TAD	DCOUNT	/SETUP COUNT OF TOTAL OUTPUT	8
9	3181	006044	3044		DCA	EXP		9
10	3182	006045	4527		JMS I	DOUBLE	/ROTATE OUT THE 1ST 4 BITS	10
11	3183	006046	2354		ISZ	OUTDG		11
12	3184	006047	5245		JMP	.-2		12
13	3185	006050	1746		TAD I	REPT	/TEST FOR 10-15, 0, 1-9	13
14	3186	006051	7450		SNA			14
15	3187	006052	5270		JMP	FG05	/IGNORE 1ST ZERO	15
16	3188	006053	1342		TAD	FM12		16
17	3189	006054	7710		SPA CLA			17
18	3190	006055	5264		JMP	.+7	/0-9	18
19	3191	006056	7001		IAC			19
20	3192	006057	3414		DCA I	FLT XR	/OUTPUT A 1	20
21	3193	006060	2044		ISZ	EXP	/COUNT THE DIGIT	21
22	3194	006061	1342		TAD	FM12	/CORRECT REMAINDER	22
23	3195	006062	2033		ISZ	T3	/BUMP DECIMAL EXPONENT	23
24	3196	006063	7000		NOP			24
25	3197	006064	1746		TAD I	REPT	/COMPUTE RESULTANT OR SECOND DIGIT	25
26	3198	006065	2033		ISZ	T3		26
27	3199	006066	7000		NOP			27
28	3200	006067	7410		SKP			28
29	3201	006070	4747	FG05,	JMS I	M10PT	/IE. .672X10=6+.72.. ETC	29
30	3202	006071	3414		DCA I	FLT XR		30
31	3203	006072	2044		ISZ	EXP	/ALL DIGITS OUTPUT??	31
32	3204	006073	5270		JMP	.-3	/NO: CONTINUE	32
33	3205	006074	1350		TAD	SADR	/INIT BUFFER POINTER	33
34	3206	006075	3014		DCA	FLT XR		34
35	3207	006076	1343		TAD	DCOUNT		35
36	3208	006077	4751		JMS I	ROUND	/OUTPUT MANTISSA	36
37	3209	006100	5600		JMP I	FLOUTP	/FIXED POINT DONE	37
38	3210	006101	1333		TAD	CHRT	/PRINT "E"	38
39	3211	006102	4551		PRINTC			39
40								40
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1	3213			/	OUTPUT THE EXPONENT		1
2	3214						2
3	3215	006103	1033	TAD	T3	/TAKE ABSOLUTE VALUE OF EXPONENT	3
4	3216	006104	7510	SPA			4
5	3217	006105	7041	CIA			5
6	3218	006106	3045	DCA	HORD	/SAVE + POWER	6
7	3219	006107	1033	TAD	T3	/PRINT SIGN	7
8	3220	006110	7700	SMA	CLA		8
9	3221	006111	1111	TAD	M2		9
10	3222	006112	1336	TAD	SMIN		10
11	3223	006113	4551	PRINTC			11
12	3224	006114	1045	TAD	HORD		12
13	3225	006115	2044	ISZ	EXP		13
14	3226	006116	1337	TAD	M144		14
15	3227	006117	7500	SMA			15
16	3228	006120	5315	JMP	.-3		16
17	3229	006121	1340	TAD	C144		17
18	3230	006122	3045	DCA	HORD	/SAVE TENS AND UNITS	18
19	3231	006123	7040	CMA		/OUTPUT HUNDREDS	19
20	3232	006124	1044	TAD	EXP		20
21	3233	006125	7440	SZA		/UNLESS ZERO	21
22	3234	006126	4354	JMS	OUTDG		22
23	3235	006127	1045	TAD	HORD	/PRINT TWO DIGITS	23
24	3236	006130	4732	JMS I	PRNTI		24
25	3237	006131	5600	JMP I	FLOUTP		25
26	3238						26
27	3239	006132	2442	PRNTI,	PRNT		27
28	3240	006133	0305	CHRT,	"E	/E (0246) - FOR AMPERSAND	28
29	3241	006134	7763	SMSP,	" - "-	/	29
30	3242	006135	0275	PEQ,	"=		30
31	3243	006136	0255	SMIN,	"-		31
32	3244	006137	7634	M144,	-144	/-100	32
33	3245	006140	0144	C144,	0144	/+100	33
34	3246	006141	7774	M4,	-4		34
35	3247	006142	7766	FM12,	-12		35
36	3248	006143	7771	DCOUNT,	-DIGITS-1	/NUMBER OF DIGITS OUTPUT	36
37	3249	006144	6275	PPTEN,	PTEN	/IEI	37
38	3250	006145	5713	DPT,	DIGIT		38
39	3251	006146	5712	REPT,	REMAIN	/OVERFLOW FROM INTEGER MULTIPLY	39
40	3252	006147	5667	M10PT,	MULT10		40
41	3253	006150	7467	SADR,	BUFFER-1		41
42	3254	006151	5400	ROUND,	TGO	/ACTUAL OUTPUT ROUTINE	42
43	3255	006152	6271	TENPT,	TEN		43
44	3256	006153	5571	ABSOL2,	ABSOLV		44
45	3257						45
46	3258	006154	0000	OUTDG,	0	/OUTPUT ONE DIGIT	46
47	3259	006155	1113	TAD	C260		47
48	3260	006156	4551	PRINTC			48
49	3261	006157	5754	JMP I	OUTDG		49
50	3262						50
51	3263			/	USED BY 8K		51
52							52
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1	3265		/	FLOATING POINT INPUT		1
2	3266					2
3	3267	*6200		PAGE		3
4	3268					4
5	3269	006200	0000	FLINTP, 0	/IF C(AC) = 0, USE CHAR	5
6	3270	006201	7640	SZA CLA	/IF C(AC) NON-ZERO, GET NEXT	6
7	3271	006202	4706	JMS I XIN	/GET FIRST CHAR	7
8	3272	006203	1066	TAD CHAR	/IGNORE LEADING SPACES	8
9	3273	006204	1114	TAD M240		9
10	3274	006205	7650	SNA CLA		10
11	3275	006206	5202	JMP .-4		11
12	3276	006207	4702	JMS I DPCVPT	/READ FIRST DIGIT GROUP	12
13	3277	006210	1066	TAD CHAR	/AND SET "SIGNF"	13
14	3278	006211	1115	TAD MPER		14
15	3279	006212	7640	SZA CLA	/ENDED BY PERIOD?	15
16	3280	006213	5221	JMP FIG01		16
17	3281	006214	4706	JMS I XIN	/YES, READ 2AND GROUP	17
18	3282	006215	3705	DCA I DPN		18
19	3283	006216	4703	JMS I DCONP		19
20	3284	006217	1705	TAD I DPN	/SAVE NUMBER OF DIGITS IN T3	20
21	3285	006220	7041	CMA IAC		21
22	3286	006221	3033	FIG01, DCA T3	/NO,	22
23	3287	006222	1310	TAD P43		23
24	3288	006223	3044	DCA EXP		24
25	3289	006224	4704	JMS I RESOL5		25
26	3290	006225	4707	JMS I INORM	/NORMALIZE FIRST, THEN	26
27	3291	006226	4407	FINT		27
28	3292	006227	6430	FPUT I PT1	/SAVE NUMBER	28
29	3293	006230	0000	FEXT		29
30	3294	006231	1066	TAD CHAR		30
31	3295	006232	1301	TAD MINUSE		31
32	3296	006233	7640	SZA CLA	/"E" READ IN?	32
33	3297	006234	5246	JMP ENDFI+3	/NO	33
34	3298	006235	4706	JMS I XIN	/YES, READ 3RD DIGIT GROUP	34
35	3299	006236	4702	JMS I DPCVPT	/I.E. CONVERT DECIMAL EXPONENT	35
36	3300	006237	4704	JMS I RESOL5		36
37	3301	006240	1047	TAD OVER2		37
38	3302	006241	1033	TAD T3	/C(SEXP) PLACES TO RIGHT	38
39	3303	006242	3033	DCA T3	/OF LAST DIGIT	39
40						40
41						41
42						42
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1	3305		/	COMPENSATE FOR DECIMAL EXPONENTS	1
2	3306				2
3	3307	006243	4407	ENDFI, FINT /RESTORE MANTISSA	3
4	3308	006244	0430	FGET I PT1	4
5	3309	006245	0000	FEXT	5
6	3310	006246	1033	TAD T3 /TEST DECIMAL EXPONENT	6
7	3311	006247	7450	SNA	7
8	3312	006250	5600	JMP I FLINTP /FINISHED	8
9	3313	006251	7700	SMA CLA	9
10	3314	006252	5261	JMP FIG04	10
11	3315	006253	4407	FINT /. IS TO THE LEFT:	11
12	3316	006254	4275	FMUL PTEN /TIMES .1000	12
13	3317	006255	6430	FPUT I PT1	13
14	3318	006256	0000	FEXT	14
15	3319	006257	7001	IAC	15
16	3320	006260	5266	JMP .+6	16
17	3321				17
18	3322	006261	4407	FIG04, FINT /. IS TO THE RIGHT:	18
19	3323	006262	4271	FMUL TEN /MULTIPLY BY 10	19
20	3324	006263	6430	FPUT I PT1	20
21	3325	006264	0000	FEXT	21
22	3326	006265	7040	CMA	22
23	3327	006266	1033	TAD T3	23
24	3328	006267	3033	DCA T3	24
25	3329	006270	5246	JMP ENDFI+3	25
26	3330				26
27	3331	006271	0004	TEN, 0004	27
28	3332	006272	2400	2400	28
29	3333	006273	0000	0000	29
30	3334	006274	0000	0000	30
31	3335				31
32	3336	006275	7775	PTEN, 7775	32
33	3337	006276	3146	3146	33
34	3338	006277	3147	3147 / (3146) - FOR 4-WORD	34
35	3339	006300	3150	3150	35
36	3340				36
37	3341	006301	7473	MINUSE, -"E / (7532) - FOR AMPERSAND	37
38	3342				38
39	3343	006302	5600	DPCVPT, DECONV	39
40	3344	006303	5627	DCONP, DECON	40
41	3345	006304	7173	RESOL5, RESOLV	41
42	3346	006305	5714	DPN, DNUMBR	42
43	3347	006306	0756	XIN, INPUT	43
44	3348	006307	7335	INORM, DNORM	44
45	3349	006310	0043	P43, 43	45
46	3350				46
47	3351		/	END OF FLOATING POINT INPUT	47
48	3352				48
49	3353		/	7 FREE	49
50	3354				50
51	3355		/	USED BY H.S. READER	51
52					52
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1	3357	*6400		PAGE					1
2	3358								2
3	3359	/		FLOATING-POINT INTERPRETER FOR FOCAL.					3
4	3360								4
5	3361	006400	0000	FPNT,	0				5
6	3362	006401	7300		CLA CLL				6
7	3363	006402	3047		DCA	OVER2	/(NOP) - FOR 4-WORD		7
8	3364	006403	3043		DCA	OVER1	/(NOP) - FOR 4-WORD.		8
9	3365	006404	1600		TAD I	FPNT	/GET NEXT INSTRUCTION		9
10	3366	006405	7450		SNA				10
11	3367	006406	5600		JMP I	FPNT	/FAST EXIT		11
12	3368	006407	3262		DCA	JUMP			12
13	3369	006410	1262		TAD	JUMP			13
14	3370	006411	0123		AND	C200	/GET PAGE BIT		14
15	3371	006412	7650		SNA CLA		/PAGE ZERO?		15
16	3372	006413	5216		JMP	.+3	/YES		16
17	3373	006414	1104		TAD	P7600	/NO		17
18	3374	006415	0200		AND	FPNT	/C(FPNT)0-4 CONTAINS PAGE BITS		18
19	3375	006416	3040		DCA	ADDR			19
20	3376	006417	1106		TAD	P177	/GET 7 BIT ADDRESS		20
21	3377	006420	0262		AND	JUMP			21
22	3378	006421	1040		TAD	ADDR			22
23	3379	006422	3040		DCA	ADDR			23
24	3380	006423	1263		TAD	INDRCT	/INDIRECT BIT=1?		24
25	3381	006424	0262		AND	JUMP			25
26	3382	006425	7650		SNA CLA				26
27	3383	006426	5231		JMP	LOOP01	/NO-GO ON		27
28	3384	006427	1440		TAD I	ADDR	/YES, DEFER, W/O AUTO-INDEX		28
29	3385	006430	3040		DCA	ADDR			29
30	3386	006431	2200	LOOP01,	ISZ	FPNT			30
31	3387	006432	7040		CMA				31
32	3388	006433	1040		TAD	ADDR			32
33	3389	006434	3015		DCA	FLT XR2			33
34	3390	006435	1262		TAD	JUMP	/GET COMMAND		34
35	3391	006436	7106		CLL RTL				35
36	3392	006437	7006		RTL				36
37	3393	006440	0107		AND	P17	/GET BITS 0-2, IE OPCODE		37
38	3394	006441	7450		SNA				38
39	3395	006442	5267		JMP	FLGT			39
40	3396	006443	1264		TAD	TABLE	/LOOKUP IN TABLE		40
41	3397	006444	3262		DCA	JUMP			41
42	3398	006445	1662		TAD I	JUMP			42
43	3399	006446	7450		SNA				43
44	3400	006447	5265		JMP	FLPT			44
45	3401	006450	3262		DCA	JUMP			45
46	3402	006451	1304		TAD	CEX1	/SAVE FLOATING ARGUMENT, UNLESS 'GET' OR 'PUT'		46
47	3403	006452	3014		DCA	FLT XR			47
48	3404	006453	1117		TAD	MFLT			48
49	3405	006454	3057		DCA	CNTR			49
50	3406	006455	1415		TAD I	FLT XR2			50
51	3407	006456	3414		DCA I	FLT XR			51
52	3408	006457	2057		ISZ	CNTR			52
53	3409	006460	5255		JMP	.-3			53
54	3410	006461	5662		JMP I	JUMP	/GO THERE		54
55									55
56									56
57									57
58									58

1	3448		/	INTERPRETIVE POWER				1
2	3449							2
3	3450	006515	1045	FLEX,	TAD	HORD	/ZERO?	3
4	3451	006516	7640		SZA CLA			4
5	3452	006517	5325		JMP	.+6		5
6	3453	006520	3044	ZERO,	DCA	EXP	/YES	6
7	3454	006521	3045		DCA	HORD		7
8	3455	006522	3046		DCA	LORD		8
9	3456	006523	3047		DCA	OVER2		9
10	3457	006524	5201		JMP	FPNT+1		10
11	3458							11
12	3459	006525	4543		PUSHF		/AC TO A + POWER	12
13	3460	006526	0044		FLAC			13
14	3461	006527	4543		PUSHF		/SETUP ARGUMENT (THE EXPONENT)	14
15	3462	006530	0040		EX1			15
16	3463	006531	4544		POPF			16
17	3464	006532	0044		FLAC			17
18	3465	006533	4453		JMS I	INTEGER	/ONLY POSITIVE, INTEGER EXPONENTS	18
19	3466	006534	7510		SPA			19
20	3467	006535	5342		JMP	.+5	/(COULD DIVIDE)	20
21	3468	006536	7040		CMA			21
22	3469	006537	3262		DCA	JUMP	/TEMP STORAGE	22
23	3470	006540	3043		DCA	OVER1	/(NOP) - FOR 4-WORD	23
24	3471	006541	1045		TAD	HORD		24
25	3472	006542	7640		SZA CLA			25
26	3473	006543	4566		ERROR2		/TOO LARGE OR NEGATIVE EXPONENT	26
27	3474	006544	4543		PUSHF		/INITIALIZE TO ONE.	27
28	3475	006545	2405		FLTONE			28
29	3476	006546	4544		POPF			29
30	3477	006547	0044		FLAC			30
31	3478	006550	4544		POPF			31
32	3479	006551	7470		ITER1			32
33	3480	006552	5360		JMP	.+6		33
34	3481	006553	4543		PUSHF			34
35	3482	006554	7470		ITER1			35
36	3483	006555	4544		POPF			36
37	3484	006556	0040		EX1			37
38	3485	006557	4766		JMS I	MULT	/ "MULT"	38
39	3486	006560	2262		ISZ	JUMP		39
40	3487	006561	5353		JMP	. - 6		40
41	3488	006562	5201		JMP	FPNT+1		41
42								42
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1	3490	006563	4766	FLMY,	JMS I	MULT	/MULTIPLY	1
2	3491	006564	5201		JMP	FPNT+1		2
3	3492							3
4	3493							4
5	3494	006565	7153	OPMINS,	MINUS2			5
6	3495	006566	7004	MULT,	DMULT			6
7	3496	006567	7335	NORM,	DNORM			7
8	3497	006570	6623	ALGN,	ALIGN			8
9	3498	006571	5754	RAR1,	DIV1			9
10	3499	006572	6757	RAR2,	DIV2			10
11	3500	006573	5733	TRAD,	DUBLAD			11
12	3501							12
13	3502		6573		ITABLE=	.-1		13
14	3503							14
15	3504	006574	6506		FLAD			15
16	3505	006575	6505		FLSU			16
17	3506	006576	7107		FLDV			17
18	3507	006577	6563		FLMY			18
19	3508	006600	6515		FLEX			19
20	3509	006601	0000		0000			20
21	3510	006602	6513		NORF			21
22	3511							22
23	3512	006603	0000	ACMINS,	0		/ROUTINE TO COMPLEMENT FLAC - VIA "MINSKI"	23
24	3513	006604	7300		CLA CLL			24
25	3514	006605	1047		TAD	OVER2	/TRIPLE PRECISION NEGATION	25
26	3515	006606	7041		CMA IAC		/OF FLOATING AC	26
27	3516	006607	3047		DCA	OVER2		27
28	3517	006610	1046		TAD	LORD		28
29	3518	006611	7040		CMA			29
30	3519	006612	7430		SZL			30
31	3520	006613	7101		CLL IAC			31
32	3521	006614	3046		DCA	LORD		32
33	3522	006615	1045		TAD	HORD		33
34	3523	006616	7040		CMA			34
35	3524	006617	7430		SZL			35
36	3525	006620	7101		CLL IAC			36
37	3526	006621	3045		DCA	HORD		37
38	3527	006622	5603		JMP I	ACMINS		38
39								39
40								40
41								41
42								42
43								43
44								44
45								45
46								46
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57								57
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1	3529	006623	0000	ALIGN, 0		/SUBROUTINE TO ALIGN	1
2	3530	006624	1045	TAD	HORD	/BINARY POINTS	2
3	3531	006625	7450	SNA			3
4	3532	006626	1046	TAD	LORD	/IS MANTISSA ZERO?	4
5	3533	006627	7650	SNA	CLA		5
6	3534	006630	5311	JMP	NOX1	/YES, RESULT=OPERAND	6
7	3535	006631	1041	TAD	AC1H	/NO, IS OPERAND ZERO?	7
8	3536	006632	7450	SNA			8
9	3537	006633	1042	TAD	AC1L		9
10	3538	006634	7450	SNA			10
11	3539	006635	1043	TAD	OVER1		11
12	3540	006636	7650	SNA	CLA		12
13	3541	006637	5623	JMP	I ALIGN	/YES, EXIT.	13
14	3542	006640	1040	TAD	EX1		14
15	3543	006641	7041	CMA	IAC		15
16	3544	006642	1044	TAD	EXP		16
17	3545	006643	7450	SNA		/ARE EXPONENTS EQUAL?	17
18	3546	006644	5273	JMP	ADONE	/YES	18
19	3547	006645	3203	DCA	ACMINS		19
20	3548	006646	1203	TAD	ACMINS		20
21	3549	006647	7500	SMA		/NO	21
22	3550	006650	7041	CIA		/NEGATE AND	22
23	3551	006651	3322	DCA	AMOUNT	/SAVE THE DIFFERENCE	23
24	3552	006652	1322	TAD	AMOUNT		24
25	3553	006653	1336	TAD	TEST2		25
26	3554	006654	7710	SPA	CLA	/CAN THE EXPONENTS BE ALIGNED?	26
27	3555	006655	5275	JMP	NOX	/NO, USE LARGER OF THE TWO.	27
28	3556	006656	1203	TAD	ACMINS	/YES, SHIFT THE SMALLER	28
29	3557	006657	7700	SMA	CLA		29
30	3558	006660	5265	JMP	ASHFT		30
31	3559	006661	4357	JMS	DIV2		31
32	3560	006662	2322	ISZ	AMOUNT		32
33	3561	006663	5261	JMP	. -2		33
34	3562	006664	5273	JMP	ADONE		34
35							35
36							36
37							37
38							38
39							39
40							40
41							41
42							42
43							43
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[illegible]

1	3598		/				LEAVE 12 BIT ANSWER IN AC UPON RETURN	1
2	3599		/				LEAVE FLAC AS AN INTEGER,	2
3	3600							3
4	3601	006724	0000	FIX,	0		/VIA (INTEGER)	4
5	3602	006725	4751		JMS I	ABSOL		5
6	3603	006726	1044		TAD	EXP	/TEST FOR FRACTION	6
7	3604	006727	7750		SPA SNA	CLA		7
8	3605	006730	5353		JMP	FIXM	/DOUBLE CHECK FOR MINUS ONE.	8
9	3606	006731	7001		IAC			9
10	3607	006732	3043		DCA	OVER1		10
11	3608	006733	1350		TAD	P27	/INIT ALIGNMENT	11
12	3609	006734	3040		DCA	EX1		12
13	3610	006735	4223		JMS	ALIGN	/DO THE ALIGNMENT TO AN INTEGER	13
14	3611	006736	0027	TEST2,	0027		/ALREADY DONE; (43)-FOR 4-WORD	14
15	3612	006737	2047		ISZ	OVER2		15
16	3613	006740	5344		JMP	.+4		16
17	3614	006741	2046		ISZ	LORD		17
18	3615	006742	7410		SKP			18
19	3616	006743	2045		ISZ	HORD		19
20	3617	006744	3047		DCA	OVER2	/CLEAR THE FRACTION	20
21	3618	006745	4752		JMS I	RESOL		21
22	3619	006746	1046		TAD	LORD	/EXIT WITH LOW ORDER RESULT IN AC.	22
23	3620	006747	5724		JMP I	FIX		23
24	3621							24
25	3622	006750	0027	P27,	27			25
26	3623	006751	5571	ABSOL,	ABSOLV			26
27	3624	006752	7173	RESOL,	RESOLV			27
28	3625							28
29	3626	006753	3044	FIXM,	DCA	EXP	/CLEAR EXPONENT	29
30	3627	006754	3045		DCA	HORD		30
31	3628	006755	3046		DCA	LORD		31
32	3629	006756	5344		JMP	TEST2+6		32
33	3630							33
34	3631	006757	0000	DIV2,	0		/SHIFT FLAC RIGHT	34
35	3632	006760	7300		CLA CLL			35
36	3633	006761	1045		TAD	HORD		36
37	3634	006762	7510		SPA			37
38	3635	006763	7020		CML			38
39	3636	006764	7010		RAR			39
40	3637	006765	3045		DCA	HORD		40
41	3638	006766	1046		TAD	LORD		41
42	3639	006767	7010		RAR			42
43	3640	006770	3046		DCA	LORD		43
44	3641	006771	1047		TAD	OVER2		44
45	3642	006772	7010		RAR			45
46	3643	006773	3047		DCA	OVER2		46
47	3644	006774	2044		ISZ	EXP		47
48	3645	006775	5757		JMP I	DIV2		48
49	3646	006776	5757		JMP I	DIV2		49
50								50
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1	3648	6777	SPECIAL=.	/INPUT CHARACTERS	1
2	3649				2
3	3650	006777 0337	337	/LEFT ARROW	3
4	3651	007000 0377	377	/RUBOUT	4
5	3652	007001 0212	"J&237	/L.F.	5
6	3653	007002 0375	375	/ALT MODE	6
7	3654	007003 7777	-1		7
8	3655				8
9	3656	/	(A+B+C)*(D+E+F)=A*D, A*E, B*D, B*E		9
10	3657				10
11	3658	007004 0000	DMULT, 0	/N- PRECISION MULTIPLY WITH	11
12	3659	007005 7001	IAC	/PRODUCT IN TRIPLE PRECISION	12
13	3660	007006 1040	TAD	EX1	13
14	3661	007007 4324	JMS	SIGN	14
15	3662	007010 7710	SPA CLA		15
16	3663	007011 4353	JMS	MINUS2	16
17	3664	007012 3301	DCA	DATUM-1	17
18	3665	007013 3300	DCA	DATUM-2	18
19	3666	007014 3277	DCA	DATUM-3	19
20	3667	007015 3276	DCA	DATUM-4	20
21	3668	007016 1045	TAD	A	21
22	3669	007017 3751	SAVE		22
23	3670	007020 1041	TAD	D	23
24	3671	007021 4752	MULTY		24
25	3672	007022 0002	2	/ACCUMULATE STARTING IN #2 DATA WORD	25
26	3673	007023 1042	TAD	E	26
27	3674	007024 4752	MULTY		27
28	3675	007025 0003	3		28
29	3676	007026 1046	TAD	B	29
30	3677	007027 3751	SAVE		30
31	3678	007030 1041	TAD	D	31
32	3679	007031 4752	MULTY		32
33	3680	007032 0003	3		33
34	3681	007033 1042	TAD	E	34
35	3682	007034 4752	MULTY		35
36	3683	007035 0004	4		36
37	3684	007036 5263	DMULT4, JMP	DMDONE	37
38	3685	007037 3274	DCA	DATUM-6	38
39	3686	007040 1043	TAD	F	39
40	3687	007041 3751	SAVE		40
41	3688	007042 1045	TAD	A	41
42	3689	007043 4752	MULTY		42
43	3690	007044 0004	4		43
44	3691	007045 1046	TAD	B	44
45	3692	007046 4752	MULTY		45
46	3693	007047 0005	5		46
47	3694	007050 1047	TAD	C	47
48	3695	007051 3751	SAVE		48
49	3696	007052 1041	TAD	D	49
50	3697	007053 4752	MULTY		50
51	3698	007054 0004	4		51
52	3699	007055 1042	TAD	E	52
53	3700	007056 4752	MULTY		53
54	3701	007057 0005	5		54
55					55
56					56
57					57
58					58

1	3703	007060	1043		TAD F		/C*F	1
2	3704	007061	4752		MULTY			2
3	3705	007062	0006		6			3
4	3706	007063	1301	DMDONE,	TAD	DATUM-1	/COPY RESULT	4
5	3707	007064	3045		DCA	HORD		5
6	3708	007065	1300		TAD	DATUM-2		6
7	3709	007066	3046		DCA	LORD		7
8	3710	007067	1277		TAD	DATUM-3		8
9	3711	007070	3047		DCA	OVER2		9
10	3712	007071	4301		JMS	MULDIV		10
11	3713	007072	3047		DCA	OVER2	/(NOP) - FOR 4-WORD	11
12	3714	007073	5604		JMP I	DMULT		12
13	3715							13
14	3716		7102		DATUM=	+.6	/INTERMEDIATE STORAGE	14
15	3717							15
16	3718			/	#6-LOW ORDER RESULT			16
17	3719			/	#5			17
18	3720			/	#4			18
19	3721			/	#3			19
20	3722			/	#2			20
21	3723			/	#1-HIGH ORDER RESULT			21
22	3724							22
23	3725		*7101		*DATUM-1			23
24	3726							24
25	3727	007101	0000	MULDIV,	0		/TERMINATE MULTIPLY AND DIVIDE.	25
26	3728	007102	2050		ISZ	SIGNF	/CORRECT FOR SIGN	26
27	3729	007103	4451		JMS I	MINSKI		27
28	3730	007104	4747		JMS I	NORMF	/SHIFT LEFT	28
29	3731	007105	2047		ISZ	OVER2	/ROUNDUP LAST BIT; (NOP) - FOR 4-WORD	29
30	3732	007106	5701		JMP I	MULDIV		30
31	3733							31
32	3734	007107	1041	FLDV,	TAD	AC1H	/4:DIVIDE	32
33	3735	007110	7650		SNA CLA			33
34	3736	007111	4566		ERROR2		/DIVISION BY ZERO	34
35	3737	007112	1040		TAD	EX1	/SUBTRACT EXPONENTS+1	35
36	3738	007113	7041		CMA IAC			36
37	3739	007114	7001		IAC			37
38	3740	007115	4324		JMS	SIGN	/SET UP SIGNS	38
39	3741	007116	7700		SMA CLA			39
40	3742	007117	4353		JMS	MINUS2	/NEGATE DIVISOR	40
41	3743	007120	4750		JMS I	DIVIDE	/DIVIDE	41
42	3744	007121	4301		JMS	MULDIV		42
43	3745	007122	5723		JMP I	+.1;	FPNT+1	43
44	3746	007123	6401					44
45								45
46								46
47								47
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1	3748		/				THIS SUBROUTINE PREPARES MULTIPLY AND DIVIDE	1
2	3749		/				FOR ANY COMBINATION OF SIGNED ARGUMENTS AND FOR ZERO.	2
3	3750		/				THE RESULT OF EITHER IS ZERO IF FLAC = 0.	3
4	3751		/				RESULT OF MULTIPLY IS ZERO IF EITHER IS ZERO;	4
5	3752		/				DIVISION BY ZERO IS CHECKED BEFORE THIS	5
6	3753		/				ROUTINE IS CALLED.	6
7	3754							7
8	3755		/				THE CALLING AC CONTAINS AN UPDATE VALUE FOR THE	8
9	3756		/				EXPONENT. THE RETURNING AC CONTAINS THE SIGN OF	9
10	3757		/				THE ARGUMENT FOR FURTHER TESTING BY EACH ROUTINE.	10
11	3758							11
12	3759							12
13	3760	007124	0000	SIGN,	0		/TEST AND SAVE SIGN OF RESULT	13
14	3761	007125	1044		TAD	EXP	/COMPUTE NEW EXPONENT FOR MUL-DIV.	14
15	3762	007126	3044		DCA	EXP		15
16	3763	007127	1124		TAD	P4000	/LOAD 4000 TO XOR THE SIGN BITS	16
17	3764	007130	0045		AND	HORD		17
18	3765	007131	1041		TAD	AC1H		18
19	3766	007132	7700		SMA	CLA	/RESULT MAY BE ZERO	19
20	3767	007133	7040		CMA			20
21	3768	007134	3050		DCA	SIGNF		21
22	3769	007135	1045		TAD	HORD		22
23	3770	007136	7450		SNA			23
24	3771	007137	5746		JMP	I REVIT	/ANSWER IS ZERO.	24
25	3772	007140	7710		SPA	CLA	/TAKE ABSOLUTE VALUE OF FLAC	25
26	3773	007141	4451		JMS	I MINSKI		26
27	3774	007142	1041		TAD	AC1H		27
28	3775	007143	7450		SNA		/RESULT OF EITHER MAY BE ZERO	28
29	3776	007144	5746		JMP	I REVIT		29
30	3777	007145	5724		JMP	I SIGN		30
31	3778							31
32	3779			/			SIGN OF RESULT = SIGNF	32
33	3780			/			+=-1	33
34	3781			/			--=0	34
35	3782							35
36	3783	007146	6520	REVIT,	ZERO			36
37	3784	007147	7335	NORMF,	DNORM			37
38	3785	007150	7261	DIVIDE,	DUBDIV			38
39	3786							39
40	3787		3751	SAVE=	DCA	I ;MP2		40
41	3788	007151	7256					41
42	3789		4752	MULTY=	JMS	I ;MP4		42
43	3790	007152	7200					43
44	3791							44
45	3792		0045	A=	FLAC+1			45
46	3793		0046	B=	FLAC+2			46
47	3794		0047	C=	FLAC+3			47
48	3795		0041	D=	AC1H			48
49	3796		0042	E=	AC1L			49
50	3797		0043	F=	OVER1			50
51								51
52								52
53								53
54								54
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56								56
57								57
58								58

1	3799	007153	0000	MINUS2,	0	/NEGATE OPERAND		1
2	3800	007154	7300		CLA CLL	/TRIPLE PRECISION		2
3	3801	007155	1043		TAD	OVER1		3
4	3802	007156	7041		CMA	IAC		4
5	3803	007157	3043		DCA	OVER1		5
6	3804	007160	1042		TAD	AC1L		6
7	3805	007161	7040		CMA			7
8	3806	007162	7430		SZL			8
9	3807	007163	7101		CLL	IAC		9
10	3808	007164	3042		DCA	AC1L		10
11	3809	007165	1041		TAD	AC1H		11
12	3810	007166	7040		CMA			12
13	3811	007167	7430		SZL			13
14	3812	007170	7101		CLL	IAC		14
15	3813	007171	3041		DCA	AC1H		15
16	3814	007172	5753		JMP	I	MINUS2	16
17	3815							17
18	3816	007173	0000	RESOLV,	0			18
19	3817	007174	1050		TAD	SIGNF		19
20	3818	007175	7710		SPA	CLA		20
21	3819	007176	4451		JMS	I	MINSKI	21
22	3820	007177	5773		JMP	I	RESOLV	22
23	3821							23
24	3822		*7200		PAGE			24
25	3823							25
26	3824	007200	0000	MP4,	0	/SINGLE PRECISION, UNSIGNED MULTIPLY - "MULTY"		26
27	3825	007201	7450		SNA	/NO RESULT ADDED IF ZERO		27
28	3826	007202	5600		JMP	I	MP4	28
29	3827							29
30	3828			/	FOR EAE INSERT THE FOLLOWING:			30
31	3829			/				31
32	3830			/	7203	3206	DCA .+3	32
33	3831			/	7204	1256	TAD MP2	33
34	3832			/	7205	7425	MQL MUY	34
35	3833			/	7206	0000	0	35
36	3834			/	7207	3253	DCA MP5	36
37	3835			/	7210	7501	MQA	37
38	3836			/	7211	3255	DCA MP3	38
39	3837			/	7212	5227	JMP .+15	39
40								40
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1	3839	007203	3254		DCA	MP1	/12 BITS BY 12 BITS	1
2	3840	007204	3253		DCA	MP5		2
3	3841	007205	1257		TAD	THIR		3
4	3842	007206	3255		DCA	MP3		4
5	3843	007207	7100		CLL			5
6	3844	007210	1254	MP6,	TAD	MP1		6
7	3845	007211	7010		RAR			7
8	3846	007212	3254		DCA	MP1		8
9	3847	007213	1253		TAD	MP5		9
10	3848	007214	7420		SNL			10
11	3849	007215	5220		JMP	.+3		11
12	3850	007216	7100		CLL			12
13	3851	007217	1256		TAD	MP2		13
14	3852	007220	7010		RAR			14
15	3853	007221	3253		DCA	MP5	/SAVE HIGH ORDER RESULT	15
16	3854	007222	2255		ISZ	MP3		16
17	3855	007223	5210		JMP	MP6		17
18	3856	007224	1254		TAD	MP1	/CORRECT LOW ORDER RESULT	18
19	3857	007225	7010		RAR			19
20	3858	007226	3255		DCA	MP3		20
21	3859	007227	1600		TAD I	MP4	/PICKUP SCALE FACTOR	21
22	3860	007230	7041		CIA			22
23	3861	007231	1252		TAD	DATUMA	/COMPUTE ADDRESS	23
24	3862	007232	3254		DCA	MP1	/TEMP	24
25	3863	007233	1255		TAD	MP3	/LOW ORDER PART	25
26	3864	007234	7100		CLL			26
27	3865	007235	1654		TAD I	MP1	/ACCUMULATE	27
28	3866	007236	3654		DCA I	MP1		28
29	3867	007237	2254		ISZ	MP1		29
30	3868	007240	7004		RAL			30
31	3869	007241	1253		TAD	MP5		31
32	3870	007242	1654		TAD I	MP1		32
33	3871	007243	3654		DCA I	MP1		33
34	3872	007244	7420		SNL			34
35	3873	007245	5600		JMP I	MP4	/NO CARRY	35
36	3874	007246	2254		ISZ	MP1		36
37	3875	007247	2654		ISZ I	MP1		37
38	3876	007250	5600		JMP I	MP4		38
39	3877	007251	5246		JMP	.-3	/CARRY AGAIN	39
40	3878							40
41	3879	007252	7102	DATUMA,	DATUM			41
42	3880	007253	0000	MP5,	0		/PRODUCT	42
43	3881	007254	0000	MP1,	0		/MULTIPLIER	43
44	3882	007255	0000	MP3,	0			44
45	3883	007256	0000	MP2,	0		/MULTIPLICAND	45
46	3884	007257	7764	THIR,	-14		/12 BITS	46
47	3885	007260	7751	MIF,	-27		/(-43) - FOR 4-WORD (=7735)	47
48								48
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1	3887	007261	0000	DUBDIV, 0		/2 OR 3 PRECISION DIVIDE	1
2	3888	007262	3200	DCA	MP4		2
3	3889	007263	3254	DCA	MP1		3
4	3890	007264	1260	TAD	MIF	/INIT BIT COUNTER	4
5	3891	007265	3255	DCA	MP3		5
6	3892	007266	7410	SKP			6
7	3893	007267	4527	DV3, JMS I	DOUBLE	/SHIFT FLAC LEFT	7
8	3894	007270	7100	CLL			8
9	3895	007271	1042	TAD	AC1L	/COMBINE ONE POSITION AND (4-WORD)	9
10	3896	007272	1046	TAD	LORD		10
11	3897	007273	3256	DCA	MP2	/SAVE RESULT	11
12	3898	007274	7004	RAL			12
13	3899	007275	1045	TAD	HORD	/ADD OVERFLOW	13
14	3900	007276	1041	TAD	AC1H		14
15	3901	007277	7420	SNL		/SKIP IF OVERFLOW	15
16	3902	007300	5304	JMP	.+4		16
17	3903	007301	3045	DCA	HORD	/UPDATE FLAC	17
18	3904	007302	1256	TAD	MP2		18
19	3905	007303	3046	DCA	LORD		19
20	3906	007304	7200	CLA		/CLEAR ACCUMULATOR	20
21	3907	007305	1254	TAD	MP1	/SAVE OVERFLOW BITS CIRCULARLY	21
22	3908	007306	7004	RAL			22
23	3909	007307	3254	DCA	MP1		23
24	3910	007310	1200	TAD	MP4		24
25	3911	007311	7004	RAL			25
26	3912	007312	3200	DCA	MP4		26
27	3913	007313	2255	ISZ	MP3	/TEST FOR END OF DIVIDE	27
28	3914	007314	5267	JMP	DV3		28
29	3915	007315	1254	TAD	MP1	/LOAD RESULTS	29
30	3916	007316	3046	DCA	LORD		30
31	3917	007317	1200	TAD	MP4		31
32	3918	007320	3045	DCA	HORD		32
33	3919	007321	5661	JMP I	DUBDIV	/(NOP)	33
34	3920	007322	7004	RAL		/EXTRA FOR 4-WORD	34
35	3921	007323	3335	DCA	DNORM		35
36	3922	007324	2255	ISZ	MP3	/TEST FOR END OF DIVIDE	36
37	3923	007325	5267	JMP	DV3		37
38	3924	007326	1335	TAD	DNORM		38
39	3925	007327	3045	DCA	HORD		39
40	3926	007330	1200	TAD	MP4		40
41	3927	007331	3046	DCA	LORD		41
42	3928	007332	1254	TAD	MP1		42
43	3929	007333	3047	DCA	OVER2		43
44	3930	007334	5661	JMP I	DUBDIV		44
45							45
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1	3932	007335	0000	DNORM,	0		/SUBROUTINE TO NORMALIZE FLAC	1
2	3933	007336	4775		JMS I	ABSOL3		2
3	3934	007337	4366		JMS	TEST4		3
4	3935	007340	1045		TAD	HORD		4
5	3936	007341	7450		SNA		/IS MANTISSA=0?	5
6	3937	007342	1047		TAD	OVER2		6
7	3938	007343	7450		SNA			7
8	3939	007344	1046		TAD	LORD		8
9	3940	007345	7650		SNA CLA			9
10	3941	007346	5363		JMP	EXIT3	/YES	10
11	3942	007347	1045		TAD	HORD		11
12	3943	007350	7104		CLL RAL			12
13	3944	007351	7710		SPA CLA		/WILL SHIFT BE TOO FAR?	13
14	3945	007352	5360		JMP	.+6		14
15	3946	007353	4527		JMS I	DOUBLE		15
16	3947	007354	7140		CLL CMA			16
17	3948	007355	1044		TAD	EXP		17
18	3949	007356	3044		DCA	EXP		18
19	3950	007357	5347		JMP	.-10		19
20	3951							20
21	3952	007360	4776		JMS I	RESOL3		21
22	3953	007361	4366		JMS	TEST4	/DON'T LEAVE 4000	22
23	3954	007362	5735		JMP I	DNORM		23
24	3955							24
25	3956	007363	3044	EXIT3,	DCA	EXP	/SET TO ZERO	25
26	3957	007364	5735		JMP I	DNORM	/RETURN	26
27	3958							27
28	3959	007365	6757	XRAR2,	DIV2			28
29	3960							29
30	3961	007366	0000	TEST4,	0			30
31	3962	007367	1045		TAD	HORD	/TEST FOR 4000	31
32	3963	007370	7510		SPA			32
33	3964	007371	7041		CIA			33
34	3965	007372	7710		SPA CLA			34
35	3966	007373	4765		JMS I	XRAR2	/SHIFT BACK	35
36	3967	007374	5766		JMP I	TEST4		36
37	3968							37
38	3969	007375	5571	ABSOL3,	ABSOLV			38
39	3970	007376	7173	RESOL3,	RESOLV			39
40								40
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[illegible]

1								1
2								2
3	A	=0045	CHIN	2155	DMULT4	7036	EXP 0044 FNEG 5163	3
4	ABSOL	6751	CHRT	6133	DNORM	7335	EXTR 2313 FNOR =7000	4
5	ABSOLV	5571	CIA	=7041	DNUMBR	5714	EX1 0040 FNPT 4554	5
6	ABSOL2	6153	CLA	=7200	DO	0420	F =0043 FNTABF=0374	6
7	ABSOL3	7375	CLCU	7427	DOK	2111	FADD =1000M FNTABL=2165	7
8	ACMINS	6603	CLF	0076	DONE	2127	FCONT 1101 FOR 1041	8
9	AC1H	0041	CLL	=7100	DOONE	0463	FCOS 5200 FOUTPU 0130	9
10	AC1L	0042	CMA	=7040	DOUBLE	0127	FCOUNT 5535 FPAC1 7474	10
11	ADD	0061	CML	=7020	DPCVPT	6302	FDIV =3000M FPNT 6400	11
12	ADDR	=0040	CNTR	0057	DPN	6305	FEND3 2267 FPOW =5000M	12
13	ADONE	6673	COL	1255	DPT	6145	FEXP 4620 FPRNT 5465	13
14	AF	4677	COMBOT	0226	DSAVE	5640	FEXT =0000 FPUT =6000M	14
15	ALFZ	4755	COMBUF	0132	DTST	5647	FGET =0000M FRST 3206	15
16	ALF1	4760	COMEIN=3140		DUBDIV	7261	FG02 6011 FRSTX 3215	16
17	ALF2	4763	COMEOU=3206		DUBLAD	5733	FG03 6027 FSIN 5205	17
18	ALGN	6570	COMGO =1163		DV3	7267	FG04 6034 FSUB =2000M	18
19	ALIGN	6623	COMLST=0774		E	=0042	FG05 6070 FXIT =0000	19
20	ALIST	=1372	COMMEN=0614		ECALL	1601	FIG01 6221 GECALL 1460	20
21	AMOUNT	6722	CON1 5037		ECHOLS	1624	FIG04 6261 GEND 2334	21
22	AND	=0000M	CSTAR 0225		EFOP	=0056	FINCR 1065 GERR 0340	22
23	ARCALG	4732	C100 0006		EFUN	1743	FINDLN=4555 GETARG 1403	23
24	ARCRTN	5024	C140 2554		EFUN2	1754	FINDN 2246 GETC =4545	24
25	ARGNXT	1723	C144 6140		EFUN3	2017	FINFIN 1137 GETLN =4554	25
26	ARTN	5000	C200 0123		EFUN3I	0136	FINKP 1133 GETSGN=1045	26
27	ASHFT	6665	C260 0113		ELPAR	1763	FINPUT 0131 GETVAR 1407	27
28	ASK	1202	C3 5346		END	0134	FINT =4407 GET1 2330	28
29	ATEI	4465	C5 5342		ENDFI	6243	FISW 0052 GET3 2345	29
30	ATES	4513	C7 5336		ENDLN	=4556	FIX 6724 GEXIT 0352	30
31	ATLIST=1570		C9 5332		ENDT	0135	FIXM 6753 GFND1 1505	31
32	ATSW	0056	D =0041		ENUM	1732	FLAC =0044 GINC 0070	32
33	AXIN	0010	DATUM =7102		EPAR	1710	FLAD 6506 GLIST =1377	33
34	AXOUT	0017	DATUMA 7252		EPAR2	1765	FLAG1 5162 GO 5021	34
35	B	=0046	DCA =3000M		ERASE	2204	FLAG2 4725 GONE 0232	35
36	BACK	5503	DCONP 6303		ERG	2225	FLARG 2030 GOTO 0603	36
37	BEGIN	4371	DCONT 0471		ERL	2222	FLARGP 0125 GRPTST 0744	37
38	BETZ	4766	DCOUNT 6143		ERROR2=4566		FLDV 7107 GSERCH 1426	38
39	BET1	4771	DDTJR 0004		ERROR3=4566		FLEX 6515 GS1 1437	39
40	BET2	4774	DEBGSW 0026		ERROR4=4566		FLGT 6467 GS2 1461	40
41	BF	4702	DECON 5627		ERROR5 2725		FLIMIT 1075 GS3 1441	41
42	BFX	4557	DECONV 5600		ERR2 2726		FLINTP 6200 GS4 1454	42
43	BFXX	4556	DECP 5533		ERT	2214	FLIST1 0577 GTEM 0021	43
44	BOTTOM	0035	DECR 5521		ERV	2217	FLIST2 0574 GZERR 0362	44
45	BUFBEG=3217		DELETE=4565		ERVX	2237	FLMY 6563 G8L 4466	45
46	BUFFER=7470		DF 4710		ER5	4555	FLOG 5040 HINBUF 0037	46
47	BUFR	0060	DGRP 0425		ESCA	2532	FLOP 1674 HLT =7402	47
48	BUFST	5531	DGRP1 0441		ETERM	1647	FLOUT 5556 HORD 0045	48
49	C	=0047	DIG 5543		ETERMN	1644	FLOUTP 6000 HREAD 6321	49
50	CCR	0077	DIGIT 5713		ETERM1	1627	FLPT 6465 HREAD2 6324	50
51	CDF	=7000	DIGITS=0006		ETERM2	1655	FLSU 6505 HSGO 6364	51
52	CEXP	6503	DIVIDE 7150		EVAL	1613	FLTONE 2405 HSPSW 6375	52
53	CEX1	6504	DIV1 5754		EXIT	2646	FLT XR 0014 HSPX 6361	53
54	CF	4705	DIV2 6757		EXITJ	2661	FLT XR2 0015 HSWITC 6343	54
55	CFRS	0133	DMDONE 7063		EXIT1	5034	FLTZER 2407 HTST 6376	55
56	CFRSX	0137	DMPSW 0100		EXIT2	5302	FMUL =4000M IAC =7001	56
57	CHAR	0066	DMULT 7004		EXIT3	7363	FM12 6142 IBAR 0212	57
58								58

1											1
2											2
3	IECALL	1037	L1	5126	NEGP	4724	PI	5312	RESOL	6752	3
4	IF	1013	L2	5131	NOP	=7000	PIOT	5316	RESOLV	7173	4
5	IF1	1035	L3	5134	NORF	6513	PI2	5036	RESOL3	7376	5
6	IF3	1025	L4	5137	NORM	6567	PLCE	=5536	RESOL5	6304	6
7	IGNOR	0217	L8A	4550	NORMF	7147	PLS	=6026	RESTR	6377	7
8	ILIST	0771	L8AX	4553	NOX	6675	POPA	=1413	RET	5452	8
9	IN	5513	L8AY	4552	NOX1	6711	POPF	=4544	RETRN	1563	9
10	INBUF	0034	L8B	4551	NOX2	6704	POPJ	=5541	RETURN	=5536	10
11	INDEV	0064	MBREAK	2602	OM12	5530	PPTEN	6144	REVIT	7146	11
12	INDRCT	6463	MCOM	1136	ONE	4716	PRINTC	=4551	RFC	=6014	12
13	INFIX	=2401	MCR	0116	OOUT	4544	PRINTD	7550	RMF	=6244	13
14	INLIST	=0570	MD	5526	OP	3115	PRNT	2442	RND2	5527	14
15	INORM	6307	MEQ	1135	OPMINS	6565	PRNTI	6132	ROOTGO	7461	15
16	INPUT	0756	MF	0602	OPNEXT	1622	PRNTLN	=4553	ROT	2557	16
17	INPUTX	0271	MFLT	0117	OPTABL	1731	PRNT2	3114	ROUND	6151	17
18	INSUB	0036	MIF	7260	OPTRI	2665	PRNT8	7527	RRB	=6012	18
19	INTEGE	0053	MINE	5662	OPTR0	2664	PROC	0611	RSF	=6011	19
20	INTRPT	2603	MINSKI	0051	OPTR0	2663	PROCES	0610	RTL	=7006	20
21	IOBUF	=3120	MINUSA	0112	OPUT	5532	PSIN	0165	RTL6	=4557	21
22	IOF	=6002	MINUSE	6301	OUT	2465	PTCH	0126	RTR	=7012	22
23	ION	=6001	MINUSZ	5663	OUTA	5536	PTEN	6275	RUBIT	2555	23
24	IPART	1040	MINUS2	7153	OUTCR	2476	PTEST	1457	RUB1	3004	24
25	IRETN	0227	MOD	5215	OUTDEV	0063	PT1	0030	RUB2	3042	25
26	ISZ	=2000M	MODIFY	1256	OUTDG	6154	PUSHA	=4542	RUB3	3030	26
27	ITABLE	=6573	MPER	0115	OUTL	1354	PUSHF	=4543	RUB4	3037	27
28	ITER1	7470	MPLUS	5664	OUTX	2475	PUSHJ	=4540	RUB5	3041	28
29	I33	2414	MP1	7254	OVER1	0043	P13	0005	R6	5441	29
30	JMP	=5000M	MP2	7256	OVER2	0047	P17	0107	SADR	6150	30
31	JMS	=4000M	MP3	7255	01	4370	P177	0106	SAVAC	2600	31
32	JUMP	6462	MP4	7200	02	4561	P2	4566	SAVE	=3751	32
33	KINT	2625	MP5	7253	04	4412	P2000	0373	SAVLK	2601	33
34	KRB	=6036	MP6	7210	05	4563	P27	6750	SBAR	1302	34
35	KSF	=6031	MSPACE	5665	06	4564	P277	0110	SCHAR	1273	35
36	K5	5525	MULDIV	7101	P	=0000	P3	2034	SCONT	1270	36
37	LASTLN	0025	MULT	6566	PACBUF	2502	P337	0075	SCOUNT	5534	37
38	LASTOP	0055	MULTY	=4752	PACKC	=4546	P377	2553	SET	=1041	38
39	LASTV	0031	MULT10	5667	PACKST	0027	P40	2552	SEX	1340	39
40	LCON	0371	MULT2	5715	PACX	2530	P4000	0124	SEXC	0740	40
41	LG2E	4713	M10PT	6147	PALG	5261	P43	6310	SFOUND	1306	41
42	LIBRAR	7503	M100	=0101	PARTES	2047	P7	4565	SGOT	1312	42
43	LINENO	0067	M11	0121	PA1	2524	P7600	0104	SIGN	7124	43
44	LISTG0	=1370	M12	2413	PC	0022	P77	0122	SIGNF	0050	44
45	LIST3	=0077	M137	2357	PCHECK	5245	P7700	0101	SIN	2662	45
46	LIST6	=0072	M140	2556	PCHK	0510	P7740	0372	SKP	=7410	46
47	LIST7	=0074	M144	6137	PCK1	2535	RAL	=7004	SMA	=7500	47
48	LOG2	5157	M2	0111	PC1	0614	RANO	1530	SMIN	6136	48
49	LOG5	5142	M20	0105	PDLXR	0013	RAR	=7010	SMP	=6101	49
50	LOG6	5145	M240	0114	PDP	4562	RAR1	6571	SMSP	6134	50
51	LOG7	5150	M260	1526	PDP5	4570	RAR2	6572	SNA	=7450	51
52	LOG8	5153	M271	1527	PDP5X	4463	RDIV	0152	SNL	=7420	52
53	LOOKUP	4571	M4	6141	PDP8I	4567	READC	=4552	SORTB	1314	53
54	LOOP01	6431	M40	2356	PD2	0534	RECOVR	2740	SORTC	=4550	54
55	LORD	0046	M5	0120	PD3	0554	RECOVX	2761	SORTCN	0054	55
56	LPRTST	2035	M77	0103	PEQ	6135	REMAIN	5712	SORTJ	=4547	56
57	LP7	7556	NAGSW	0065	PER	0102	REPT	6146	SPA	=7510	57
58											58

1				1
2				2
3	SPECIA=6777	T12	4426	3
4	SPLAT 3051	T2	0071	4
5	SPNOR =4560	T3	0033	5
6	SQCON1 7467	UTE	2276	6
7	SQEND 7465	UTQ	2305	7
8	SRETN 0261	UTRA	2274	8
9	SRNLST=1363	UTX	2316	9
10	START 0177	VAL	=0032	10
11	STARTL 5064	WALL	0664	11
12	STARTV=0060	WORDS	=0003	12
13	SUBS =1517	WRITE	0635	13
14	SZA =7440	WTESTG	0667	14
15	SZL =7430	WTEST2	0653	15
16	T =0000	WX	0673	16
17	TABLE 6464	X	5322	17
18	TAD =1000M	XABS	2014	18
19	TAG1 6723	XADC	1343	19
20	TASK 1204	XCT	0020	20
21	TASK4 1253	XCTIN	0062	21
22	TCF =6042	XDELET	2062	22
23	TCRLF 1251	XDYS	1142	23
24	TCRLF2 1246	XENDLN	2360	24
25	TDUMP 3052	XF	4560	25
26	TELSW 0016	XFIND	2242	26
27	TEM 5156	XGETLN	0302	27
28	TEMP 4726	XIN	6306	28
29	TEN 6271	XINPUT	5666	29
30	TENPT 6152	XINT	1160	30
31	TERMS =1770	XI33	2666	31
32	TESTA 0322	XOUTL	2676	32
33	TESTC =4564	XPOPJ	1565	33
34	TESTN =4561	XPRNT	2425	34
35	TEST2 6736	XPUSHA	0477	35
36	TEST4 7366	XPUSHJ	0521	36
37	TEXTP =0017	XRAN	1553	37
38	TGO 5400	XRAR2	7365	38
39	THIR 7257	XRT	0011	39
40	THISLN 0023	XRTL6	0413	40
41	THISOP 0024	XRT2	0012	41
42	TINTR 1241	XSGN	2010	42
43	TLIST =1400	XSORTC	0721	43
44	TLIST2 1404	XSPNOR	1517	44
45	TLIST3=2377	XSQR	5326	45
46	TLS =6046	XSQRT	7400	46
47	TPC =6044	XSQ2	4676	47
48	TQUOT 1232	XTESTC	0700	48
49	TRAD 6573	XTESTN	1533	49
50	TSF =6041	XT3	0717	50
51	TSTGRP=4563	XYZ	2451	51
52	TSTLPR=4562	X1	5035	52
53	TWO 4721	X2	4675	53
54	TWOPI 5306	ZERO	6520	54
55	TYPE 1203			55
56	TYPE2 1226			56
57	T1 0032			57
58				58

1		1
2		2
3	NO ERRORS DETECTED	3
4		4
5	NO LINKS GENERATED	5
6		6
7	7K MEMORY UTILIZED	7
8		8
9	2 FILES CREATED	9
10		10
11	657 SYMBOLS	11
12		12
13		13
14		14
15		15
16		16
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3										3
4										4
5	00000	01111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	5
6	00100	11111111	11111111	11111111	11111111	11111111	11111111	11111110	00000011	6
7										7
8	00200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	8
9	00300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	9
10										10
11	00400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11
12	00500	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	12
13										13
14	00600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	14
15	00700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	15
16										16
17	01000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	17
18	01100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	18
19										19
20	01200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	20
21	01300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	21
22										22
23	01400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	23
24	01500	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	24
25										25
26	01600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	26
27	01700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	27
28										28
29	02000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	29
30	02100	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	30
31										31
32	02200	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	32
33	02300	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	33
34										34
35	02400	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	35
36	02500	11111111	11111111	11111111	11111111	11111111	11111111	11110000	00000000	36
37										37
38	02600	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	38
39	02700	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	39
40										40
41	03000	11111111	11111111	11111111	11111111	11111111	11111111	11111111	11111111	41
42	03100	11111111	11111111	00000000	00000000	00000000	00000000	00000000	00000000	42
43										43
44	03200	00000011	11111110	00000000	00000000	00000000	00000000	00000000	00000000	44
45	03300									45
46										46
47	03400									47
48	03500									48
49										49
50	03600									50
51	03700									51
52										52
53										53
54										54
55										55
56										56
57										57
58										58

[illegible]